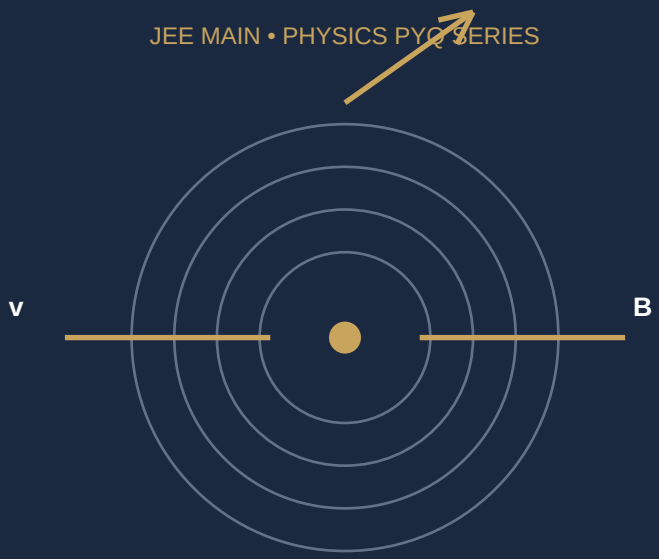
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JEE MAIN • PHYSICS PYQ SERIES



MOVING CHARGES AND MAGNETISM

PREVIOUS YEAR QUESTIONS

JEE Main • Questions with Solutions

PRACTICE. ANALYSE. IMPROVE.

- JEE Main PYQs from multiple exam sessions
- Concept-based worked solutions
- Lorentz force, magnetic fields and charged-particle motion

Moving Charges and Magnetism

Question1

1 μC charge moving with velocity $\vec{v} = (\hat{i} - 2\hat{j} + 3\hat{k})$ m/s in the region of magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 5\hat{k})$ T. The magnitude of force acting on it is $\sqrt{\alpha} \times 10^{-6}$ N. The value of α is _____.

JEE Main 2026 (Online) 2nd April Morning Shift

Answer: 171

Solution:

The magnetic force acting on a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is given by :

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$\vec{F} = q \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_x & v_y & v_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\vec{F} = 1 \times 10^{-6} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 2 & 3 & -5 \end{vmatrix}$$

Expanding the determinant:

$$\vec{F} = 1 \times 10^{-6}(\hat{i}[(-2)(-5) - (3)(3)] - \hat{j}[(1)(-5) - (3)(2)] + \hat{k}[(1)(3) - (-2)(2)])$$

$$\vec{F} = 1 \times 10^{-6}[\hat{i}(10 - 9) - \hat{j}(-5 - 6) + \hat{k}(3 + 4)]$$

$$\vec{F} = 1 \times 10^{-6}[1\hat{i} + 11\hat{j} + 7\hat{k}]$$

So, the magnitude $|\vec{F}|$ is:

$$|\vec{F}| = 10^{-6}\sqrt{1^2 + 11^2 + 7^2}$$

$|\vec{F}| = 10^{-6}\sqrt{1+121+49}$

$|\vec{F}| = \sqrt{171} \times 10^{-6}\ \text{N}$

$\Rightarrow \sqrt{\alpha} \times 10^{-6}\ \text{N} = \sqrt{171} \times 10^{-6}\ \text{N}$

$\Rightarrow \alpha = 171$

Therefore, the value of α is 171. Thus, the correct answer is 171.

Question2

A circular coil of radius 2 cm and 125 turns carries a current of 1 A . The coil is placed in a uniform magnetic field of magnitude 0.4 T . The axis of the coil makes an angle of 30° with the direction of the magnetic field. The torque acting on the coil is $\alpha \times 10^{-4}\ \text{N.m}$. The value of α is ____ .

$(\pi = 3.14)$

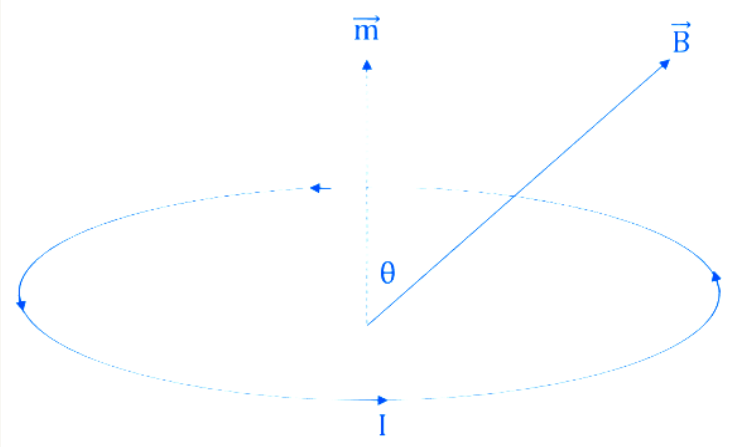
JEE Main 2026 (Online) 4th April Evening Shift

Answer: 314

Solution:

The magnetic dipole moment of a single loop of wire carrying current I and enclosing area A is given by

$\text{m} = \text{IA}$



For a coil consisting of N turns, the total magnetic moment is :

$\mathrm{m}=\mathrm{NIA}$

Since the coil is circular with radius r , the area A is $\pi \mathrm{r}^2$.

$\mathrm{m}=\mathrm{NI}\left(\pi \mathrm{r}^2\right)$

The torque experienced by a magnetic dipole in a uniform magnetic field $\overrightarrow{\mathrm{B}}$ is given by the vector cross product :

$\vec{\tau}=\overrightarrow{\mathrm{m}} \times \overrightarrow{\mathrm{B}}$

The magnitude of this torque is :

$\tau=\mathrm{mB} \sin \theta$

Where θ is the angle between the magnetic moment (which is along the axis of the coil) and the magnetic field direction.

The number of turns is $\mathrm{N}=125$.

The current in the coil is $\mathrm{I}=1 \mathrm{~A}$.

The radius of circular coil is $\mathrm{r}=2 \mathrm{~cm}=0.02 \mathrm{~m}$.

The magnitude of magnetic field strength is, $\mathrm{B}=0.4 \mathrm{~T}$.

The angle between the magnetic field and magnetic moment is $\theta=30^{\circ}$.

Substituting the values into the torque formula:

$\tau=\left(\mathrm{NI} \pi \mathrm{r}^2\right) \mathrm{B} \sin \theta$

$\Rightarrow \tau=\left(125 \times 1 \times 3.14 \times(0.02)^2\right) \times 0.4 \times \sin 30^{\circ}$

$\Rightarrow \tau=(125 \times 3.14 \times 0.0004) \times 0.4 \times 0.5$

$\Rightarrow \tau=314 \times 10^{-4} \mathrm{~N} . \mathrm{m}=\alpha \times 10^{-4} \mathrm{~N} \cdot \mathrm{~m} \Rightarrow \alpha=314$

Therefore, the value of a is 314 .

Question3

The charged particle moving in a uniform magnetic field of $(3\hat{i}+2\hat{j})\mathrm{T}$ has an acceleration $\left(4\hat{i}-\frac{x}{2}\hat{j}\right)\mathrm{m/s^2}$. The value of x is

JEE Main 2026 (Online) 5th April Morning Shift

Answer: 12

Solution:

The force ($\vec{F}$) acting on a particle with charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is given by the Lorentz force law:

$\overrightarrow{\mathrm{F}}=\mathrm{q}(\overrightarrow{\mathrm{v}} \times \overrightarrow{\mathrm{B}})$

According to Newton's second law, $\overrightarrow{\mathrm{F}}=\mathrm{m} \overrightarrow{\mathrm{a}}$.

Therefore,

$\mathrm{m} \overrightarrow{\mathrm{a}}=\mathrm{q}(\overrightarrow{\mathrm{v}} \times \overrightarrow{\mathrm{B}})$

The resulting vector is always perpendicular to both $\vec{v}$ and $\vec{B}$.

Since $\overrightarrow{\mathrm{a}}$ is in the same direction as $\overrightarrow{\mathrm{F}}$, the acceleration vector $\overrightarrow{\mathrm{a}}$ must be perpendicular to the magnetic field vector $\overrightarrow{\mathrm{B}}$.

$\overrightarrow{\mathrm{a}} \perp \overrightarrow{\mathrm{B}}$

If two vectors are perpendicular, their dot product must be zero:

$\overrightarrow{\mathrm{a}} \cdot \overrightarrow{\mathrm{B}}=0$

The magnetic field is given as $\vec{B}=3\hat{\imath}+2\hat{\jmath}$ and $\vec{a}=4\hat{\imath}-\frac{x}{2}\hat{\jmath}$:

$\left(4\hat{\imath}-\frac{x}{2}\hat{\jmath}\right) \cdot (3\hat{\imath}+2\hat{\jmath})=0$

$\Rightarrow (4 \times 3)+\left(-\frac{x}{2} \times 2\right)=0$

$\Rightarrow 12-x=0$

$\Rightarrow x=12$

Therefore, the value of x is 12 .

Question4

A moving coil of galvanometer when shunted with 2Ω resistance gives a full scale deflection for a current of 500 mA . When a resistance of 470Ω is connected in series it gives a full scale

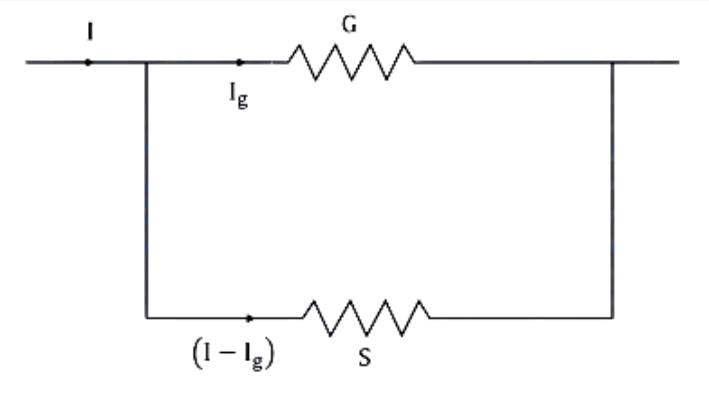
deflection for 10 V potential applied on it. The value of resistance of galvanometer coil is ____ Ω.

JEE Main 2026 (Online) 6th April Evening Shift

Answer: 50

Solution:

When a galvanometer of resistance G is shunted with resistance S, the total current I splits. The current through the galvanometer is $\mathrm{I_g}$.



The potential across the galvanometer and shunt is equal:

$$\mathrm{I_g G} = (\mathrm{I} - \mathrm{I_g})\mathrm{S}$$

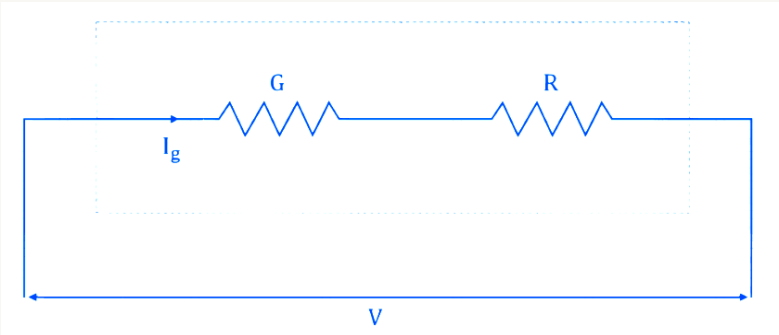
The resistance of shunt is $\mathrm{S} = 2\Omega$ and the current to measure with full deflection is $\mathrm{I} = 500\,\mathrm{mA} = 0.5\,\mathrm{A}$

$$I_g G = (0.5 - I_g) \times 2$$

$$\Rightarrow \mathrm{I_g G} = 1 - 2\mathrm{I_g}$$

$$I_g(G+2) = 1 \Rightarrow I_g = \frac{1}{G+2} \ldots (i)$$

When a resistance R is connected in series, the total potential V is :



$V = I_g(G + R)$

The resistance added in series is given as $\mathrm{R} = 470\Omega$ and $\mathrm{V} = 10\,\mathrm{V}$:

$10 = I_g(G + 470) \Rightarrow I_g = \frac{10}{G+470} \ldots (ii)$

Since the full-scale deflection current $\mathrm{I_g}$ is constant for the galvanometer,

$\frac{1}{G+2} = \frac{10}{G+470}$

$\Rightarrow \mathrm{G} + 470 = 10(\mathrm{G} + 2)$

$\Rightarrow \mathrm{G} + 470 = 10\mathrm{G} + 20$

$\Rightarrow 470 - 20 = 10\mathrm{G} - \mathrm{G}$

$\Rightarrow 450 = 9\mathrm{G}$

$\Rightarrow \mathrm{G} = \frac{450}{9} = 50\Omega$

Therefore, the value of resistance of the galvanometer coil is 50Ω.

Question5

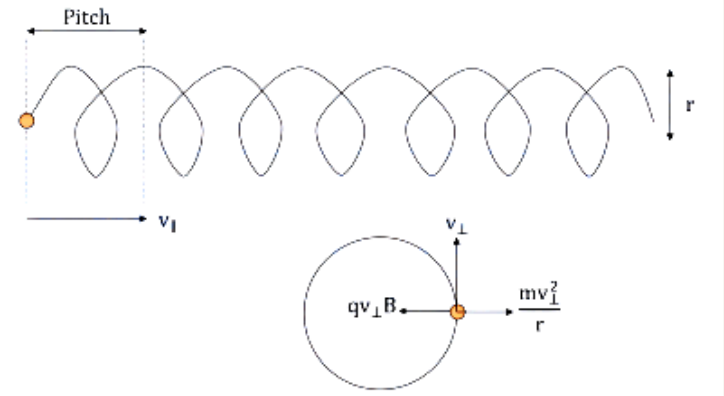
A 5 mg particle carrying a charge of $5\pi \times 10^{-6}\mathrm{C}$ is moving with velocity of $(3\hat{i} + 2\hat{k}) \times 10^{-2}\ \mathrm{m/s}$ in a region having magnetic field $\vec{B} = 0.1\hat{k}\ \mathrm{Wb/m^2}$. It moves a distance of α meter along $\hat{k}$ when it completes 5 revolutions. The value of α is ____.

JEE Main 2026 (Online) 8th April Evening Shift

Answer: 2

Solution:

When a charged particle enters a magnetic field with a velocity that has both a component perpendicular to the field and a component parallel to it, the particle follows a helical path.



The perpendicular component $(\mathrm{v}_{\perp})$ causes circular motion due to the Lorentz force $(\mathrm{F}=\mathrm{qv}_{\perp}\mathrm{B})$.

The parallel component ($\mathrm{v}_{\|}$) remains constant because the magnetic force on it is zero ($\mathrm{F}=\mathrm{qv}_{\|}\mathrm{B}\sin 0^{\circ}=0$), causing the particle to move linearly along the field lines.

The velocity is given as $\vec{v}=(3\hat{\imath}+2\hat{k})\times 10^{-2}\,\mathrm{m/s}$ and the magnetic field is $\vec{B}=0.1\hat{\mathrm{k}}\mathrm{T}$.

The component along the z -axis $(\widehat{\mathrm{k}})$ is $2\times 10^{-2}\,\mathrm{m/s}$.

The component along the x -axis (î) is $3\times 10^{-2}\,\mathrm{m/s}$.

For one revolution, the time taken is determined by the circular component of the motion.

Centripetal force is provided by the magnetic force:

$$\frac{mv_\perp^2}{r}=qv_\perp B\Rightarrow r=\frac{mv_\perp}{qB}$$

The time for one revolution is.

$$\mathrm{T}=\frac{2\pi \mathrm{r}}{\mathrm{v}_{\perp}}=\frac{2\pi\,\mathrm{m}}{\mathrm{qB}}$$

The distance moved along the field lines in 1 revolution is called the Pitch.

$$\text{Pitch}=\mathrm{v}_{\|}\times\mathrm{T}$$

For 5 revolutions, the total distance (a) is:

$$\mathrm{a}=5\times\mathrm{v}_{\|}\times\mathrm{T}=5\times\mathrm{v}_{\|}\times\left(\frac{2\pi\,\mathrm{m}}{\mathrm{qB}}\right)$$

The mass of charged particle is $\mathrm{m}=5\mathrm{mg}=5\times 10^{-6}\,\mathrm{kg}$.

The magnitude of charge on the particle is $\mathrm{q}=5\pi\times 10^{-6}\mathrm{C}$

The parallel component of velocity to magnetic field is $\mathrm{v}_{\|}=2\times 10^{-2}\,\mathrm{m/s}$

The magnitude of magnetic field in the region is $\mathrm{B}=0.1\,\mathrm{T}$

So, the distance travelled is;

$$a=5\times\left(2\times 10^{-2}\right)\times\frac{2\pi\times\left(5\times 10^{-6}\right)}{\left(5\pi\times 10^{-6}\right)\times 0.1}$$

$\Rightarrow \mathrm{a} = 10 \times 10^{-2} \times \frac{10}{5 \times 0.1}$

$\Rightarrow \mathrm{a} = 10^{-1} \times \frac{10}{0.5}$

$\Rightarrow a = 0.1 \times 20 = 2$ meters

Therefore, the distance a moved along the $\hat{k}$ direction is 2 meters.

Question6

A current carrying solenoid is placed vertically and a particle of mass m with charge Q is released from rest. The particle moves along the axis of solenoid. If g is acceleration due to gravity then the acceleration (a) of the charged particle will satisfy :

JEE Main 2026 (Online) 21st January Morning Shift

Options:

A.

$a > g$

B.

$a = 0$

C.

$0 < a < g$

D.

$a = g$

Answer: D

Solution:

A current-carrying solenoid produces a magnetic field ($\overrightarrow{\mathrm{B}}$). For an ideal solenoid, the magnetic field inside is uniform and directed parallel to its axis.

The magnetic force $\left(\vec{F}_m\right)$ on a moving charge Q with velocity $\vec{v}$ is given by the Lorentz force formula :

$\overrightarrow{\mathrm{F}}_{\mathrm{m}}=\mathrm{Q}(\overrightarrow{\mathrm{v}}\times\overrightarrow{\mathrm{B}})$

In this case, the charged particle is released from rest along the axis. The magnetic field $\vec{B}$ is also along the axis. Therefore, the velocity $\vec{v}$ (once the particle starts moving) and the magnetic field $\vec{B}$ are parallel to each other.

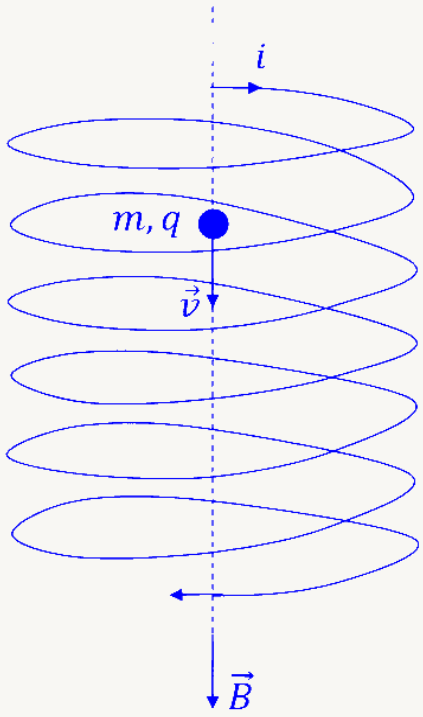
The magnitude of the magnetic force is the cross product of $\mathrm{Q}(\overrightarrow{\mathrm{v}}\times\overrightarrow{\mathrm{B}})$,

$F_m = QvB\sin\theta$

Since the angle θ between the velocity and the magnetic field is 0° :

$\mathrm{F}_{\mathrm{m}}=\mathrm{Q}v\,\mathrm{B}\sin 0^\circ$

The magnetic field exerts no force on the particle moving along the axis.



The solenoid is placed vertically. When the particle of mass m is released, the only non-zero force acting on it is the force of gravity $\left(\overrightarrow{\mathrm{F}}_{\mathrm{g}}\right)$:

$\overrightarrow{\mathrm{F}}_{\mathrm{g}}=\mathrm{mg}$

This force acts vertically downwards along the axis of the solenoid.

According to Newton's Second Law ($\mathrm{F}_{\text{net}}=\mathrm{ma}$) :

$mg = ma$

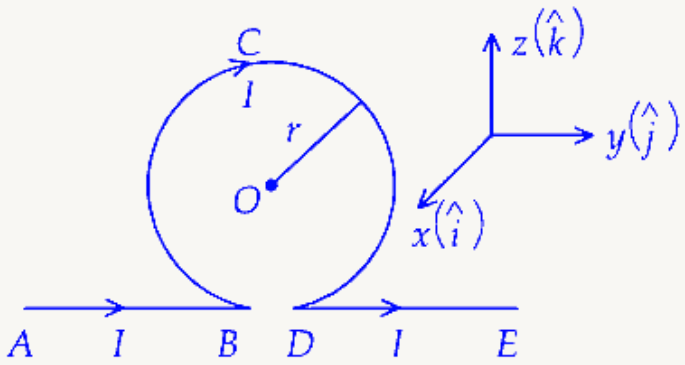
$\Rightarrow a = g$

Since the magnetic force is zero the particle falls freely under gravity. Thus, its acceleration is exactly equal to g.

Hence, the correct option is (D).

Question7

An infinitely long straight wire carrying current I is bent in a planar shape as shown in the diagram. The radius of the circular part is r. The magnetic field at the centre O of the circular loop is :



JEE Main 2026 (Online) 21st January Evening Shift

Options:

A.

$-\frac{\mu_0}{2\pi}\frac{I}{r}(\pi+1)\hat{i}$

B.

$-\frac{\mu_0}{2\pi}\frac{I}{r}(\pi-1)\hat{i}$

C.

$\frac{\mu_0}{2\pi}\frac{I}{r}(\pi+1)\hat{i}$

D.

$\frac{\mu_0}{2\pi}\frac{I}{r}(\pi-1)\hat{i}$

Answer: B

Solution:

The segments AB and DE together form an infinitely long straight wire.

For an infinite wire the magnitude of the magnetic field at distance r is given as

$B_{straight} = \frac{\mu_0 I}{2\pi r}$

Using the Right-Hand Thumb Rule, if the thumb points in the direction of current I (along +y), the fingers curl out of the page at point O , i.e., in the -x direction.

$\vec{B}_{straight} = \frac{\mu_0 I}{2\pi r}\hat{i}$

The circular part is a nearly complete loop of radius r .

For a circular loop at its centre the magnitude of the magnetic field is given as,

$B_{loop} = \frac{\mu_0 I}{2r}$

Using the Right-Hand Rule for loops, curling the fingers in the direction of the current (clockwise in the yz - plane) the thumb points into the page, which is the -x direction.

$\vec{B}_{loop} = -\frac{\mu_0 I}{2r}\hat{i}$

The total field is the vector sum of both parts :

$\vec{B}_{total} = \vec{B}_{straight} + \vec{B}_{loop}$

$\Rightarrow \vec{B}_{total} = \left(\frac{\mu_0 I}{2\pi r}\right)\hat{i} + \left(-\frac{\mu_0 I}{2r}\right)\hat{i}$

$\Rightarrow \vec{B}_{total} = \frac{\mu_0 I}{2r}\left[\frac{1}{\pi} - 1\right]\hat{i} = \frac{\mu_0 I}{2\pi r}[1-\pi]\hat{i} = -\frac{\mu_0 I}{2\pi r}[\pi - 1]\hat{i}$

So, the magnetic field at the centre O of the circular loop is $= -\frac{\mu_0 I}{2\pi r}[\pi - 1]\hat{i}$.

Hence, the correct option is (B).

Question8

The current passing through a conducting loop in the form of equilateral triangle of side $4\sqrt{3}$ cm is 2 A . The magnetic field at its centroid is $\alpha \times 10^{-5}$ T. The value of α is ____ .

(Given : $\mu_o = 4\pi \times 10^{-7}$ SI units)

JEE Main 2026 (Online) 23rd January Evening Shift

Options:

A.

$3\sqrt{3}$

B.

$2\sqrt{3}$

C.

$\sqrt{3}$

D.

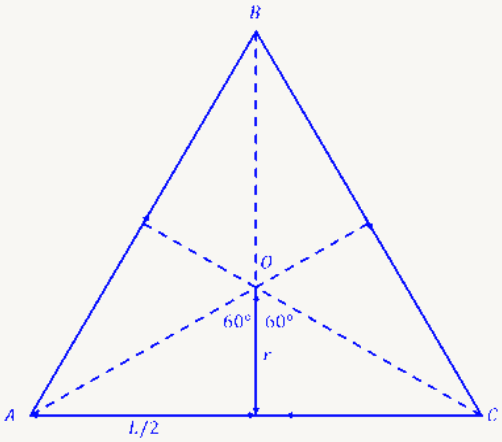
$\frac{\sqrt{3}}{2}$

Answer: A

Solution:

Let L be the side of the equilateral triangle, where $L = 4\sqrt{3}$ cm $= 4\sqrt{3} \times 10^{-2}$ m.

The centroid O is equidistant from all three sides. Let r be the perpendicular distance from the centroid to any side.



In an equilateral triangle :

$\tan 60^\circ = \sqrt{3} = \frac{L/2}{r} \Rightarrow r = \frac{L}{2\sqrt{3}}$

Substituting the value of L :

$\mathrm{r} = \frac{4\sqrt{3}\times 10^{-2}}{2\sqrt{3}} = 2 \times 10^{-2}\ \mathrm{m}$

The magnetic field B_1 due to a straight wire of finite length at a distance r is given by :

$\mathrm{B_1} = \frac{\mu_0 \mathrm{I}}{4\pi \mathrm{r}}(\sin\theta_1 + \sin\theta_2)$

Here, $\theta_1 = \theta_2 = 60^\circ$

$B_1 = \frac{\mu_0 I}{4\pi r}(\sin 60^\circ + \sin 60^\circ) = \frac{\mu_0 I}{4\pi r}(2\sin 60^\circ)$

$\Rightarrow B_1 = \frac{\mu_0 I}{4\pi r}\left(2 \times \frac{\sqrt{3}}{2}\right) = \frac{\mu_0 I\sqrt{3}}{4\pi r}$

Since there are three sides and the current flows in a way that all three fields at the centroid point in the same direction,

$B_{net} = 3 \times B_1 = \frac{3\sqrt{3}\mu_0 I}{4\pi r}$

Substituting the given values :

- 1. $\mu_0 = 4\pi \times 10^{-7}$
- 2. $\mathrm{I} = 2\ \mathrm{A}$
- 3. $\mathrm{r} = 2 \times 10^{-2}\ \mathrm{m}$

$\mathrm{B_{net}} = \frac{3\sqrt{3}\times(4\pi\times 10^{-7})\times 2}{4\pi\times(2\times 10^{-2})}\ \mathrm{T}$

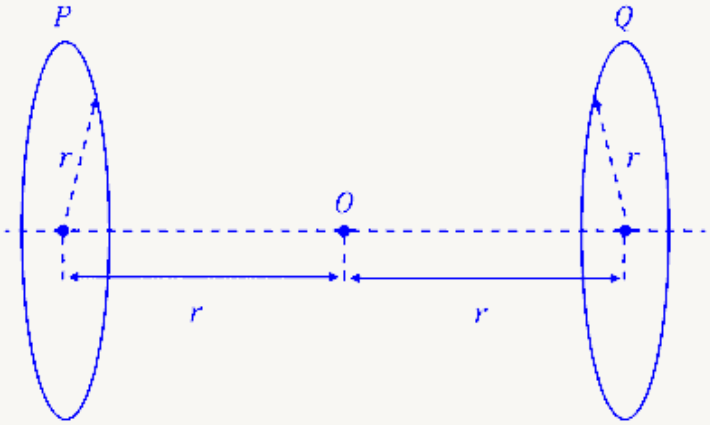
$\Rightarrow \mathrm{B_{net}} = \frac{3\sqrt{3}\times 10^{-7}\times 2}{2\times 10^{-2}}\ \mathrm{T}$

$\Rightarrow B_{net} = 3\sqrt{3} \times 10^{-5}\ T = \alpha \times 10^{-5}\ T$

$\Rightarrow \alpha = 3\sqrt{3}$

Question9

Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and $4I$ respectively in clockwise direction as seen from O. The net magnetic field at O is :



JEE Main 2026 (Online) 24th January Evening Shift

Options:

A.

$\frac{\mu_o I}{4\sqrt{2}r}$ towards Q

B.

$\frac{3\mu_o I}{4\sqrt{2}r}$ towards Q

C.

$\frac{3\mu_0 I}{4\sqrt{2}r}$ towards P

D.

$\frac{\mu_o I}{4\sqrt{2}r}$ towards P

Answer: B

Solution:

The magnetic field (B) on the axis of a circular loop of radius r at a distance x from its center is given by :

$$\mathrm{B}=\frac{\mu_0 \mathrm{Ir}^2}{2\left(\mathrm{r}^2+\mathrm{x}^2\right)^{3/2}}$$

The point O is at a distance $\mathrm{x}=\mathrm{r}$ from the center of both loops P and Q .

For loop P :

Current is $\mathrm{i}=\mathrm{I}$.

The magnetic field $\mathrm{B_P}$ at O due to loop P is :

$$\mathrm{B_P}=\frac{\mu_0 \mathrm{Ir}^2}{2\left(\mathrm{r}^2+\mathrm{r}^2\right)^{3/2}}=\frac{\mu_0 \mathrm{I}}{4\sqrt{2}\mathrm{r}}$$

Using the right-hand thumb rule, the current is clockwise as seen from O , so the field $\mathrm{B_P}$ due to loop P points towards P.

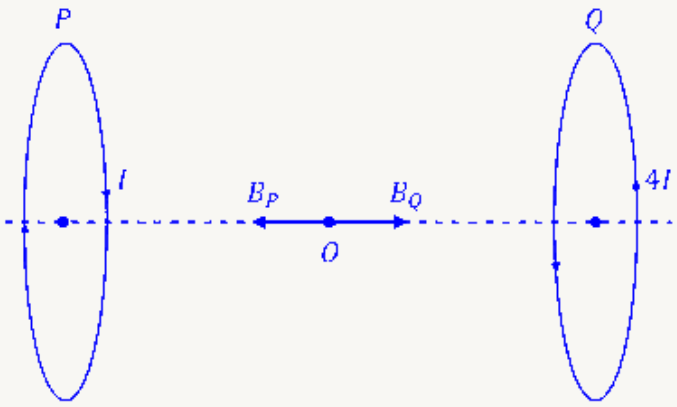
For loop Q :

Current is $\mathrm{i}=4\mathrm{I}$.

The magnetic field $\mathrm{B_Q}$ at O due to loop Q is:

$$B_Q=\frac{\mu_0(4I)r^2}{2\left(r^2+r^2\right)^{3/2}}=\frac{4\mu_0 I}{4\sqrt{2}r}$$

Using the right-hand thumb rule, the current is clockwise as seen from O , so the field $\mathrm{B_Q}$ due to loop Q points towards Q.



The net magnetic field (B_{net}) at O is the vector sum of these opposing fields. Since $B_Q > B_P$, the net field will be in the direction of $\mathrm{B_Q}$ (towards Q):

$$B_{net}=B_Q-B_P=\frac{4\mu_0 I}{4\sqrt{2}r}-\frac{\mu_0 I}{4\sqrt{2}r}=\frac{3\mu_0 I}{4\sqrt{2}r}$$

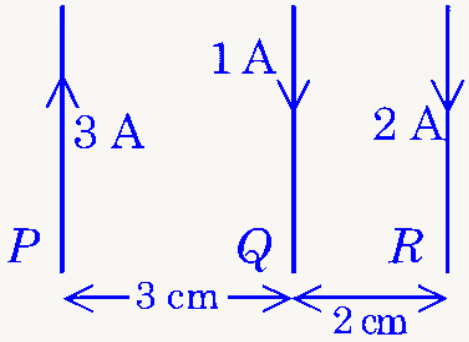
Therefore, the resulting magnetic field at the midpoint is $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards Q.

Hence, the correct option is (B).

Question10

Three long straight wires carrying current are arranged mutually parallel as shown in the figure. The force experienced by 15 cm length of wire Q is ____ .

$\left(\mu_o = 4\pi \times 10^{-7}\ \mathrm{T.m/A}\right)$



JEE Main 2026 (Online) 28th January Morning Shift

Options:

A.

6×10^{-6} N towards P

B.

6×10^{-6} N towards R

C.

6×10^{-7} N towards P

D.

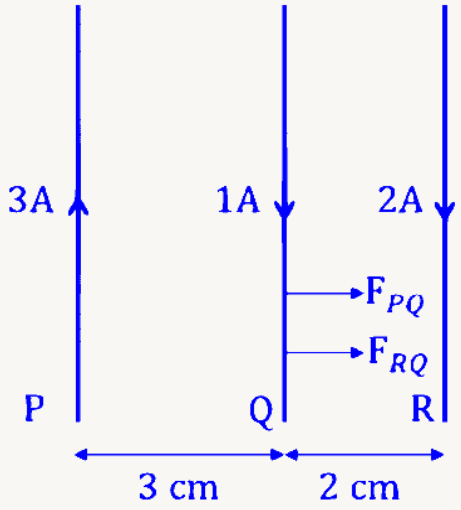
6×10^{-7} N towards R

Answer: B

Solution:

Using the formula for the force per unit length between two parallel current-carrying wires :

$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi r}$



Wires P and Q carry currents in opposite directions (3 A up and 1 A down). Parallel wires with opposite currents repel each other. Therefore, wire P pushes wire Q to the right (towards wire R) with a force F_{PQ}

Given $\mathrm{I_P} = 3\ \mathrm{A}, \mathrm{I_Q} = 1\ \mathrm{A}, \mathrm{r_{PQ}} = 3\ \mathrm{cm} = 0.03\ \mathrm{m}$

So, the magnitude of force on 15 cm of wire Q due to wire P is,

$\mathrm{F_{PQ}} = \frac{\left(4\pi \times 10^{-7}\right) \times 3 \times 1 \times 0.15}{2\pi \times 0.03}\ \mathrm{N} = 3 \times 10^{-6}\ \mathrm{N}$

Wires Q and R carry currents in the same direction (1 A down and 2 A down). Parallel wires with samedirection currents attract each other. Therefore, wire R pulls wire Q to the right (towards wire R) with force F_{RQ}.

Given that $\mathrm{I_R} = 2\ \mathrm{A}, \mathrm{I_Q} = 1\ \mathrm{A}, \mathrm{r_{RQ}} = 2\ \mathrm{cm} = 0.02\ \mathrm{m}$

$\mathrm{F_{RQ}} = \frac{\left(4\pi \times 10^{-7}\right) \times 2 \times 1 \times 0.15}{2\pi \times 0.02}$

$\Rightarrow \mathrm{F_{RQ}} = \frac{2 \times 10^{-7} \times 0.30}{0.02} = 2 \times 10^{-7} \times 15 = 3 \times 10^{-6}\ \mathrm{N}$

Since both forces acting on wire Q are directed towards the right, the net force will be towards R.

$\mathrm{F_{net}} = \mathrm{F_{PQ}} + \mathrm{F_{RQ}}$

$\Rightarrow \mathrm{F}_{\text{net}} = 3 \times 10^{-6} + 3 \times 10^{-6}$

$F_{net} = 6 \times 10^{-6}$ N

Hence, the net force acting on wire Q is 6×10^{-6} N towards R.

Hence, the correct option is (B).

Question11

The magnetic field at the centre of a current carrying circular loop of radius R is 16μ T. The magnetic field at a distance $x = \sqrt{3}R$ on its axis from the centre is

____ μT.

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Options:

A.

8

B.

2

C.

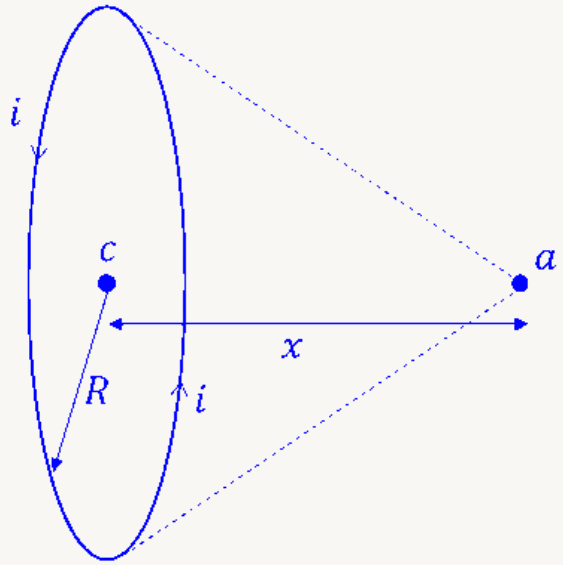
4

D.

$2\sqrt{2}$

Answer: B

Solution:



The magnetic field at the centre of a circular loop of radius R carrying current i is given by:

$\mathrm{B_c} = \frac{\mu_0 \mathrm{i}}{2\mathrm{R}}$

We are given that $\mathrm{B_c} = 16\mu$ T.

The magnetic field at a point on the axis at a distance x from the centre is given by:

$\mathrm{B_a} = \frac{\mu_0 \mathrm{iR}^2}{2(\mathrm{R}^2+\mathrm{x}^2)^{\frac{3}{2}}}$

The ratio of magnetic field at the centre to the axis,

$\frac{\mathrm{B_C}}{\mathrm{B_a}} = \frac{\left(\frac{\mu_0 \mathrm{i}}{2\mathrm{R}}\right)}{\left(\frac{\mu_0 \mathrm{iR}^2}{2\left(\mathrm{R}^2+\mathrm{x}^2\right)^{\frac{3}{2}}}\right)}$

$\Rightarrow \mathrm{B_a} = \mathrm{B_c}\left[\frac{\mathrm{R}^3}{(\mathrm{R}^2+\mathrm{x}^2)^{3/2}}\right]$

Given the distance $x = \sqrt{3}R$,

$\Rightarrow B_a = B_c\left[\frac{R^3}{\left(R^2+(\sqrt{3}R)^2\right)^{3/2}}\right] = \frac{B_c}{8}$

$\Rightarrow \mathrm{B_a} = \frac{16\mu \text{ T}}{8} = 2\mu \text{ T}$

Hence, the correct option is (B).

Question12

A long cylindrical conductor with large cross section carries an electric current distributed uniformly over its cross-section. Magnetic field due to this current is:

A. maximum at either ends of the conductor and minimum at the midpoint

B. maximum at the axis of the conductor

C. minimum at the surface of the conductor

D. minimum at the axis of the conductor

E. same at all points in the cross-section of the conductor

Choose the correct answer from the options given below:

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Options:

A.

B, C Only

B.

E Only

C.

A, D Only

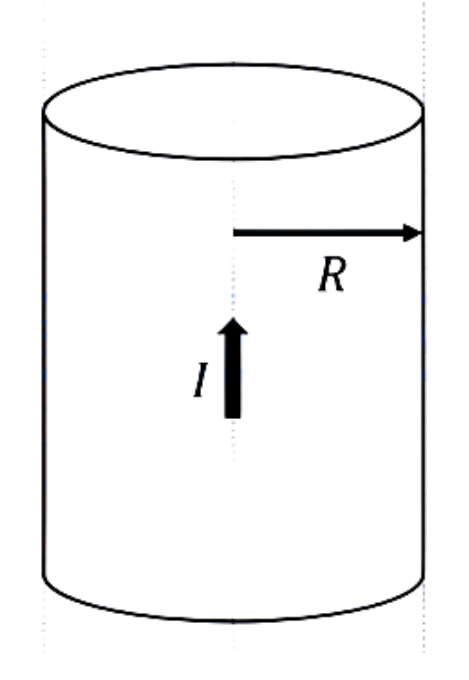
D.

D Only

Answer: D

Solution:

Consider a long, straight wire of radius R carrying a steady current I. Since the current distribution is uniform, the current density (J) is constant throughout the cross-section.



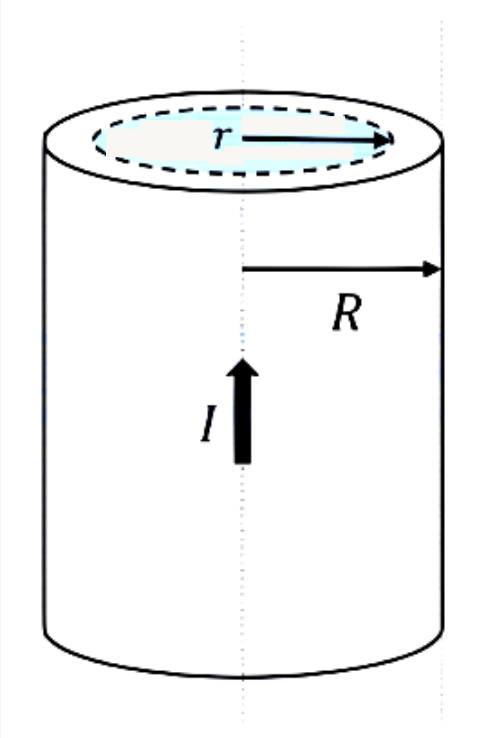
The current density J is given by :

$$\mathrm{J} = \frac{\mathrm{I}}{\text{Total Area}} = \frac{\mathrm{I}}{\pi \mathrm{R}^2}$$

Ampère's Law states that the line integral of the magnetic field B around a closed loop (Amperian loop) is equal to μ_0 times the current enclosed ($\mathrm{I}_{\mathrm{enc}}$) by that loop :

$$\oint \mathrm{B} \cdot \mathrm{dl} = \mu_0 \mathrm{I}_{\mathrm{enc}}$$

To find the field inside, consider a circular Amperian loop of radius r (where $r < R$) centered on the wire's axis.



$\oint \mathrm{Bdl} = \mathrm{B} \oint \mathrm{dl} = \mathrm{B}(2\pi \mathrm{r})$

The current enclosed is,

$I_{enc} = J \times (\text{ Area of loop}) = \left(\frac{I}{\pi R^2}\right) \times \left(\pi r^2\right)$

$\Rightarrow I_{\mathrm{enc}} = \frac{Ir^2}{R^2}$

Using Ampère's Law :

$B(2\pi r) = \mu_0 \left(\frac{Ir^2}{R^2}\right)$

$\Rightarrow \mathrm{B} = \frac{\mu_0 \mathrm{Ir}^2}{2\pi \mathrm{r}^2}$

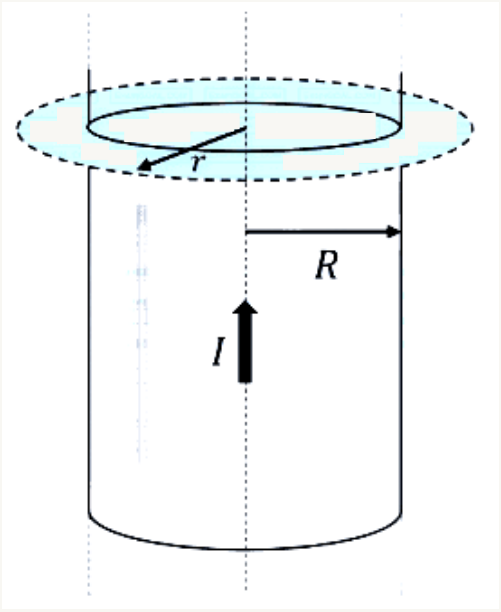
$\Rightarrow B = \frac{\mu_0 Ir}{2\pi R^2}$

The magnetic field inside the wire increases linearly with the distance r from the center.

$B \propto r$

At the axis of the conductor ($r = 0$), the magnetic field B becomes zero (minimum). Hence, statement B is incorrect and statement D is correct.

For a point outside the conductor $(\mathrm{r} > \mathrm{R})$:



Since the Amperian loop (radius r) completely surrounds the entire cross-section of the cylinder (radius R), it encloses the total current I flowing through the conductor.

$\mathrm{I}_{\mathrm{enc}} = \mathrm{I}$

Using Ampère's Law equation :

$\mathrm{B}(2\pi \mathrm{r}) = \mu_0 \mathrm{I}$

$\Rightarrow B = \frac{\mu_0 I}{2\pi r}$

This shows that B is inversely proportional to r $\left(\mathrm{B} \propto \frac{1}{\mathrm{r}}\right)$.

The field is maximum at the surface ($\mathrm{r} = \mathrm{R}$) and decreases when moving further away. Hence, statement C is incorrect.

The field distribution depends on the distance from the axis, not the length/midpoint of the conductor. Hence, statement (A) is incorrect.

Since, the field varies linearly inside and inversely outside; it is not uniform. Hence, option E is incorrect.

Therefore, only statement D is correct.

Hence, the correct option is (D).

Question13

Two identical long current carrying wires are bent into the shapes shown in the following figures. If the magnitude of magnetic fields at the centres P and Q of a semicircular arc are B_1 and B_2 respectively, then the ratio $\frac{B_1}{B_2}$ is _____.



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Options:

A.

$\frac{2+\pi}{1+\pi}$

B.

$\frac{1+\pi}{1-\pi}$

C.

$\frac{2+\pi}{1-\pi}$

D.

$\frac{1+\pi}{2-\pi}$

Answer: A

Solution:

For semi-infinite straight wire the field at a distance r from wire is

$B = \frac{\mu_0 I}{4\pi r}$

The field at the centre of a circular arc subtending angle ϕ is $B = \frac{\mu_0 I}{4\pi r}\,\phi$, where ϕ is in radians.

For a semicircle, $\phi = \pi$, so $B = \frac{\mu_0 I}{4r}$.

For system of current carrying wire (I), the direction of magnetic field using right hand rule is into the page due to semi-infinite wires and semi-circular wire.

Field from two semi-infinite wires is $2 \times \frac{\mu_0 I}{4\pi r} = \frac{\mu_0 I}{2\pi r}$

Field from semicircle is $\frac{\mu_0 I}{4r}$

As they act in the same direction, the net magnetic field is,

$B_1 = \frac{\mu_0 I}{4r} + \frac{\mu_0 I}{2\pi r} = \frac{\mu_0 I}{4\pi r}(\pi + 2)$

The system of current carrying wires (II) consists of two straight wire and one semi-circular wire.

One of the straight wire is passing through the point Q, so the magnetic field due to the wire which passes through the same point where magnetic field is to be calculated is zero.

So, the magnetic field due to semi-infinite wire is $\frac{\mu_0 \mathrm{I}}{4\pi \mathrm{r}}$

The magnetic field due to semi-circular wire is $\frac{\mu_0 \mathrm{I}}{4\mathrm{r}}$

As they act in the same direction, the net magnetic field is,

$B_2 = \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4r} = \frac{\mu_0 I}{4\pi r}(\pi + 1)$

Therefore, the ratio of magnetic fields is,

$\frac{B_1}{B_2} = \frac{\frac{\mu_0 I}{4\pi r}(\pi+2)}{\frac{\mu_0 I}{4\pi r}(\pi+1)}$

The ratio of the magnetic fields is $\frac{2+\pi}{1+\pi}$. Thus, the correct option is (A).

Question14

A particle having charge 10^{-9} C moving in x-y plane in fields of $0.4\hat{j}$ N/C and $4 \times 10^{-3}\hat{k}$ T experiences a force of $(4\hat{i} + 2\hat{j}) \times 10^{-10}$ N. The velocity of the particle at that instant is _____ m/s.

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

$50\hat{i} + 100\hat{j}$

B.

$100\hat{i} + 50\hat{j}$

C.

$-50\hat{i} + 100\hat{j}$

D.

$50\hat{i} - 100\hat{j}$

Answer: A

Solution:

The total force on the charged particle is found by adding the electric force and the magnetic force:

$\overrightarrow{\mathrm{F}} = q\overrightarrow{\mathrm{E}} + q(\overrightarrow{\mathrm{v}} \times \overrightarrow{\mathrm{B}})$

Here, the charge on the particle is $q = 10^{-9}$ C.

The force acting on it is $\overrightarrow{\mathrm{F}} = (4\hat{i} + 2\hat{j}) \times 10^{-10}$ N.

The electric field is $\vec{E} = 0.4\hat{j}\,\mathrm{N/C}$.

The magnetic field is $\vec{B} = 4 \times 10^{-3}\hat{k}\,\mathrm{T}$.

Since the particle is moving in the x-y plane, let its velocity be $\overrightarrow{v} = v_x\hat{i} + v_y\hat{j}$.

Now, calculate the cross product:

$\vec{v} \times \vec{B} = (v_x\hat{i} + v_y\hat{j}) \times (4 \times 10^{-3}\hat{k})$

Using the rules $\hat{i} \times \hat{k} = -\hat{j}$ and $\hat{j} \times \hat{k} = \hat{i}$:

$\vec{v} \times \vec{B} = 4 \times 10^{-3}v_y\hat{i} - 4 \times 10^{-3}v_x\hat{j}$

Substitute in the main force equation after dividing both sides by q:

$\frac{(4\hat{i}+2\hat{j})\times 10^{-10}}{10^{-9}} = 0.4\hat{j} + [(4 \times 10^{-3}v_y)\hat{i} - (4 \times 10^{-3}v_x)\hat{j}]$

This simplifies to:

$0.4\hat{i} + 0.2\hat{j} = (4 \times 10^{-3}v_y)\hat{i} - (4 \times 10^{-3}v_x + 0.4)\hat{j}$

Comparing components along $\hat{i}$:

$0.4 = 4 \times 10^{-3}v_y$

$v_y = \frac{0.4}{4\times 10^{-3}} = 100\,\text{m/s}$

Comparing components along $\hat{j}$:

$0.2 = 0.4 - 4 \times 10^{-3}v_x$

$4 \times 10^{-3}v_x = 0.2$

$v_x = 50\,\text{m/s}$

Hence, the velocity of the particle is:

$\vec{v} = 50\hat{i} + 100\hat{j}\,\text{m/s}$

Therefore, the correct answer is (A).

Question15

An insulated wire is wound so that it forms a flat coil with $N = 200$ turns. The radius of the innermost turn is $r_1 = 3$ cm, and of the outermost turn $r_2 = 6$ cm. If 20 mA current flows in it then the magnetic moment will be $\alpha \times 10^{-2}$ A. m^2. The value of α is ____ .

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

4.4

B.

2.64

C.

3.25

D.

1.2

Answer: B

Solution:

For a single circular turn of radius r carrying current I, the magnetic moment is :

$\mathrm{m}=\mathrm{I} \cdot \mathrm{A}=\mathrm{I}\left(\pi \mathrm{r}^2\right)$

In a flat coil where the turns are wound from r_1 to r_2, the number of turns per unit radial length is $\mathrm{n}=\frac{\mathrm{N}}{\mathrm{r}_2-\mathrm{r}_1}$. The total magnetic moment (M) is the integral of the moments of all infinitesimal turns of thickness dr :

$M=\int_{r_1}^{r_2}\left(I \pi r^2\right) \cdot n d r$

$\Rightarrow \mathrm{M}=\mathrm{I} \pi \mathrm{n} \int_{\mathrm{r}_1}^{\mathrm{r}_2} \mathrm{r}^2 \mathrm{dr}=\mathrm{I} \pi\left(\frac{\mathrm{N}}{\mathrm{r}_2-\mathrm{r}_1}\right)\left[\frac{\mathrm{r}^3}{3}\right]_{\mathrm{r}_1}^{\mathrm{r}_2}$

$\Rightarrow \mathrm{M}=\frac{\mathrm{NI} \pi}{3\left(\mathrm{r}_2-\mathrm{r}_1\right)}\left(\mathrm{r}_2^3-\mathrm{r}_1^3\right)$

Using the algebraic identity $r_2^3-r_1^3=\left(r_2-r_1\right)\left(r_2^2+r_1^2+r_1 r_2\right)$:

$\mathrm{M}=\frac{\mathrm{NI} \pi}{3}\left(\mathrm{r}_1^2+\mathrm{r}_2^2+\mathrm{r}_1 \mathrm{r}_2\right)$

Substituting the values, where number of turns is $\mathrm{N}=200$, the current is $\mathrm{I}=20 \mathrm{~mA}=20 \times 10^{-3} \mathrm{~A}$, the inner radius of coil is $\mathrm{r}_1=3 \mathrm{~cm}=0.03 \mathrm{~m}$ and the outer radius is $\mathrm{r}_2=6 \mathrm{~cm}=0.06 \mathrm{~m}$

$\mathrm{M}=\frac{200 \times 20 \times 10^{-3} \times 3.14}{3}\left(0.03^2+0.06^2+0.03 \times 0.06\right)$

$\Rightarrow M=\frac{4 \times 3.14}{3}(0.0009+0.0036+0.0018)$

$\Rightarrow \mathrm{M}=\frac{12.56}{3}(0.0063)=12.56 \times 0.0021$

$\Rightarrow M=0.026376 \approx 2.64 \times 10^{-2} A . m^2=\alpha \times 10^{-2} A . m^2 \Rightarrow \alpha=2.64$

Therefore, the value of α is 2.64 .

Thus, the correct option is (B).

Question16

A particle of charge q and mass m is projected from origin with an initial velocity $\vec{v}=\left(\frac{v_0}{\sqrt{2}}\hat{x}+\frac{v_0}{\sqrt{2}}\hat{y}\right)$. There exists a uniform magnetic field $\vec{B}=B_0\hat{z}$ and a space varying electric field $\vec{E}=E_0e^{-\lambda x}\hat{x}$ within the region $0\leqslant x\leqslant L$. After travelling a distance such that x-coordinate has changed from $x=0$ to $x=L$, the change in the kinetic energy is ____ .

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

$\frac{qE_0}{\lambda}\left[1-e^{-\lambda L}\right]$

B.

$\left(\frac{v_0qB_0}{2\lambda}\right)\left[2-e^{-2\lambda L}\right]$

C.

$\frac{qE_0}{\lambda}\left[1+e^{-\lambda L}\right]$

D.

$q\left(\frac{E_0+v_0B_0}{\lambda}\right)\left[1-e^{-\lambda L/2}\right]$

Answer: A

Solution:

The magnetic force $\vec{F}_m$ acting on a charged particle is given by $\vec{F}_m=q(\vec{v}\times\vec{B})$.

So, the magnetic force is always perpendicular to the velocity of the particle $\left(\vec{\mathrm{F}}_{\mathrm{m}}\perp\vec{\mathrm{v}}\right)$.

Since the force is perpendicular to the displacement at every instant, the work done by the magnetic field is always zero:

$W_m = \int \vec{F}_m \cdot \vec{dr} = 0$

The electric force $\vec{F}_e$ is given by $\vec{F}_e = q\vec{E}$.

The electric field in the region is given as $\vec{E} = E_o e^{-\lambda x}\hat{x}$, the force is:

$\vec{F}_e = qE_o e^{-\lambda x}\hat{x}$

The work done by the electric field as the particle moves from $x = 0$ to $x = L$ is :

$W_e = \int_0^L \vec{F}_e \cdot \vec{dx}$

$\Rightarrow W_e = \int_0^L qE_o e^{-\lambda x} \cdot dx$

$\Rightarrow W_e = eE_0 \int_0^L e^{-\lambda x} \cdot dx$

$\Rightarrow W_e = eE_0 \left[\frac{e^{-\lambda x}}{-\lambda}\right]_0^L$

$\Rightarrow W_e = -\frac{qE_o}{\lambda}\left[e^{-\lambda L} - e^0\right]$

$\Rightarrow W_e = -\frac{qE_o}{\lambda}\left[e^{-\lambda L} - 1\right]$

$\Rightarrow W_e = \frac{qE_o}{\lambda}\left[1 - e^{-\lambda L}\right]$

Using work-energy theory, the total work done on particle is equal to total change in kinetic energy.

$\Delta K = W_{\text{total}} = W_m + W_e$

$\Delta K = 0 + \frac{qE_o}{\lambda}\left[1 - e^{-\lambda L}\right]$

$\Delta K = \frac{qE_o}{\lambda}\left[1 - e^{-\lambda L}\right]$

Therefore, the change in kinetic energy is $\frac{qE_0}{\lambda}\left[1 - e^{-\lambda L}\right]$.

Thus, the correct option is (A).

Question17

A small cube of side 1 mm is placed at the centre of a circular loop of radius 10 cm carrying a current of 2 A . The magnetic energy stored inside the cube is $\alpha \times 10^{-14}$ J. The value of α is ____ .

$\left(\mu_o = 4\pi \times 10^{-7} \text{Tm/A}, \pi = 3.14\right)$

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

6.28

B.

6.28×10^{-6}

C.

628

D.

6.28×10^{-4}

Answer: A

Solution:

The magnetic field produced at the center of a circular wire of radius r carrying current I is:

$B = \frac{\mu_0 I}{2r}$

The electric current in the circular loop is $\mathrm{I} = 2$ A, the radius of the loop is $\mathrm{r} = 10$ cm $= 0.1$ m.

$\mathrm{B} = \frac{4\pi \times 10^{-7} \times 2}{2 \times 0.1} = 4\pi \times 10^{-6}$ T

Magnetic energy is stored in the space where a magnetic field exists. The energy per unit volume is:

$u_B = \frac{B^2}{2\mu_0}$

For a very small volume V where the field B is approximately uniform:

$\mathrm{U} = \mathrm{u_B} \times \mathrm{V} = \frac{\mathrm{B}^2}{2\mu_0} \times \mathrm{V}$

The volume of the cube is,

$V = (1\ \text{mm})^3 = \left(10^{-3}\ \text{m}\right)^3 = 10^{-9}\ \text{m}^3$

So, the total energy stored in the cube is,

$U = \frac{\left(4\pi \times 10^{-6}\right)^2}{2 \times 4\pi \times 10^{-7}} \times 10^{-9}$

$\Rightarrow U = \frac{16\pi^2 \times 10^{-12}}{8\pi \times 10^{-7}} \times 10^{-9} = 2\pi \times 10^{-5} \times 10^{-9}$

$\Rightarrow \mathrm{U}=2 \times 3.14 \times 10^{-14}=6.28 \times 10^{-14} \mathrm{~J}=\alpha \times 10^{-14} \mathrm{~J}$

$\Rightarrow \alpha=6.28$

Therefore, the correct option is (A).

Question18

A current of 30 A each flows in opposite directions in two conducting wires, placed parallel to each other at a distance of 8 cm . The magnetic field at the mid point between the two wires is ____ $\mu \mathrm{T}$.

$\left(\frac{\mu_{\mathrm{o}}}{4 \pi}=10^{-7} \mathrm{~N} / \mathrm{A}^2\right)$

JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

30

B.

300

C.

150

D.

0.0

Answer: B

Solution:

According to the Ampere's circuital Law, the magnetic field B at a distance r from an infinitely long straight wire carrying current I is :

$\oint \overrightarrow{\mathrm{B}} \cdot \overrightarrow{\mathrm{dl}}=\mu_0 \mathrm{I}$

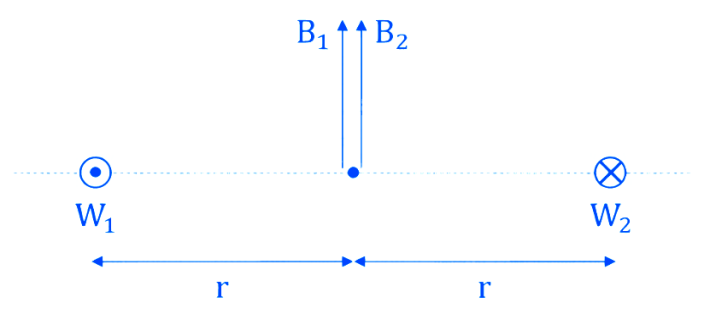
For a circular path of radius r :

$B(2\pi r) = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r}$

Let the wires be W_1 and W_2 separated by $d = 8$ cm. The midpoint is at $r = 4$ cm $= 0.04$ m from each wire.

Wire 1 carries current along out of the page. Using the right-hand rule, the field B_1 at the midpoint points along y-axis.

Wire 2 carries current along into the page. Using the right-hand rule, the field B_2 at the midpoint also points along y -axis.



Since both fields point in the same direction, the net magnetic field $B_{net} = B_1 + B_2$.

The current in the wires are given as $I_1 = I_2 = 30$ A and $r = 0.04$ m :

$B_1 = B_2 = \frac{\mu_0 I}{2\pi r} = 2 \times \left(\frac{\mu_0}{4\pi}\right)\frac{I}{r}$

$\Rightarrow B_1 = B_2 = 2 \times 10^{-7} \times \frac{30}{0.04} = 2 \times 10^{-7} \times 750 = 15 \times 10^{-5}$ T

$\Rightarrow B_{net} = 15 \times 10^{-5} + 15 \times 10^{-5} = 30 \times 10^{-5}$ T

$\Rightarrow B_{net} = 300 \times 10^{-6}$ T $= 300\mu$ T

Therefore, the magnitude of magnetic field at the midpoint is 300μ T.

Thus, the correct option is (B).

Question19

A current carrying circular loop of radius 2 cm with unit normal $\hat{n} = \frac{\hat{k}+\hat{i}}{\sqrt{2}}$ is placed in a magnetic field, $\vec{B} = B_o(3\hat{i} + 2\hat{k})$. If $B_o = 4 \times 10^{-3}$ T and current $I = 100\sqrt{2}$ A, the torque experienced by the loop is ____ Wb.A. ($\pi = 3.14$)

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

$16 \times 10^{-5}\hat{k}$

B.

$5024 \times 10^{-7}\hat{k}$

C.

$5024 \times 10^{-7}\hat{i}$

D.

$5024 \times 10^{-7}\hat{j}$

Answer: D

Solution:

When a current-carrying loop is placed in an external magnetic field, it experiences a torque $(\vec{\tau})$. The magnetic properties of the loop are represented by its magnetic dipole moment $(\overrightarrow{\mathrm{m}})$, defined as :

$$\overrightarrow{\mathrm{m}} = \mathrm{I}\,\overrightarrow{\mathrm{A}}$$

Where I is the current and $\vec{A}$ is the area vector of the loop. The direction of the area vector is perpendicular to the plane of the loop, given by the unit normal $\hat{\mathrm{n}}$.

$$\overrightarrow{\mathrm{A}} = \mathrm{A}\hat{\mathrm{n}}$$

The torque is given by the cross product of the magnetic moment and the magnetic field :

$$\vec{\tau} = \overrightarrow{\mathrm{m}} \times \overrightarrow{\mathrm{B}}$$

$$\Rightarrow \vec{\tau} = (\mathrm{IA}\hat{\mathrm{n}}) \times \overrightarrow{\mathrm{B}}$$

The radius of circular loop is $\mathrm{r} = 2\ \mathrm{cm} = 0.02\ \mathrm{m}$.

So, the area is

$$\mathrm{A} = \pi \mathrm{r}^2 = \pi(0.02)^2 = 4\pi \times 10^{-4}\ \mathrm{m}^2$$

The unit normal is $\hat{\mathrm{n}} = \frac{\hat{\mathrm{k}}+\hat{\mathrm{t}}}{\sqrt{2}}$

And the current in loop is $\mathrm{I} = 100\sqrt{2}\ \mathrm{A}$

So, the magnetic moment vector is;

$\overrightarrow{\mathrm{m}} = (100\sqrt{2}) \times \left(4\pi \times 10^{-4}\right) \times \left(\frac{\hat{\imath}+\hat{\mathrm{k}}}{\sqrt{2}}\right)$

$\Rightarrow \overrightarrow{\mathrm{m}} = 400\pi \times 10^{-4}(\hat{\imath}+\hat{\mathrm{k}}) = 0.04\pi(\hat{\imath}+\hat{\mathrm{k}})\mathrm{A}\cdot\ \mathrm{m}^2$

The magnetic field in the region is $\vec{B} = B_o(3\hat{\imath}+2\hat{k})$ with $B_o = 4\times10^{-3}$ T

$\overrightarrow{\mathrm{B}} = 4\times10^{-3}(3\hat{\imath}+2\hat{\mathrm{k}}) = (12\hat{\imath}+8\hat{\mathrm{k}})\times10^{-3}$ T

Using the formula for magnetic torque,

$\vec{\tau} = \overrightarrow{\mathrm{m}} \times \overrightarrow{\mathrm{B}}$

$\Rightarrow \vec{\tau} = [0.04\pi(\hat{\imath}+\hat{k})] \times \left[(12\hat{\imath}+8\hat{k})\times10^{-3}\right]$

$\Rightarrow \vec{\tau} = \left(0.04\pi\times10^{-3}\right)[(\hat{\imath}+\hat{k})\times(12\hat{\imath}+8\hat{k})]$

$\Rightarrow \vec{\tau} = \left(0.04\pi\times10^{-3}\right)(4\hat{\jmath})$

$\Rightarrow \vec{\tau} = 0.16\pi\times10^{-3}\hat{\jmath}$

$\Rightarrow \vec{\tau} = 0.16\times3.14\times10^{-3}\hat{\jmath} = 0.5024\times10^{-3}\hat{\jmath}$

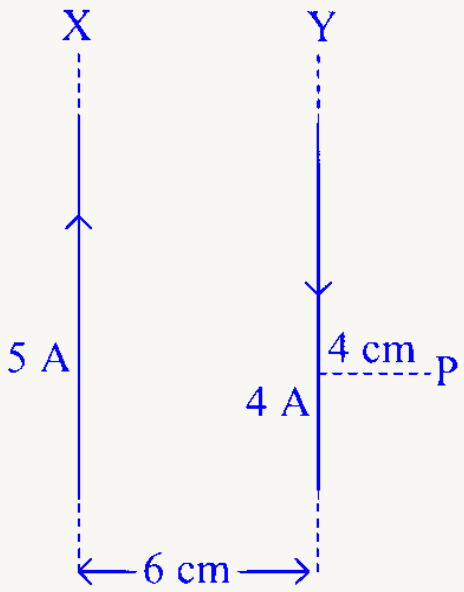
$\Rightarrow \vec{\tau} = 5024\times10^{-7}\hat{\jmath}$

Therefore, the torque experienced by the loop is $5024\times10^{-7}\hat{\mathrm{j}}$.

Hence, the correct option is (D).

Question20

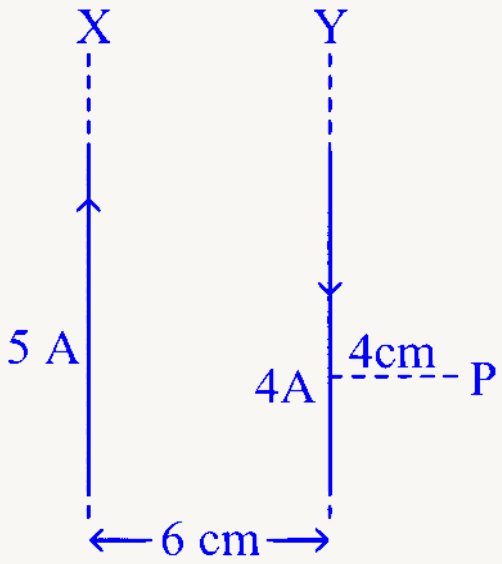
Two long parallel wires X and Y, separated by a distance of 6 cm , carry currents of 5 A and 4A, respectively, in opposite directions as shown in the figure. Magnitude of the resultant magnetic field at point P at a distance of 4 cm from wire Y is $x\times10^{-5}$ T. The value of x is _____ . Take permeability of free space as $\mu_0 = 4\pi\times10^{-7}$ SI units.



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Answer: 1

Solution:



$$
\begin{aligned}
B &= \frac{\mu_0(5)}{2\pi \times .01} - \frac{\mu_0 4}{2\pi \times 0.04} \\
&= -\frac{100\mu_0}{4\pi} \\
&= -100 \times 10^{-7} \\
&= -1 \times 10^{-5}\ \mathrm{T}
\end{aligned}
$$

Question21

A proton is moving undeflected in a region of crossed electric and magnetic fields at a constant speed of $2 \times 10^5\ \mathrm{ms}^{-1}$. When the electric field is switched off, the proton moves along a circular path of radius 2 cm . The magnitude of electric field is $x \times 10^4\ \mathrm{N/C}$. The value of x is _____. Take the mass of the proton $= 1.6 \times 10^{-27}\ \mathrm{kg}$.

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Answer: 2

Solution:

Let's break down the problem step by step:

When the proton moves undeflected in crossed electric and magnetic fields, the electric and magnetic forces balance each other. That is,

$qE = qvB$,

which simplifies to

$E = vB$.

After the electric field is switched off, the proton moves in a circular path under the action of the magnetic force. The magnetic force provides the required centripetal force:

$qvB = \frac{mv^2}{r}$.

Solving for the magnetic field B, we get:

$B = \frac{mv}{qr}$.

Now substitute this expression for B back into the equilibrium condition:

$E = vB = v\left(\frac{mv}{qr}\right) = \frac{mv^2}{qr}$.

Plug in the given values:

Proton mass, $m = 1.6 \times 10^{-27}\,\text{kg}$

Speed, $v = 2 \times 10^5\,\text{m/s}$

Radius, $r = 2\,\text{cm} = 0.02\,\text{m}$

Proton charge, $q = 1.6 \times 10^{-19}\,\text{C}$

Thus,

$E = \frac{(1.6\times10^{-27}\,\text{kg})(2\times10^5\,\text{m/s})^2}{(1.6\times10^{-19}\,\text{C})(0.02\,\text{m})}$.

Calculate the numerator:

$(2 \times 10^5)^2 = 4 \times 10^{10}$,

So, $(1.6 \times 10^{-27}) \times (4 \times 10^{10}) = 6.4 \times 10^{-17}$.

Calculate the denominator:

$(1.6 \times 10^{-19}) \times (0.02) = 3.2 \times 10^{-21}$.

Now compute the electric field:

$E = \frac{6.4\times10^{-17}}{3.2\times10^{-21}} = 2 \times 10^4\,\text{N/C}$.

The problem states that the magnitude of the electric field is $x \times 10^4\,\text{N/C}$. Since we found

$E = 2 \times 10^4\,\text{N/C}$,

it follows that

$x = 2$.

Question22

A current of 5 A exists in a square loop of side $\frac{1}{\sqrt{2}}$ m. Then the magnitude of the magnetic field B at the centre of the square loop will be $p \times 10^{-6}$ T. where, value of p is _____ [Take $\mu_0 = 4\pi \times 10^{-7}$ T mA^{-1}].

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Answer: 8

Solution:

$B = \frac{2\sqrt{2}\,\mu_0 I}{\pi L}$

For a square loop of side length

$L = \frac{1}{\sqrt{2}}$ m,

the perpendicular distance from the center to any side is

$d = \frac{L}{2} = \frac{1}{2\sqrt{2}}$ m.

For a finite straight wire, the magnetic field at a point at distance d from the wire is given by the Biot–Savart expression

$B_{\text{side}} = \frac{\mu_0 I}{4\pi d}(\sin\theta_1 + \sin\theta_2)$

where θ_1 and θ_2 are the angles between the extended wire and the line joining the ends of the wire with the observation point. For one side of the square, by symmetry, the midpoint of the side and the center of the square give

$\tan\theta = \frac{L/2}{L/2} = 1 \quad \Rightarrow \quad \theta = 45^\circ.$

Thus,

$\sin\theta_1 = \sin\theta_2 = \sin 45^\circ = \frac{1}{\sqrt{2}}.$

Then the field due to one side is

$$B_{\text{side}} = \frac{\mu_0 I}{4\pi\left(\frac{L}{2}\right)}\left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\right) = \frac{\mu_0 I}{4\pi\left(\frac{L}{2}\right)}\left(\frac{2}{\sqrt{2}}\right) = \frac{\mu_0 I\sqrt{2}}{2\pi L}.$$

Since there are 4 sides with contributions that add vectorially in the same direction at the center, the total magnetic field is

$$B = 4 \cdot B_{\text{side}} = \frac{4\mu_0 I\sqrt{2}}{2\pi L} = \frac{2\sqrt{2}\,\mu_0 I}{\pi L}.$$

Substitute

$$L = \frac{1}{\sqrt{2}},$$

to obtain

$$B = \frac{2\sqrt{2}\,\mu_0 I}{\pi\left(\frac{1}{\sqrt{2}}\right)} = \frac{2\sqrt{2}\,\mu_0 I\sqrt{2}}{\pi} = \frac{4\,\mu_0 I}{\pi}.$$

Using the provided values

$I = 5\ \text{A}$ and $\mu_0 = 4\pi \times 10^{-7}$ T\,m/A,

the magnetic field becomes

$$B = \frac{4\times\left(4\pi\times10^{-7}\right)\times5}{\pi} = \frac{80\pi\times10^{-7}}{\pi} = 80 \times 10^{-7}\ \text{T} = 8 \times 10^{-6}\ \text{T}.$$

Thus, the value of p is

$p = 8$.

Question23

A tightly wound long solenoid carries a current of 1.5 A . An electron is executing uniform circular motion inside the solenoid with a time period of 75 ns . The number of turns per metre in the solenoid is _____.

[Take mass of electron $\mathrm{m_e} = 9 \times 10^{-31}$ kg, charge of electron $|\mathrm{q_e}| = 1.6 \times 10^{-19}\mathrm{C}$,$\mu_0 = 4\pi \times 10^{-7}\frac{\mathrm{N}}{\mathrm{A}^2}, 1\ \mathrm{ns} = 10^{-9}\ \mathrm{s}$]

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Answer: 250

Solution:

The problem involves an electron executing uniform circular motion inside a solenoid. The objective is to find the number of turns per meter in the solenoid.

Given:

Current in the solenoid, $I = 1.5\,\text{A}$

Time period of the electron's circular motion, $T = 75\,\text{ns} = 75 \times 10^{-9}\,\text{s}$

Mass of electron, $m_e = 9 \times 10^{-31}\,\text{kg}$

Charge of electron, $|q_e| = 1.6 \times 10^{-19}\,\text{C}$

Permeability of free space, $\mu_0 = 4\pi \times 10^{-7}\,\text{N/A}^2$

Explanation:

The time period T for a revolving charge in a magnetic field is given by:

$T = \frac{2\pi m}{qB}$

The magnetic field B inside a solenoid is:

$B = \mu_0 n I$

where n is the number of turns per meter. Thus, substituting B in the expression for T, we get:

$T = \frac{2\pi m}{q(\mu_0 n I)}$

Plugging in the known values:

$75 \times 10^{-9} = \frac{(2\pi)(9\times10^{-31})}{1.6\times10^{-19}\times4\pi\times10^{-7}\times n\times1.5}$

Solving for n, we find:

$n = 250$

Thus, the number of turns per meter in the solenoid is 250.

Question24

The magnetic field inside a 200 turns solenoid of radius 10 cm is 2.9×10^{-4} Tesla. If the solenoid carries a current of 0.29 A , then the length of the solenoid is _____ πcm.

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Answer: 8

Solution:

We know, magnetic field due to solenoid, $B = \mu_0 nI$

where, I = current, $n = \frac{N}{L}$ = no. of turns per unit length

$\Rightarrow B = \frac{\mu_0 NI}{L}$

$\Rightarrow L = \frac{\mu_0 NI}{B} = \frac{4\pi \times 10^{-7} \times 200 \times 0.29}{2.9 \times 10^{-4}}$

$\Rightarrow L = 8\pi \times 10^{-2}\, m$

$\Rightarrow L = 8\pi\, cm$

Hence, answer = 8.

Question25

A 4.0 cm long straight wire carrying a current of 8 A is placed perpendicular to a uniform magnetic field of strength 0.15 T . The magnetic force on the wire is _____mN .

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Answer: 48

Solution:

To calculate the magnetic force on a wire, we use the formula:

$F = I\ell B$

Where:

I is the current in the wire (in amperes),

ℓ is the length of the wire (in meters),

B is the magnetic field strength (in teslas).

Given that:

The current I is 8 A,

The length of the wire ℓ is 4.0 cm, which is 0.04 m (since 1 cm = 0.01 m),

The magnetic field strength B is 0.15 T,

Substituting these values into the formula gives:

$F = 8 \times 0.04 \times 0.15$

Calculating this:

$F = 0.048\,N$

To convert this to millinewtons (mN), recall that $1\,N = 1000\,mN$. Therefore:

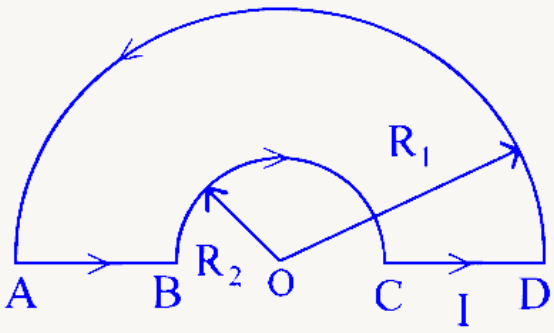
$F = 48\,mN$

Thus, the magnetic force on the wire is 48 mN.

Question26

A loop ABCDA , carrying current $I = 12$ A, is placed in a plane, consists of two semi-circular segments of radius $R_1 = 6\pi$ m and $R_2 = 4\pi$ m. The magnitude of the resultant magnetic field at center O is $k \times 10^{-7}$ T. The value of k is_____.

(Given $\mu_0 = 4\pi \times 10^{-7} TmA^{-1}$)



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Answer: 1

Solution:

Magnetic Field from Semi-Circular Segments:

The magnetic field at the center due to a semi-circular loop is given by:

$B = \frac{\mu_0 I}{4R}$

Calculating B_{R_1} and B_{R_2}:

For the semi-circle with radius R_1:

$B_{R_1} = \frac{\mu_0 I}{4R_1} = \frac{4\pi \times 10^{-7} \times 12}{4 \times 6\pi}$

For the semi-circle with radius R_2:

$B_{R_2} = \frac{\mu_0 I}{4R_2} = \frac{4\pi \times 10^{-7} \times 12}{4 \times 4\pi}$

Net Magnetic Field at the Center (O):

The total magnetic field B_0 at the center is the difference between these two fields (since they are in opposite directions):

$B_0 = |B_{R_1} - B_{R_2}|$

Calculate:

$B_0 = \frac{4\pi \times 10^{-7} \times 12}{4} \left(\frac{1}{4\pi} - \frac{1}{6\pi} \right)$

$B_0 = 12\pi \times 10^{-7} \left(\frac{1}{12\pi} \right)$

$B_0 = 1 \times 10^{-7} \text{ T}$

Value of k:

Given that the magnitude of the resultant magnetic field is $k \times 10^{-7}$ T, we find $k = 1$.

Thus, the value of k is 1.

Question27

If an optical medium possesses a relative permeability of $\frac{10}{\pi}$ and relative permittivity of $\frac{1}{0.0885}$, then the velocity of light is greater in vacuum than that in this medium by _____ times.

$$\left(\mu_0 = 4\pi \times 10^{-7} \text{H/m}, \epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}, \text{c} = 3 \times 10^8 \text{ m/s}\right)$$

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Answer: 6

Solution:

Since velocity of light in terms of $\mu \& E$ is

$$V = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_0\mu_r}} \times \frac{1}{\sqrt{\epsilon_0\epsilon_r}}$$
$$= \frac{1}{\sqrt{\mu_r\epsilon_r}} \times \frac{1}{\sqrt{\mu_0\epsilon_0}}$$
$$= \frac{C}{\sqrt{\mu_r\epsilon_r}} = \frac{C}{\sqrt{\frac{10}{\pi} \times \frac{1}{0.0885}}}$$
$$= \frac{C}{\sqrt{36}} = \frac{C}{6}$$

$$V = \frac{C}{6}$$
$$C = 6V$$

Velocity of light in vacuum is greater by 6 times the velocity of light in mediumAnswer is 6

Question28

An electron projected perpendicular to a uniform magnetic field B moves in a circle. If Bohr's quantization is applicable, then the radius of the electronic orbit in the first excited state is :

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Options:

A. $\sqrt{\frac{h}{\pi eB}}$

B. $\sqrt{\frac{4h}{\pi eB}}$

C. $\sqrt{\frac{\mathrm{h}}{2\pi e\mathrm{eB}}}$

D. $\sqrt{\frac{2h}{\pi eB}}$

Answer: A

Solution:

When an electron is projected perpendicular to a uniform magnetic field B, it travels in a circular path. According to Bohr's quantization rule, the radius of the electron's orbit in the first excited state can be determined using the following approach:

The radius r in a magnetic field is expressed as:

$r = \frac{mv}{eB}$

where m is the mass of the electron, v is its velocity, e is its charge, and B is the magnetic field strength.

According to Bohr's quantization condition:

$mvr = \frac{nh}{2\pi}$

Substituting the expression for r:

$(eBr)r = \frac{nh}{2\pi}$

This simplifies to the expression for the radius:

$r = \sqrt{\frac{nh}{2\pi eB}}$

In the first excited state, the principal quantum number n is 2. So, the radius becomes:

$r = \sqrt{\frac{2h}{2\pi eB}} = \sqrt{\frac{h}{\pi eB}}$

Thus, for the first excited state, the radius of the electron's orbit is $\sqrt{\frac{h}{\pi eB}}$.

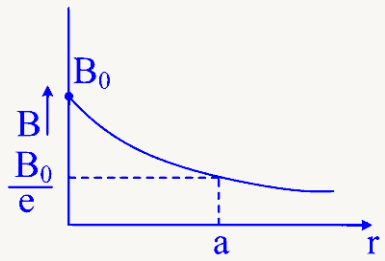
Question29

A long straight wire of a circular cross-section with radius ' a ' carries a steady current I . The current I is uniformly distributed across this cross-section. The plot of magnitude of magnetic field B with distance r from the centre of the wire is given by

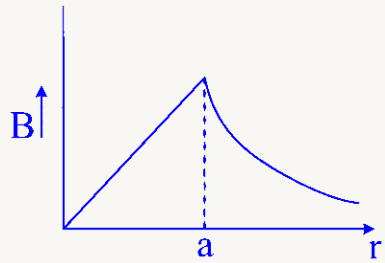
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Options:

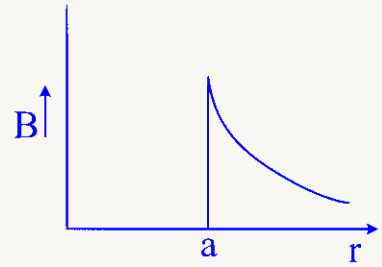
A.



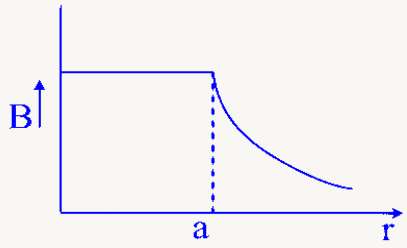
B.



C.

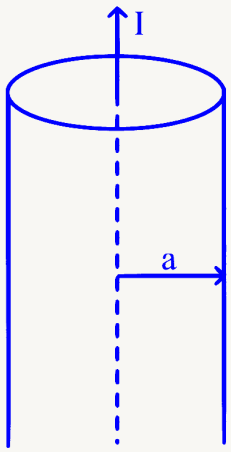


D.



Answer: B

Solution:



$$B_{in} = \frac{\mu_0(\bar{J} \times \bar{r})}{2}$$

$$B_{in} \propto r$$

$$B_{out} = \frac{\mu_0 I}{2\pi r}$$

$$B_{net} \propto \frac{1}{r}$$

Question30

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : A electron in a certain region of uniform magnetic field is moving with constant velocity in a straight line path.

Reason (R) : The magnetic field in that region is along the direction of velocity of the electron. In the light of the above statements, choose the correct answer from the options given below :

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Options:

A. (A) is true but (R) is false

B. Both (A) and (R) are true and (R) is the correct explanation of (A)

C. Both $(\mathbf{A})$ and $(\mathbf{R})$ are true but $(\mathbf{R})$ is NOT the correct explanation of (A)

D. (A) is false but (R) is true

Answer: B

Solution:

According to the Lorentz force law, the force experienced by a charged particle, such as an electron, moving in a magnetic field is given by the equation:

$\overline{\mathrm{F}} = \mathrm{q}(\overline{\mathrm{v}} \times \overline{\mathrm{B}})$

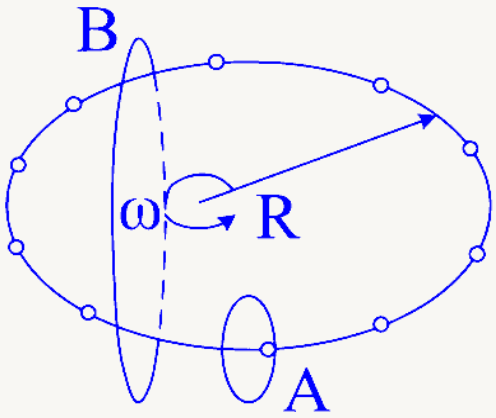
Here, $\overline{\mathrm{F}}$ is the force exerted on the particle, q is the charge of the particle, $\overline{\mathrm{v}}$ is the velocity vector of the particle, and $\overline{\mathrm{B}}$ is the magnetic field vector. The symbol $\times$ indicates the cross product, meaning the force is perpendicular to both the velocity and magnetic field vectors.

Given this, for the force $\overline{\mathrm{F}}$ to be zero:

The velocity $\overline{\mathrm{v}}$ and the magnetic field $\overline{\mathrm{B}}$ must be parallel (or anti-parallel). This condition will result in the angle $\theta = 0^\circ$ or 180°.

Under these circumstances, the electron will not experience any magnetic force and will continue to move in a straight line with constant velocity, as the cross product becomes zero. Therefore, the reason (R) is indeed a valid explanation: the magnetic field being along the direction of the electron's velocity means no perpendicular force is applied to change its path.

Question31



N equally spaced charges each of value q , are placed on a circle of radius R . The circle rotates about its axis with an angular velocity ω as shown in the figure. A bigger Amperian loop B encloses the whole circle where as a smaller Amperian loop A encloses a small segment. The difference between enclosed currents, $I_A - I_B$, for the given Amperian loops is

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Options:

A. $\frac{N^2}{2\pi} q\omega$

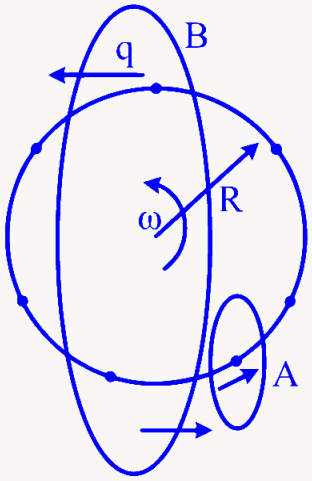
B. $\frac{N}{2\pi} q\omega$

C. $\frac{2\pi}{N} q\omega$

D. $\frac{N}{\pi} q\omega$

Answer: B

Solution:



$I_A = \frac{Nq}{\frac{2\pi}{\omega}}$

$I_A = \frac{Nq\omega}{2\pi}, I_B = 0$

$I_A - I_B = \frac{Nq\omega}{2\pi}$

Question32

Consider a long thin conducting wire carrying a uniform current I. A particle having mass "M" and charge " q " is released at a distance " a " from the wire with a speed v_0 along the direction of current in the wire. The particle gets attracted to the wire due to magnetic force. The particle turns round when it is at distance x from the wire. The value of x is [μ_0 is vacuum permeability]

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Options:

A. $a\left[1-\frac{mv_o}{2q\mu_0 \mathrm{I}}\right]$

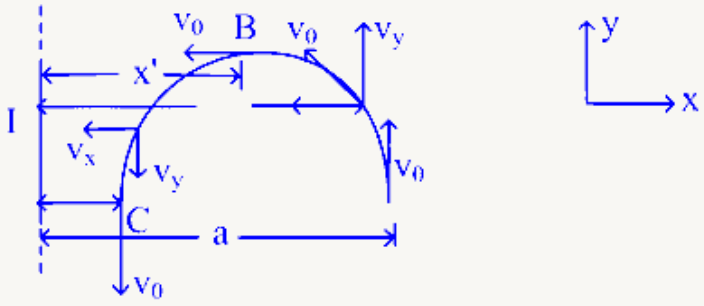
B. $ae^{-\frac{4\pi \mathrm{mv}_o}{q\mu_o \mathrm{I}}}$

C. $a\left[1-\frac{\mathrm{mv}}{\mathrm{q}}\mu_{\mathrm{o}}\mathrm{I}\right]$

D. $\frac{a}{2}$

Answer: B

Solution:



For $A \rightarrow B$

$\overrightarrow{V}=-v_x\hat{i}+v_y\hat{j}$

$\overrightarrow{B}=\frac{\mu_0 I}{2\pi r}\left(-\widehat{k}\right)$

$\because \overrightarrow{F}=q\left(\overrightarrow{v}\times\overrightarrow{B}\right)$

So, $\overrightarrow{F}=\frac{q\mu_0 I}{2\pi r}\left[\left[(-v_x)\hat{i}+v_y\hat{j}\right]\times\left(-\widehat{k}\right)\right]$

$=\frac{q\mu_0 I}{2\pi r}\left[v_x\left(\hat{i}\times\widehat{k}\right)-v_y\left(\hat{j}\times\widehat{k}\right)\right]$

$\overrightarrow{F}=\frac{q\mu_0 I}{2\pi r}\left[-v_x\hat{j}-v_y\hat{i}\right]$

$\Rightarrow F_x=\frac{-\mu_0 Iq}{2\pi r}v_y$

$\Rightarrow ma_x=\frac{-\mu_0 Iq}{2\pi r}v_y$

$\Rightarrow v_x \frac{dv_x}{dr} = \frac{-\mu_0 Iq}{2\pi m} \frac{v_y}{r}$

$\Rightarrow \frac{v_x dv_x}{v_y} = \frac{-\mu_0 Iq}{2\pi m} \frac{dr}{r}$

Since, $v_x^2 + v_y^2 = v_0^2$

$\Rightarrow v_y = \sqrt{v_0^2 - v_x^2}$

$\Rightarrow \int_0^{v_0} \frac{v_x dv_x}{\sqrt{v_0^2 - v_x^2}} = \frac{-\mu_0 Iq}{2\pi m} \int_a^{x'} \frac{dr}{r}$

Let $v_0^2 - v_x^2 = z^2$

$\Rightarrow -2v_x dv_x = 2zdz$

when $v_x = 0,\ z = v_0$

$v_x = v_0,\ z = 0$

$\Rightarrow \int_{v_0}^0 \frac{zdz}{z} = -\frac{\mu_0 Iq}{2\pi m} \int_a^{x'} \frac{dr}{r}$

$\Rightarrow -v_0 = \frac{\mu_0 Iq}{2\pi m} \ln\left(\frac{x'}{a}\right)$

$\Rightarrow \ln\left(\frac{x'}{a}\right) = -\frac{2\pi m v_0}{\mu_0 Iq}$

$\Rightarrow x' = ac\frac{-2\pi m v_0}{\mu_0 Iq}$ .... (1)

Now, for $B \to C$

$\overrightarrow{v} = -v_x\hat{i} - v_y\hat{j}$

$\overrightarrow{B} = \frac{\mu_0 I}{2\pi r}\left(-\widehat{k}\right)$

$\overrightarrow{F} = q\left(\overrightarrow{v} \times \overrightarrow{B}\right) = \frac{q\mu_0 I}{2\pi r}\left[v_x\left(\hat{i} \times \widehat{k}\right) + v_y\left(\hat{j} \times \widehat{k}\right)\right]$

$\Rightarrow \overrightarrow{F} = \frac{\mu_0 Iq}{2\pi r}\left[-v_x\hat{j} + v_y\hat{i}\right]$

$\Rightarrow F_x = \frac{\mu_0 Iq}{2\pi r} v_y$

$\Rightarrow v_x \frac{dv_x}{dr} = \frac{\mu_0 Iq}{2\pi m} \frac{v_y}{r}$

$\int_{v_0}^0 \frac{v_x dv_x}{\sqrt{v_0^2 - v_x^2}} = \frac{\mu_0 Iq}{2\pi m} \int_{x'}^x \frac{dr}{r}$

Let $v_0^2 - v_x^2 = z^2$

$-2v_x dv_x = 2zdz$

$v_x = v_0 \Rightarrow z = 0$

$v_x = 0 \Rightarrow z = v_0$

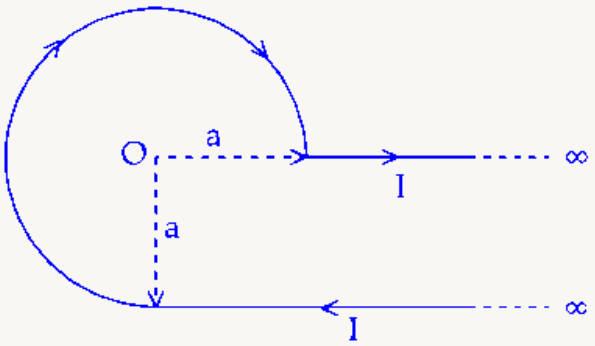
$\Rightarrow -\int_0^{v_0} \frac{zdz}{z} = \frac{\mu_0 Iq}{2\pi m}[\ln r]_{x'}^{x}$

$\Rightarrow -v_0 = \frac{\mu_0 Iq}{2\pi m} \ln\left(\frac{x}{x'}\right)$

$\Rightarrow x = x' e^{-\frac{2\pi m v_0}{\mu_0 Iq}}$ .... (2)

From (1) & (2), $x = ae^{-\frac{4\pi m v_0}{\mu_0 Iq}}$

Question33



An infinite wire has a circular bend of radius *a*, and carrying a current *I* as shown in the figure. The magnitude of magnetic field at the origin O of the arc is given by:

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Options:

A. $\frac{\mu_0}{2\pi}\frac{\mathrm{I}}{\mathrm{a}}\left[\frac{\pi}{2}+2\right]$

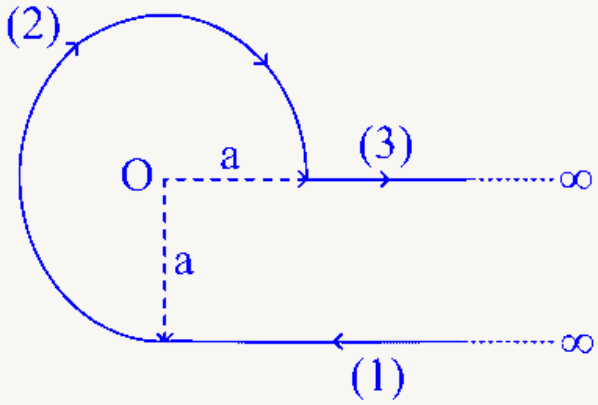
B. $\frac{\mu_0}{4\pi}\frac{\mathrm{I}}{\mathrm{a}}\left[\frac{3\pi}{2}+1\right]$

C. $\frac{\mu_0}{4\pi}\frac{\mathrm{I}}{\mathrm{a}}\left[\frac{\pi}{2}+1\right]$

D. $\frac{\mu_0}{4\pi}\frac{\mathrm{I}}{\mathrm{a}}\left[\frac{3\pi}{2}+2\right]$

Answer: B

Solution:



$B_1 = \frac{\mu_0 i}{4\pi a} \otimes$

$B_2 = \frac{\mu_0}{4\pi} \frac{i}{a} \left(\frac{3\pi}{2} \right) \otimes$

$B_3 = 0$

$B = \frac{\mu_0}{4\pi} \frac{i}{a} \left(1 \frac{3\pi}{2} \right) \otimes$

Question34

Consider a long straight wire of a circular cross-section (radius *a*) carrying a steady current *I*. The current is uniformly distributed across this cross-section. The distances from the centre of the wire's cross-section at which the magnetic field [inside the wire, outside the wire] is half of the maximum possible magnetic field, any where due to the wire, will be :

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Options:

A.

[$a/2$, $3a$]

B.

[$a/4$, $3a/2$]

C.

[$a/2$, $2a$]

D.

[$a/4$, $2a$]

Answer: C

Solution:

Maximum possible magnetic field is at the surface

$$\mathrm{B_{max}} = \frac{\mu_0 \mathrm{I}}{2\pi \mathrm{a}}$$
$$\frac{\mathrm{B_{max}}}{2} = \frac{\mu_0 \mathrm{I}}{4\pi \mathrm{a}}$$

It can be obtained inside as well as outside the wire

For inside,

$$\frac{\mu_0 \mathrm{I}}{4\pi \mathrm{a}} = \frac{\mu_0 \mathrm{Ir}}{2\pi \mathrm{a}^2}$$
$$\Rightarrow \mathrm{r} = \frac{\mathrm{a}}{2}$$

For outside

$$\frac{\mu_0 \mathrm{I}}{4\pi \mathrm{a}} = \frac{\mu_0 \mathrm{I}}{2\pi \mathrm{r}}$$
$$\Rightarrow \mathrm{r} = 2\mathrm{a}$$

Correct answer $\left[\frac{\mathrm{a}}{2}, 2\mathrm{a}\right]$

Question35

Let B_1 be the magnitude of magnetic field at center of a circular coil of radius R carrying current I. Let B_2 be the magnitude of magnetic field at an axial distance ' x ' from the center. For $x : \mathrm{R} = 3 : 4$, $\frac{\mathrm{B}_2}{\mathrm{B}_1}$ is :

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Options:

A. 64 : 125

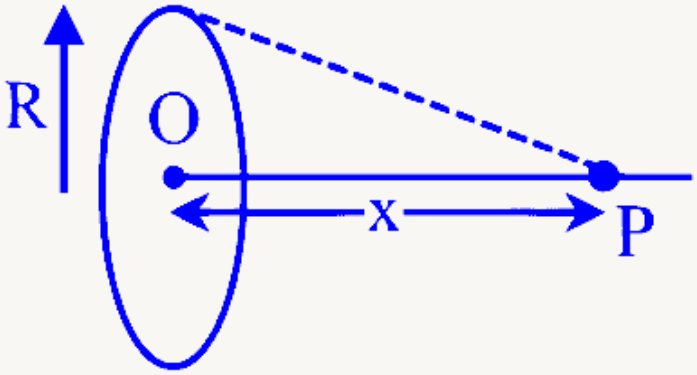
B. 25 : 16

C. 4 : 5

D. 16 : 25

Answer: A

Solution:



$B_1 = \frac{\mu_0 i}{2R} \qquad B_2 = B_1 \sin^3\theta$

$\therefore \frac{B_2}{B_1} = \sin^3\theta = \left(\frac{4}{5}\right)^3 = \frac{64}{125}$

Question36

A solenoid having area A and length ' l ' is filled with a material having relative permeability2. The magnetic energy stored in the solenoid is :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. $\frac{\mathrm{B}^2\ \mathrm{A}l}{4\mu_0}$

B. $\frac{\mathrm{B}^2\ \mathrm{A}l}{2\mu_0}$

C. $\frac{\mathrm{B}^2\ \mathrm{A}l}{\mu_0}$

D. $\mathrm{B}^2\mathrm{Al}$

Answer: A

Solution:

To determine the magnetic energy stored in a solenoid filled with a material of relative permeability 2, we start with the expression for energy density in a magnetic field:

$\frac{U}{V} = \frac{B^2}{2\mu_r\mu_0}$

Given that the relative permeability μ_r is 2, this becomes:

$\frac{U}{V} = \frac{B^2}{2\times 2\mu_0} = \frac{B^2}{4\mu_0}$

The energy U stored in the solenoid can be expressed as:

$U = \frac{B^2}{4\mu_0} \times V$

Here, V is the volume of the solenoid, calculated as $A \times \ell$ (where A is the cross-sectional area and ℓ is the length). Therefore, substituting for V, we get:

$U = \frac{B^2}{4\mu_0} \times A\ell$

This formula gives us the magnetic energy stored in the solenoid.

Question37

In a moving coil galvanometer, two moving coils M_1 and M_2 have the following particulars :

$$\mathrm{R}_1 = 5\Omega,\ \mathrm{N}_1 = 15,\ \mathrm{A}_1 = 3.6 \times 10^{-3}\ \mathrm{m}^2,\ \mathrm{B}_1 = 0.25\ \mathrm{T}$$
$$\mathrm{R}_2 = 7\Omega,\ \mathrm{N}_2 = 21,\ \mathrm{A}_2 = 1.8 \times 10^{-3}\ \mathrm{m}^2,\ \mathrm{B}_2 = 0.50\ \mathrm{T}$$

Assuming that torsional constant of the springs are same for both coils, what will be the ratio of voltage sensitivity of M_1 and M_2 ?

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Options:

A. 1 : 1

B. 1 : 3

C. 1 : 2

D. 1 : 4

Answer: A

Solution:

In a moving coil galvanometer, the voltage sensitivity is given by the formula:

Voltage sensitivity $= \frac{\theta}{V} = \frac{N \cdot A \cdot B}{c \cdot R}$

Here, θ is the deflection angle, V is the voltage, N is the number of turns, A is the coil area, B is the magnetic field, c is the torsional constant of the spring, and R is the resistance of the coil.

For the two moving coils M_1 and M_2, we want to determine the ratio of their voltage sensitivities. Given that the torsional constant c is the same for both coils, we can express the ratio as:

Ratio of voltage sensitivities $= \left(\frac{N_1 \cdot A_1 \cdot B_1}{N_2 \cdot A_2 \cdot B_2} \right) \cdot \frac{R_2}{R_1}$

Plugging in the given values:

$N_1 = 15$

$A_1 = 3.6 \times 10^{-3}\,\mathrm{m}^2$

$B_1 = 0.25\,\mathrm{T}$

$R_1 = 5\,\Omega$

$N_2 = 21$

$A_2 = 1.8 \times 10^{-3}\,\mathrm{m}^2$

$B_2 = 0.50\,\mathrm{T}$

$R_2 = 7\,\Omega$

We calculate:

Ratio $= \left(\frac{15 \times 3.6 \times 0.25}{21 \times 1.8 \times 0.5} \right) \times \frac{7}{5} = \frac{1}{1}$

Thus, the ratio of the voltage sensitivities of M_1 to M_2 is 1:1.

Question38

Given below are two statements: one is labelled as $\mathbf{Assertion A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : If Oxygen ion $\left(O^{-2}\right)$ and Hydrogen ion $\left(H^{+}\right)$enter normal to the magnetic field with equal momentum, then the path of O^{-2} ion has a smaller curvature than that of H^{+}.

Reason R : A proton with same linear momentum as an electron will form a path of smaller radius of curvature on entering a uniform magnetic field perpendicularly.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A. Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

B. $\mathbf{A}$ is false but $\mathbf{R}$ is true

C. $\mathbf{A}$ is true but $\mathbf{R}$ is false

D. Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

Answer: C

Solution:

The explanation involves analyzing the motion of charged particles in a magnetic field using the concept of radius of curvature. When a charged particle moves perpendicularly to a magnetic field, the radius of curvature r of its path is given by:

$r = \frac{mv}{qB} = \frac{p}{qB}$

Where:

m is the mass of the particle.

v is the velocity.

q is the charge.

B is the magnetic field strength.

p is the momentum of the particle.

From this formula, we can see that the radius of curvature r is directly proportional to the momentum p and inversely proportional to the charge q.

In the given scenario:

The Oxygen ion O^{-2} and the Hydrogen ion H^{+} both enter the magnetic field with equal momentum.

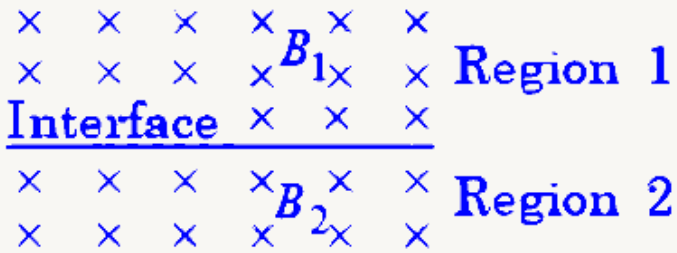
Since O^{-2} has a charge of $-2e$ and H^{+} has a charge of $+e$, and given that $r \propto \frac{1}{q}$, the path of O^{-2} will indeed have a smaller radius of curvature compared to that of H^{+}.

Therefore, the Assertion **A** is true.

The Reason **R** states that a proton and an electron with the same linear momentum will form paths of different radii upon entering a magnetic field, with the proton having a smaller radius of curvature. This reasoning is based on the same principle, since the mass and charge of the electron and proton are different. However, this explanation does not directly justify the assertion about Oxygen and Hydrogen ions, so Reason **R** is false in the context of explaining Assertion **A**. Thus, the correct conclusion is that **A** is true, but **R** is false.

Question39

Uniform magnetic fields of different strengths $(B_1$ and $B_2)$, both normal to the plane of the paper exist as shown in the figure. A charged particle of mass m and charge q, at the interface at an instant, moves into the region 2 with velocity v and returns to the interface. It continues to move into region 1 and finally reaches the interface. What is the displacement of the particle during this movement along the interface?



(Consider the velocity of the particle to be normal to the magnetic field and $B_2 > B_1$)

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Options:

A. $\frac{mv}{qB_1}\left(1-\frac{B_1}{B_2}\right)\times 2$

B. $\frac{mv}{qB_1}\left(1-\frac{B_2}{B_1}\right)\times 2$

C. $\frac{mv}{qB_1}\left(1-\frac{B_2}{B_1}\right)$

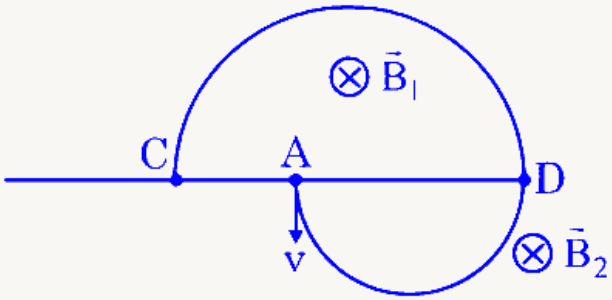
D. $\frac{mv}{qB_1}\left(1-\frac{B_1}{B_2}\right)$

Answer: A

Solution:

As $\vec{v}$ is $\perp$ to $\vec{B}$, so charge particle will move in circular path, whose radius is given by

$R=\frac{mv}{qB}$



Starting point $\rightarrow$ A

Ending point $\rightarrow$ C

$\therefore$ Net displacement $=$ AC

$$AC = CD - AD$$
$$AC = \frac{2mv}{qB_1} - \frac{2mv}{qB_2}$$
$$AC = \frac{2mv}{qB_1}\left[1-\frac{B_1}{B_2}\right]$$

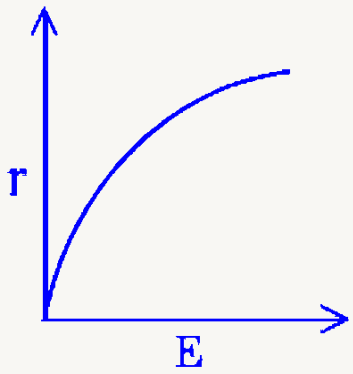
Question40

A particle of charge q, mass m and kinetic energy E enters in magnetic field perpendicular to its velocity and undergoes a circular arc of radius (r). Which of the following curves represents the variation of r with E ?

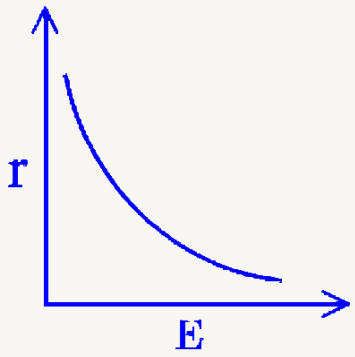
JEE Main 2025 (Online) 7th April Morning Shift

Options:

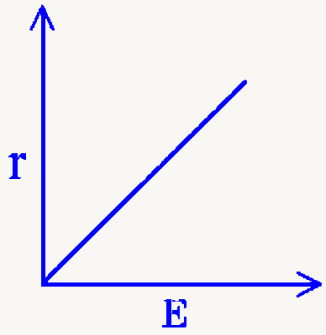
A.



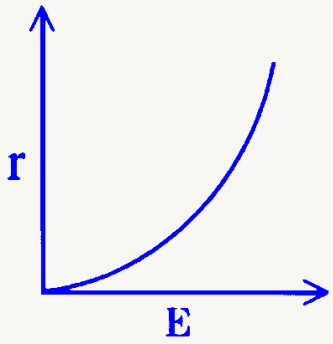
B.



C.

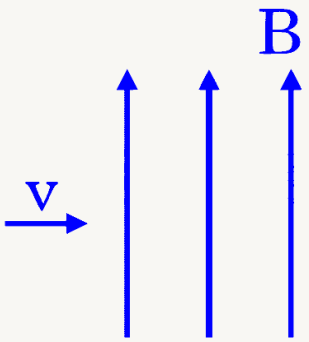


D.



Answer: A

Solution:



$$\frac{mv^2}{r} = qvB$$

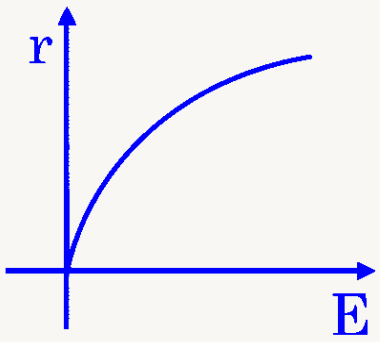
$$mv = qBr$$

$$E = \frac{1}{2}mv^2$$

$$E = \frac{1}{2}\,m\left(\frac{q^2\,B^2r^2}{m^2}\right) = \frac{q^2\,B^2r^2}{2\,m}$$

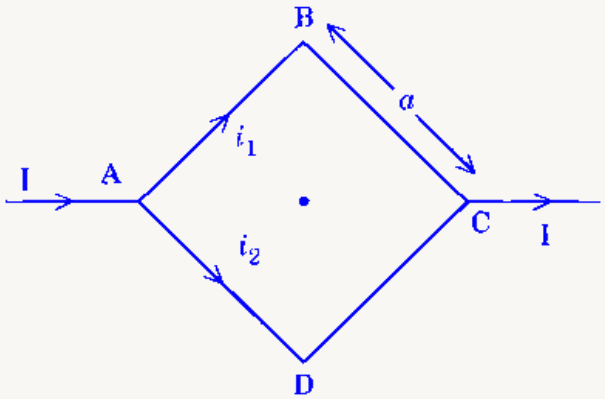
$$E = \left(\frac{q^2\,B^2}{2\,m}\right)r^2$$

$$r^2 \propto E$$



Question41

Figure shows a current carrying square loop ABCD of edge length is 'a' lying in a plane. If the resistance of the ABC part is r and that of ADC part is 2r, then the magnitude of the resultant magnetic field at centre of the square loop is



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Options:

A.

$\frac{2\mu_0 I}{3\pi a}$

B.

$\frac{\sqrt{2}\mu_0 I}{3\pi a}$

C.

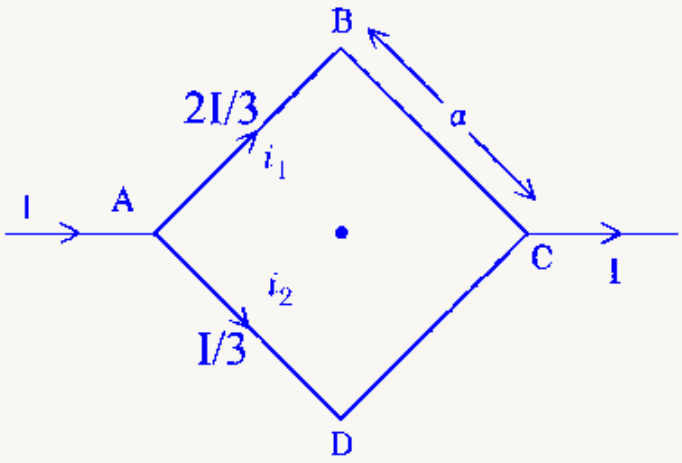
$\frac{3\pi\mu_0 I}{\sqrt{2}a}$

D.

$\frac{\mu_0 I}{2\pi a}$

Answer: B

Solution:



$$\vec{B} = \vec{B}_{AB} + \vec{B}_{BC} + \vec{B}_{CD} + \vec{B}_{DA}$$

$$\vec{B} = \left[\frac{-\mu_0(2I/3)}{4\pi(a/2)}\sqrt{2} - \frac{\mu_0(2I/3)}{4\pi(a/2)}\sqrt{2} + \frac{\mu_0(I/3)}{4\pi(a/2)}\sqrt{2} + \frac{\mu_0(I/3)}{4\pi(a/2)}\sqrt{2}\right]\hat{k}$$

$$\vec{B} = \left[\frac{-2\sqrt{2}\mu_0 I}{3\pi a} + \frac{\sqrt{2}\mu_0 I}{3\pi a}\right]\hat{k}$$

$$\vec{B} = \frac{-\sqrt{2}\mu_0 I}{3\pi a}\hat{k}$$

Question42

Two long straight wires P and Q carrying equal current 10A each were kept parallel to each other at 5 cm distance. Magnitude of magnetic force experienced by 10 cm length of wire P is F_1. If distance between wires is halved and currents on them are doubled, force F_2 on 10 cm length of wire P will be :
[24-Jan-2023 Shift 1]

Options:

A. $8F_1$

B. $10F_1$

C. $F_1/8$

D. $F_1/10$

Answer: A

Solution:

Force per unit length between two parallel straight wires $= \frac{\mu_0 i_1 i_2}{2\pi d}$

$$\frac{F_1}{F_2} = \frac{\frac{\mu_0 (10)^2}{2\pi (5\text{ cm})}}{\frac{\mu_0 (20)^2}{2\pi \left(\frac{5\text{ cm}}{2}\right)}} = \frac{1}{8}$$

$$\Rightarrow F_2 = 8F_1$$

Question43

A circular loop of radius r is carrying current I A. The ratio of magnetic field at the centre of circular loop and at a distance r from the center of the loop on its axis is : [24-Jan-2023 Shift 1]

Options:

A. $1 : 3\sqrt{2}$

B. $3\sqrt{2} : 2$

C. $2\sqrt{2} : 1$

D. $1 : \sqrt{2}$

Answer: C

Solution:

Magnetic field due to current carrying circular loop on its axis is given as

$$\frac{\mu_0 i r^2}{2(r^2 + x^2)^{3/2}}$$

At centre, $x = 0, B_1 = \frac{\mu_0 i}{2r}$

At $x = r, B_2 = \frac{\mu_0 i}{2 \times 2\sqrt{2}r}$

$\frac{B_1}{B_2} = 2\sqrt{2}$

Question44

A long solenoid is formed by winding 70 turns cm^{-1}. If 2.0A current flows, then the magnetic field produced inside the solenoid is___
($\mu_0 = 4\pi \times 10^{-7} TmA^{-1}$)
[24-Jan-2023 Shift 2]

Options:

A. $1232 \times 10^{-4} T$

B. $176 \times 10^{-4} T$

C. $352 \times 10^{-4} T$

D. $88 \times 10^{-4} T$

Answer: B

Solution:

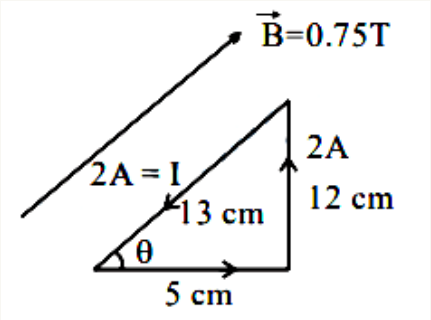
$B = \mu_0 nI$
$= 4\pi \times 10^{-7} \times 70 \times 10^2 \times 2$
$= 56\pi \times 10^{-4} T$
$= 176 \times 10^{-4} T$

Question45

A single turn current loop in the shape of a right angle triangle with sides 5 cm, 12 cm, 13 cm is carrying a current of 2A. The loop is in a uniform magnetic field of magnitude 0.75T whose direction is parallel to the current in the 13 cm side of the loop. The magnitude of the magnetic force on the 5 cm side will be $\frac{x}{130}$N. The value of x is____
[24-Jan-2023 Shift 2]

Answer: 9

Solution:



Force on 5 cm side is

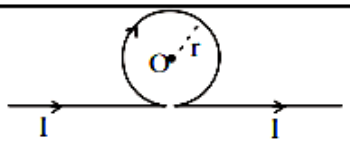
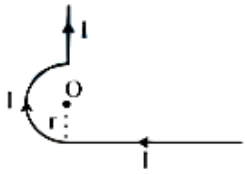
$|\vec{F}| = ILB\sin\theta$

$= (2)(5\times10^{-2})\times\frac{3}{4}\times\frac{12}{13} = \frac{9}{130}N$

So, $x = 9$

Question46

Math List I with List II

List – I (Current configuration)		List – II (Magnetic field at point O)	
A		I.	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 2]$
B		II.	$B_0 = \frac{\mu_0}{4}\frac{I}{r}$
C		III.	$B_0 = \frac{\mu_0 I}{2\pi r}[\pi - 1]$
D		IV	$B_0 = \frac{\mu_0 I}{4\pi r}[\pi + 1]$

Choose the correct answer from the option given below: [25-Jan-2023 Shift 1]

Options:

A. A-III, B-IV, C-I, D-II

B. A-I, B-III, C-IV, D-II

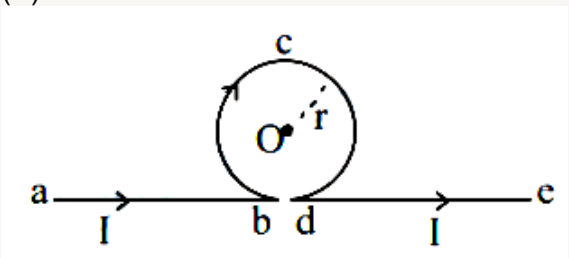
C. A-III, B-I, C-IV, D-II

D. A-II, B-I, C-IV, D-III

Answer: C

Solution:

(A)



$B_{ab} = \frac{\mu_0}{4\pi}\frac{I}{r}$ (out of the plane)

$B_{bcd} = \frac{\mu_0}{4\pi}\frac{I}{r}(2\pi)$ (in the plane)

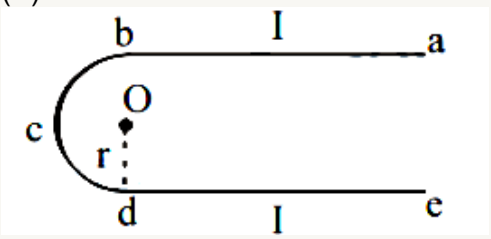
$B_{de} = \frac{\mu_0}{4\pi}\frac{I}{r}$ (out of the plane)

Hence magnetic field at O is

$B_0 = -\frac{\mu_0}{4\pi}\frac{I}{r} + \frac{\mu_0}{4\pi}\frac{I}{r}(2\pi) - \frac{\mu_0}{4\pi}\frac{I}{r}$

$B_0 = \frac{\mu_0}{2\pi}\frac{I}{r}(\pi - 1)$..... (III)

(B)



$B_{ab} = \frac{\mu_0}{4\pi}\frac{I}{r}$ (out of the plane)

$B_{bcd} = \frac{\mu_0}{4\pi}\frac{I}{r}(\pi)$ (out of the plane)

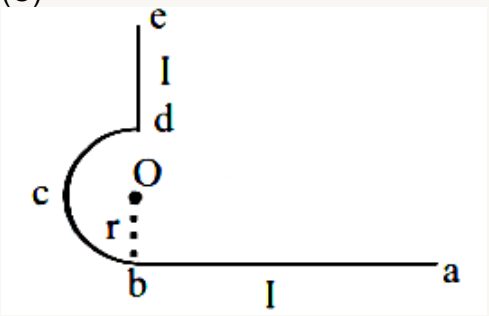
$B_{de} = \frac{\mu_0}{4\pi}\frac{I}{r}$ (out of theplane)

Hence magnetic field at O is

$B_0 = \frac{\mu_0}{4\pi}\frac{I}{r} + \frac{\mu_0}{4\pi}\frac{I}{r}(\pi) + \frac{\mu_0}{4\pi}\frac{I}{r}$

$B_0 = \frac{\mu_0}{4\pi}\frac{I}{r}(\pi + 2)$... (I)

(C)



$B_{ab} = \frac{\mu_0}{4\pi}\frac{I}{r}$ (in the plane)

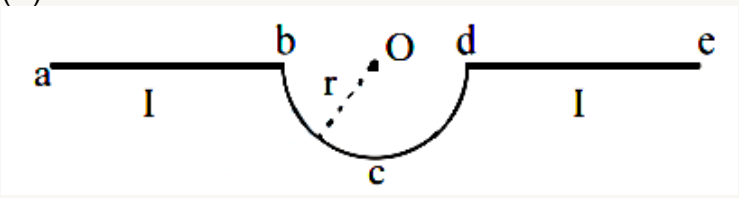
$B_{bcd} = \frac{\mu_0}{4\pi}\frac{I}{r}(\pi)$ (in the plane)

$B_{de} = 0$ (at the axis)

Hence magnetic field at O is

$B_0 = \frac{\mu_0}{4\pi}\frac{I}{r}(1+\pi)\ldots$

(D)



$B_{ab} = 0$ (at the axis)

$B_{bcd} = \frac{\mu_0}{4\pi}\frac{I}{r}(\pi)$ (out of the plane)

$B_{de} = 0$ (at the axis)

Hence magnetic field at O is

$B_0 = \frac{\mu_0 I}{4r}\ldots$

Question47

A solenoid of 1200 turns is wound uniformly in a single layer on a glass tube 2m long and 0.2m in diameter. The magnetic intensity at the center of the solenoid when a current of 2 A flows through it is: [25-Jan-2023 Shift 1]

Options:

A. $2.4 \times 10^3 Am^{-1}$

B. $1.2 \times 10^3 Am^{-1}$

C. $1Am^{-1}$

D. $2.4 \times 10^{-3} Am^{-1}$

Answer: B

Solution:

Magnetic field at centre inside the solenoid is given by

$B = \mu_0 nI$

So magnetic intensity at centre

$$H = \frac{B}{\mu_0} = nI = \left(\frac{1200}{2}\right)(2)$$
$$H = 1.2 \times 10^3 Am^{-1}$$

Question48

For a moving coil galvanometer, the deflection in the coil is 0.05 rad when a current of 10 mA is passed through it. If the torsional constant of suspension wire is 4.0×10^{-5} Nm rad^{-1}, the magnetic field is 0.01T and the number of turns in the coil is 200 , the area of each turn (in cm^2) is :
[25-Jan-2023 Shift 2]

Options:

A. 2.0

B. 1.0

C. 1.5

D. 0.5

Answer: B

Solution:

$\tau = K\theta$
$NiAB = K\theta$
$$A = \frac{K\theta}{NiB} = \frac{4 \times 10^{-5} \times 0.05}{200 \times 10 \times 10^{-3} \times 0.01}$$
On solving $A = 10^{-4} m^2 = 1 cm^2$

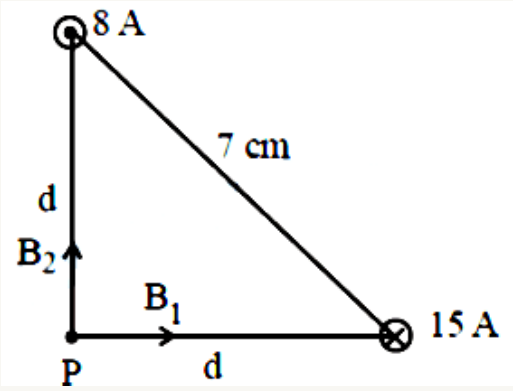
Question49

Two long parallel wires carrying currents 8A and 15A in opposite directions are placed at a distance of 7 cm from each other. A point P is at equidistant from both the wires such that the lines joining the point P to the wires are perpendicular to each other. The magnitude

of magnetic field at P is _____ $\times 10^{-6}$T. (Given : $\sqrt{2} = 1.4$)
[25-Jan-2023 Shift 2]

Answer: 68

Solution:



Magnetic fields due to both wires will be perpendicular to each other.

$$B_1 = \frac{\mu_0 i_1}{2\pi d} \quad B_2 = \frac{\mu_0 i_2}{2\pi d}$$

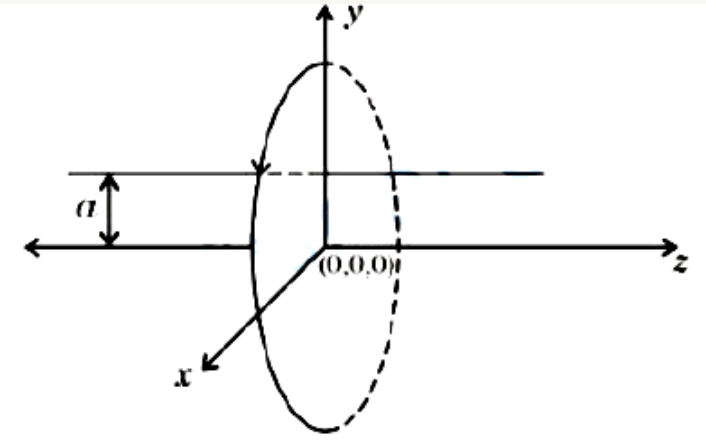
$$B_{net} = \sqrt{B_1^2 + B_2^2} \Rightarrow \frac{\mu_0}{2\pi d}\sqrt{i_1^2 + i_2^2}$$

$$\Rightarrow \frac{4\pi \times 10^{-7}}{2\pi \times (7/\sqrt{2}) \times 10^{-2}} \times \sqrt{8^2 + 15^2}\left(d = \frac{7}{\sqrt{2}}\text{ cm}\right)$$

$$\Rightarrow 68 \times 10^{-6}\text{T}$$

Question50

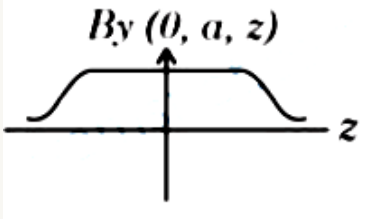
A single current carrying loop of wire carrying current I flowing in anticlockwise direction seen from +ve z direction and lying in xy plane in shown in figure. The plot of $\hat{j}$ component of magnetic field (By) at a distance ' a ' (less than radius of the coil) and on yz plane vs z coordinate look like



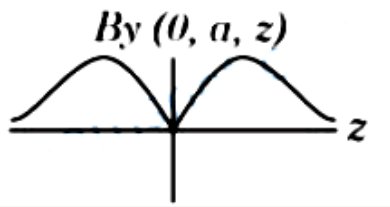
[29-Jan-2023 Shift 1]

Options:

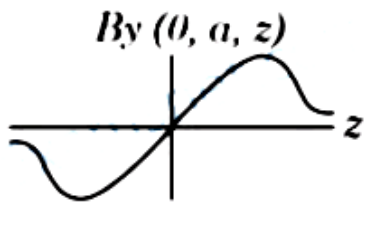
A.



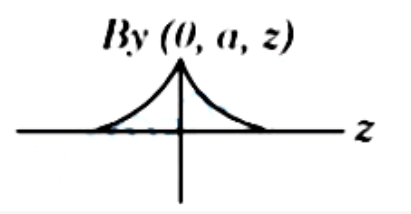
B.



C.

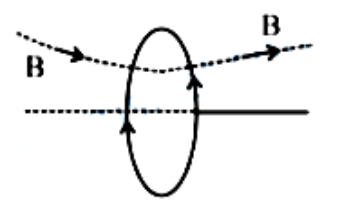


D.



Answer: C

Solution:

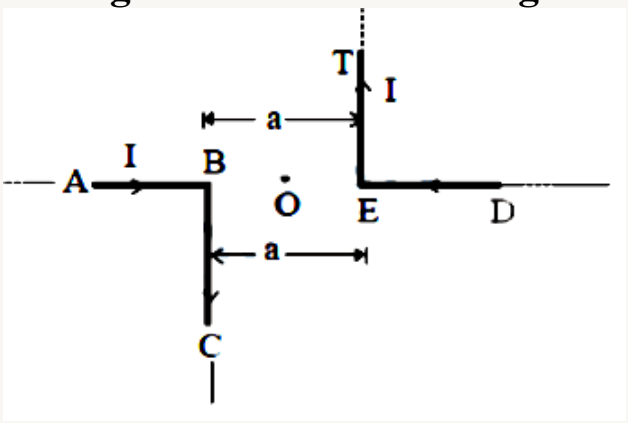


$B_y = 0$ in plane of coil

B_y is opposite of each other in $-z$ and $+z$ positions.

Question51

The magnitude of magnetic induction at mid-point O due to current arrangement as shown in Fig will be :



[29-Jan-2023 Shift 1]

Options:

A. $\frac{\mu_0 I}{2\pi a}$

B. 0

C. $\frac{\mu_0 I}{4\pi a}$

D. $\frac{\mu_0 I}{\pi a}$

Answer: D

Solution:

Magnetic field due to current in BC and ET are outward at point ' O '

$$B_0 = \frac{\mu_0 i}{4\pi r} + \frac{\mu_0 i}{4\pi r} = \frac{\mu_0 i}{2\pi r} = \frac{\mu_0 i}{\pi a}$$

Question52

A square loop of area $25cm^2$ has a resistance of 10Ω. The loop is placed in uniform magnetic field of magnitude 40.0 T. The plane of loop is perpendicular to the magnetic field. The work done in pulling the loop out of the magnetic field slowly and uniformly in 1.0 sec, will be
[29-Jan-2023 Shift 2]

Options:

A. $2.5 \times 10^{-3}J$

B. $1.0 \times 10^{-3}J$

C. $1.0 \times 10^{-4}J$

D. $5 \times 10^{-3}J$

Answer: B

Solution:

Sol. $\ell = 50\,cm$

$t = 1\ sec$

$$\therefore V = \frac{0.05}{1} = 0.05 m/s.$$

$$i = \frac{40 \times 0.05 \times 0.05}{10} = 0.01A$$

$$\therefore\ F = B_1 \ell = 40 \times 0.01 \times 0.05$$

$$F = 0.02N$$

$$\therefore\ W = 0.02 \times \ell = 0.02 \times .05$$

$$\therefore\ W = 1 \times 10^{-3}J$$

Question53

The electric current in a circular coil of four turns produces a magnetic induction 32T at its centre. The coil is unwound and is rewound into a circular coil of single turn, the magnetic induction at the centre of the coil by the same current will be :
[29-Jan-2023 Shift 2]

Options:

A. 8T

B. 4T

C. 2T

D. 16T

Answer: C

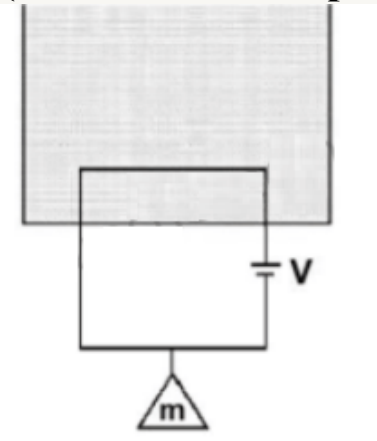
Solution:

$$B = \frac{\mu_0 i}{2R} \times 4$$
$$B' = \frac{\mu_0 i}{2R'}$$
$$R' = 4R$$
$$B' = \frac{\mu_0 i}{8R}$$
$$\frac{B'}{B} = \frac{1}{16}$$
$$B' = 2T$$

Question54

A massless square loop, of wire of resistance 10Ω. supporting a mass of I g. hangs vertically with one of its sides in a uniform magnetic field of 10^3G, directed outwards in the shaded region. A dc voltage V is applied to the loop. For what value of V. the magnetic force will exactly balance the weight of the supporting mass of 1g ?

(If sides of the loop = 10 cm, g = 10ms^{-2})



[30-Jan-2023 Shift 1]

Options:

A. $\frac{1}{10}$V

B. 100V

C. 1V

D. 10V

Answer: D

Solution:

$F_m = mg$

$\therefore \quad ILB = mg$

$\therefore \quad \left(\frac{V}{R}\right) LB = mg$

$\therefore \quad V = \frac{mgR}{LB}$

$= \frac{(1 \times 10^{-3}\,kg)(10m/s^2)(10\Omega)}{(0.1m)(10^3 \times 10^{-4}T)} = 10V$

Question55

The magnetic moments associated with two closely wound circular coils A and B of radius r_A = 10 cm and r_B = 20 cm respectively are equal if: (Where N_A, I_A and N_B, I_B are number of turn and current

of A and B respectively)
[30-Jan-2023 Shift 1]

Options:

A. $2N_AI_A = N_BI_B$

B. $N_A = 2N_B$

C. $N_AI_A = 4N_BI_B$

D. $4N_AI_A = N_BI_B$

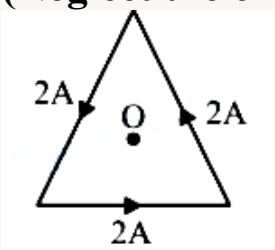
Answer: C

Solution:

$M = NIA$
$M_A = M_B$
$\therefore N_AI_AA_A = N_BI_BA_B$
$\therefore N_AI_A\pi(0.1)^2 = N_BI_B\pi(0.2)^2$
$\therefore N_AI_A = 4N_BI_B$

Question56

As shown in the figure, a current of 2A flowing in an equilateral triangle of side $4\sqrt{3}$ cm. The magnetic field at the centroid O of the triangle is:
(Neglect the effect of earth's magnetic field.)



[30-Jan-2023 Shift 2]

Options:

A. $4\sqrt{3} \times 10^{-4}$T

B. $4\sqrt{3} \times 10^{-5}$T

C. $\sqrt{3} \times 10^{-4}$T

D. $3\sqrt{3} \times 10^{-5}$T

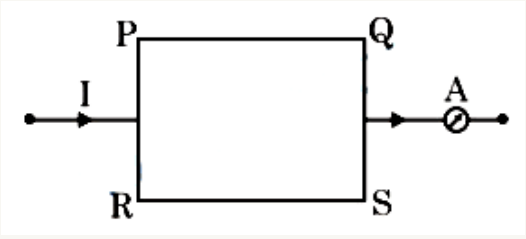
Answer: D

Solution:

$$d \tan 60^\circ = 2\sqrt{3}$$
$$d = 2 \text{ cm}$$
$$B = 3 \times \frac{\mu_0 i}{2\pi d} \sin 60^\circ$$
$$= 3 \times \frac{2 \times 10^{-7} \times 2}{2 \times 10^{-2}} \times \frac{\sqrt{3}}{2}$$
$$= 3\sqrt{3} \times 10^{-5}$$

Question57

A current carrying rectangular loop PQRS is made of uniform wire. The length PR = QS = 5 cm and PQ = RS = 100 cm. If ammeter current reading changes from I to 2I, the ratio of magnetic forces per unit length on the wire PQ due to wire RS in the two cases respectively $f_{PQ}^{I} : f_{PQ}^{2I}$ is :



[30-Jan-2023 Shift 2]

Options:

A. 1 : 2

B. 1 : 4

C. 1 : 5

D. 1 : 3

Answer: B

Solution:

$F \propto I_1 I_2$
$F_1 : F_{21} = 1 : 4$

Question58

A rod with circular cross-section area $2cm^2$ and length 40 cm is wound uniformly with 400 turns of an insulated wire. If a current of 0.4A flows in the wire windings, the total magnetic flux produced inside windings is $4\pi \times 10^{-6}$ Wb. The relative permeability of the rod is
(Given : Permeability of vacuum
$\mu_0 = 4\pi \times 10^{-7} NA^{-2}$)
[31-Jan-2023 Shift 1]

Options:

A. 12.5

B. $\frac{32}{5}$

C. 125

D. $\frac{5}{16}$

Answer: C

Solution:

$\varphi = \mu_r \mu_0 \frac{N}{\ell} I \, xA$
$\mu_r = 125$
Option 3.

Question59

A bar magnet with a magnetic moment $5.0 Am^2$ is placed in parallel position relative to a magnetic field of 0.4T. The amount of required work done in turning the magnet form parallel to antiparallel position relative to the field direction is _____.
[31-Jan-2023 Shift 1]

Options:

A. 4J

B. 1J

C. 2J

D. Zero

Answer: A

Solution:

$$u = -MB\cos\theta$$
$$W = \Delta u$$
$$W = -MB\cos 180^{\circ} - (-mB\cos 0^{\circ})$$
$$W = 2MB = 2 \times 5 \times 0.4 = 4J$$
Option 1

Question60

A long conducting wire having a current I flowing through it, is bent into a circular coil of N turns. Then it is bent into a circular coil of n turns. The magnetic field is calculated at the centre of coils in both the cases. The ratio of the magnetic field in first case to that of second case is :
[31-Jan-2023 Shift 2]

Options:

A. N : n

B. $n^2 : N^2$

C. $N^2 : n^2$

D. n : N

Answer: C

Solution:

$I = (2\pi r)n$

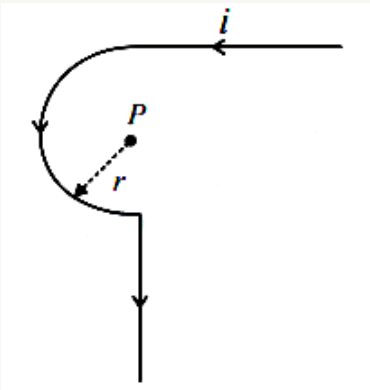
$r \propto \left(\frac{I}{n}\right)$

$B = n\left(\frac{\mu_0 i}{2r}\right) \propto \left(\frac{\mu_0 i}{2L}\right) n^2$

$\frac{B_1}{B_2} = \left(\frac{N^2}{n^2}\right)$

Question61

Find the magnetic field at the point P in figure. The curved portion is a semicircle connected to two long straight wires.



[1-Feb-2023 Shift 1]

Options:

A. $\frac{\mu_0 i}{2r}\left(1 + \frac{2}{\pi}\right)$

B. $\frac{\mu_0 i}{2r}\left(1 + \frac{1}{\pi}\right)$

C. $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{2\pi}\right)$

D. $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{\pi}\right)$

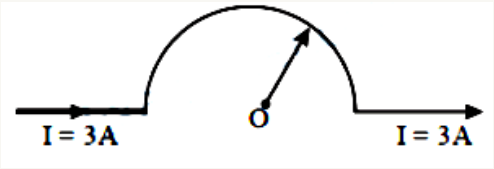
Answer: C

Solution:

$B_P = \left(\frac{\mu_0 i}{4r}+\frac{\mu_0 i}{4\pi r}\right) = \frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{2\pi}\right)$

Question62

As shown in the figure, a long straight conductor with semicircular arc of radius $\pi/10$m is carrying current I = 3A. The magnitude of the magnetic field. at the center O of the arc is:
(The permeability of the vacuum $= 4\pi \times 10^{-7} NA^{-2}$)



[1-Feb-2023 Shift 2]

Options:

A. 6μT

B. 1μT

C. 4μT

D. 3μT

Answer: D

Solution:

$B_C = \frac{\mu_0 I}{4\pi R}(\pi)$(B at centre of circular arc)

$= \frac{\mu_0 I}{4R} = \frac{4\pi \times 10^{-7} \times 3}{4 \times \frac{\pi}{10}}$

$= 3 \times 10^{-6} T = 3\mu T$

Question63

A coil is placed in magnetic field such that plane of coil is perpendicular to the direction of magnetic field. The magnetic flux through a coil can be changed:
A. By changing the magnitude of the magnetic field within the coil.
B. By changing the area of coil within the magnetic field.
C. By changing the angle between the direction of magnetic field and the plane of the coil.
D. By reversing the magnetic field direction abruptly without changing its magnitude.
Choose the most appropriate answer from the options given below: [1-Feb-2023 Shift 2]

Options:

A. A and B only

B. A, B and C only

C. A, B and D only

D. A and C only

Answer: B

Solution:

$\varphi = \vec{B} \cdot \vec{A}$

$= BA \cdot \cos\theta$

Most suitable ans is 2 [Otherwise ABCD]

Question64

A square shaped coil of area $70cm^2$ having 600 turns rotates in a magnetic field of $0.4wbm^{-2}$, about an axis which is parallel to one of the side of the coil and perpendicular to the direction of field. If the coil completes 500 revolution in a minute, the instantaneous emf when the plane of the coil is inclined at 60° with the field, will be _____ V. (Take $\pi = \frac{22}{7}$)
[1-Feb-2023 Shift 2]

Answer: 44

Solution:

$N = 600, A = 70 \times 10^{-4} m^2, B = 0.4T$

$\omega = \frac{500 \times 2\pi}{60} = \frac{100\pi}{6} rad/s$

$E = NAB\omega sin\omega t \quad \omega t \text{ is angle b/w } \vec{A} \& \vec{B}$

$= 600 \times 70 \times 10^{-4} \times 0.4 \times \frac{100\pi}{6} \times \frac{1}{2}$

$= 44V$

Question65

A long straight wire of circular cross-section (radius a) is carrying steady current I. The current I is uniformly distributed across this cross-section. The magnetic field is
[6-Apr-2023 shift 1]

Options:

A. zero in the region r < a and inversely proportional to r in the region r > a

B. inversely proportional to r in the region r < a and uniform throughout in the region r > a

C. directly proportional to r in the region r < a and inversely proportional to r in the region r > a

D. uniform in the region r < a and inversely proportional to distance r from the axis, in the region r > a

Answer: C

Solution:

It is a case of solid infinite current carrying wire.
Using ampere circuital law,
CASE I: if $r \leq R$

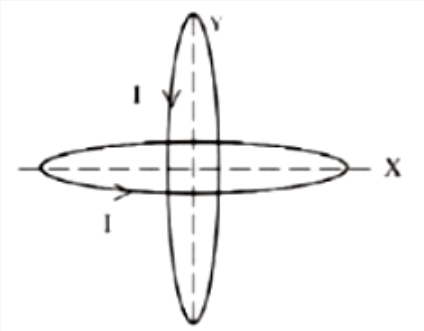
$$B = \frac{\mu_0 i}{2\pi R^2} r$$

CASE II: $r > R$

$$B = \frac{\mu_0 i}{2\pi r}$$

Question66

Two identical circular wires of radius 20 cm and carrying current $\sqrt{2}$A are placed in perpendicular planes as shown in figure. The net magnetic field at the centre of the circular wires is _____ $\times 10^{-8}$T. (Take $\pi = 3.14$)



[6-Apr-2023 shift 1]

Answer: 628

Solution:

$$\vec{B}_{net} = \frac{\mu_0 i}{2r}\hat{i} + \frac{\mu_0 i}{2r}\hat{j}$$

$$\Rightarrow B_{net} = \frac{\mu_0 i}{2r}\sqrt{2} = 4\pi \times 10^{-7} \times \sqrt{2} \times \sqrt{2} \times \frac{1}{2 \times 0.2} = 2 \times 3.14 \times 10^{-6} = 628 \times 10^{-8} T$$

Question67

A proton with a kinetic energy of 2.0 eV moves into a region of uniform magnetic field of magnitude $\frac{\pi}{2} \times 10^{-3}$T. The angle between the direction of magnetic field and velocity of proton is 60°. The pitch of the helical path taken by the proton is _____ cm. (Take, mass of proton $= 1.6 \times 10^{-27}$ kg and Charge on proton $= 1.6 \times 10^{-19}$C) .
[6-Apr-2023 shift 2]

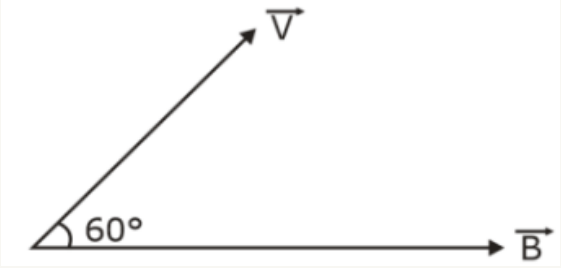
Answer: 40

Solution:

$$B = \frac{\pi}{2} \times 10^{-3}$$

$$K.E. = \frac{1}{2}mV^2$$

$$\Rightarrow V = \sqrt{\frac{2KE}{m}}$$



$$\text{PItch} = v\cos 60^\circ \times \text{ time period of one rotation}$$

$$= v\cos 60^\circ \times \frac{2\pi m}{eB}$$

$$= \sqrt{\frac{2\times 2\times 1.6\times 10^{-9}}{1.6\times 10^{-27}}} \times \cos 60^\circ \times \frac{2\pi \times 1.6\times 10^{-27}}{1.6\times 10^{-19}\times \frac{\pi}{2}\times 10^{-3}}$$

$$= 2\times 10^{4}\times \frac{1}{2}\times 4\times 10^{-5}$$

$$= 4\times 10^{-1}\text{m} = 40\,\text{cm}$$

Question68

A charge particle moving in magnetic field B, has the components of velocity along B as well as perpendicular to B. The path of the charge particle will be
[8-Apr-2023 shift 1]

Options:

A. Helical path with the axis along magnetic field B

B. Straight along the direction of magnetic field B

C. Helical path with the axis perpendicular to the direction of magnetic field B

D. Circular path

Answer: A

Solution:

Path will be helical with axis along uniform $\vec{B}$ –.

Question69

The magnetic intensity at the center of a long current carrying solenoid is found to be $1.6\times 10^{3}\text{Am}^{-1}$. If the number of turns is 8 per cm, then the current flowing through the solenoid is _____ A.
[8-Apr-2023 shift 1]

Answer: 2

Solution:

$H = 1.6 \times 10^3 A/m$, n = 8 per cm = 800 per m

$H = nI \Rightarrow I = \frac{H}{n}$

$I = \frac{1.6 \times 10^3}{8 \times 10^2} \Rightarrow I = 2A$

Question70

The ratio of magnetic field at the centre of a current carrying coil of radius r to the magnetic field at distance r from the centre of coil on its axis is $\sqrt{x}$: 1. The value of x is _____.
[8-Apr-2023 shift 2]

Answer: 8

Solution:

$$B_{axis} = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}$$

$$\frac{(B_{axis})x = R}{(B_{axis})x = 0} = \frac{\frac{\mu_0 i R^2}{2(R^2 + R^2)^{3/2}}}{\frac{\mu_0 i R^2}{2(R^2)^{3/2}}} = \frac{R^3}{2^{3/2} R^3} = \frac{1}{\sqrt{8}}$$

$$\frac{(B)_{At\ centrs}}{(B)_{At} x = R} = \frac{\sqrt{8}}{1}$$

$x = 8$

Question71

Given below are two statements :
Statements I : If the number of turns in the coil of a moving coil

galvanometer is doubled then the current sensitivity becomes double.
Statements II : Increasing current sensitivity of a moving coil galvanometer by only increasing the number of turns in the coil will also increase its voltage sensitivity in the same ratio
In the light of the above statements, choose the correct answer from the options given below :
[10-Apr-2023 shift 1]

Options:

A. Both Statement I and Statement II are true

B. Both Statement I and Statement II are false

C. Statement I is true but Statement II is false

D. Statement I is false but Statement II is true

Answer: C

Solution:

$$I = \frac{k\theta}{NBA}$$

$$C \cdot S = \frac{\theta}{I} = \frac{NBA}{K}$$

$N \rightarrow 2N \quad C \cdot S \rightarrow 2CS$

But V.S. $= \frac{\theta}{V} = \frac{NBA}{KR}$

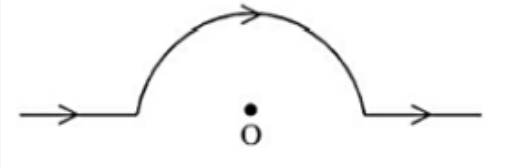
$N \rightarrow 2N C \cdot S \rightarrow 2CS$

But V.S. $= \frac{\theta}{v} = \frac{\theta}{IR} = \frac{NBA}{RK}$

As $N \rightarrow 2N, R \rightarrow 2R$ So V.S = constant

Question72

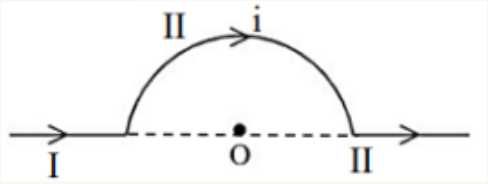
A straight wire carrying a current of 14A is bent into a semicircular arc of radius 2.2 cm as shown in the figure. The magnetic field produced by the current at the centre (O) of the arc. is _____ $\times 10^{-4}$T



[10-Apr-2023 shift 2]

Answer: 2

Solution:



$B_{total} = B_I + B_{II} + B_{III}$

$B_I = 0$

$B_{III} = 0$

Because $\vec{dl} \times \vec{r} = 0$

Now, magnetic field due to semicircular ring at its center is given by

$$B_{II} = \frac{\mu_0 i}{4R}$$

$$= \frac{4\pi \times 10^{-7} \times 14}{4 \times 2.2 \times 10^{-2}}$$

$$= \frac{22}{7} \times \frac{10^{-7} \times 14}{22 \times 10^{-3}}$$

$$= 2 \times 10^{-7}$$

$$= 2$$

Question73

An electron is allowed to move with constant velocity along the axis of current carrying straight solenoid.
A. The electron will experience magnetic force along the axis of the solenoid.
B. The electron will not experience magnetic force.
C. The electron will continue to move along the axis of the solenoid.
D. The electron will be accelerated along the axis of the solenoid.

E. The electron will follow parabolic path-inside the solenoid.
Choose the correct answer from the options given below:
[11-Apr-2023 shift 2]

Options:

A. A and D only

B. B, C and D only

C. B and E only

D. B and C only

Answer: D

Solution:

We know that
In the solenoid magnetic field along the axis of solenoid.
When charge particle moving inside solenoid along the axis $F = 0$

$$\vec{F}_m = q\left(\vec{V} \times \vec{B}\right)$$

So, $\vec{F}_m = 0$

And it moves with constant velocity.

Question74

An electron is moving along the positive x-axis. If the uniform magnetic field is applied parallel to the negative z-axis, then
A. The electron will experience magnetic force along positive y-axis
B. The electron will experience magnetic force along negative y-axis
C. The electron will not experience any force in magnetic field
D. The electron will continue to move along the positive x-axis
E. The electron will move along circular path in magnetic field
Choose the correct answer from the options given below :
[13-Apr-2023 shift 2]

Options:

A. B and E only

B. A and E only

C. B and D only

D. C and D only

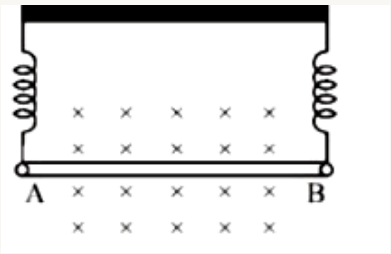
Answer: A

Solution:

$\vec{F} = q(\vec{V} \times \vec{B})$
$\vec{F} = -e(\vec{V} \times \vec{B})$
Force will be along -ve yaxis As magnetic force is $\perp_r$ to velocity, path of electron must be circle.

Question75

A straight wire AB of mass 40g and length 50 cm is suspended by a pair of flexible leads in uniform magnetic field of magnitude 0.40T as shown in the figure. The magnitude of the current required in the wire to remove the tension in the supporting leads is _____ A. (. Take g = 10ms^{-2})



[13-Apr-2023 shift 2]

Answer: 2

Solution:

For equilibrium :
$Mg = IlB$

$$I = \frac{Mg}{IB} = \frac{40 \times 10^{-3} \times 10}{50 \times 10^{-2} \times 0.4} = 2A$$

Question76

An election in a hydrogen atom revolves around its nucleus with a speed of $6.76 \times 10^{6} ms^{-1}$ in an orbit of radius 0.52A°. The magnetic field produced at the nucleus of the hydrogen atom is _____ T. [15-Apr-2023 shift 1]

Answer: 40

Solution:

Biot-Savart law,

$$B = \frac{\mu_0}{4\pi} \times \frac{qV}{r^2}$$

$$B = 10^{-7} \times \frac{1.6 \times 10^{-19} \times 6.76 \times 10^{6}}{(0.52 \times 10^{-10})^2}$$

$$B = 40T$$

Question77

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In an uniform magnetic field, speed and energy remains the same for a moving charged particle.
Reason (R): Moving charged particle experiences magnetic force perpendicular to its direction of motion.
[24-Jun-2022-Shift-1]

Options:

A. Both (A) and (R) are true and (R) is the correct explanation of (A).

B. Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

C. (A) is true but (R) is false.

D. (A) is false but (R) is true.

Answer: A

Solution:

Magnetic force $\vec{F} \perp \vec{v}$
$\Rightarrow W_b = 0$
$\Rightarrow \Delta KE = 0$ and speed remains constant.

Question78

The magnetic field at the centre of a circular coil of radius r, due to current I flowing through it, is B. The magnetic field at a point along the axis at a distance $\frac{r}{2}$ from the centre is :
[24-Jun-2022-Shift-1]

Options:

A. $B/2$

B. $2B$

C. $\left(\frac{2}{\sqrt{5}}\right)^3 B$

D. $\left(\frac{2}{\sqrt{3}}\right)^3 B$

Answer: C

Solution:

$$B = \frac{\mu_0 I}{2r}$$

$$B_a = \frac{\mu_0 I r^2}{2\left(r^2 + \frac{r^2}{4}\right)}$$

$$\Rightarrow \frac{B_a}{B} = \left(\frac{2}{\sqrt{5}}\right)^3$$

$$\Rightarrow B_a = \left(\frac{2}{\sqrt{5}}\right)^3 B$$

Question79

A proton, a deutron and an α-particle with same kinetic energy enter into a uniform magnetic field at right angle to magnetic field. The ratio of the radii of their respective circular paths is : [24-Jun-2022-Shift-2]

Options:

A. $1 : \sqrt{2} : \sqrt{2}$

B. $1 : 1 : \sqrt{2}$

C. $\sqrt{2} : 1 : 1$

D. $1 : \sqrt{2} : 1$

Answer: D

Solution:

$$\therefore r = \frac{mv}{qB} = \frac{\sqrt{2m(KE)}}{qB}$$

$$\Rightarrow r_1 : r_2 : r_3 = \frac{\sqrt{m_1}}{q_1} : \frac{\sqrt{m_2}}{q_2} : \frac{\sqrt{m_3}}{q_3}$$

$$= \frac{\sqrt{1}}{1} : \frac{\sqrt{2}}{1} : \frac{\sqrt{4}}{2}$$

$$= 1 : \sqrt{2} : 1$$

Question80

A long straight wire with a circular cross-section having radius R, is carrying a steady current I. The current I is uniformly distributed across this cross-section. Then the variation of magnetic field due to current I with distance r(r < R) from its centre will be : [25-Jun-2022-Shift-1]

Options:

A. $B \propto r^2$

B. $B \propto r$

C. $B \propto \frac{1}{r^2}$

D. $B \propto \frac{1}{r}$

Answer: B

Solution:

$\int \overline{B} \cdot \overline{dl} = \mu_0 I_{in}$

$\Rightarrow B \times 2\pi r = \frac{\mu_0 I}{\pi R^2} \times \pi r^2$

$\Rightarrow B \propto r$

Question81

A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of magnetic field will be equal to [25-Jun-2022-Shift-2]

Options:

A. B

B. 2B

C. 4B

D. $\frac{B}{2}$

Answer: A

Solution:

$B = \mu_0 ni$
Now $i \rightarrow 2i$
And $n \rightarrow \frac{n}{2}$
$B' = \mu_0 \frac{n}{2} \times 2i = \mu_0 ni = B$

Question82

A proton and an alpha particle of the same velocity enter in a uniform magnetic field which is acting perpendicular to their direction of motion. The ratio of the radii of the circular paths described by the alpha particle and proton is :
[26-Jun-2022-Shift-1]

Options:

A. 1 : 4

B. 4 : 1

C. 2 : 1

D. 1 : 2

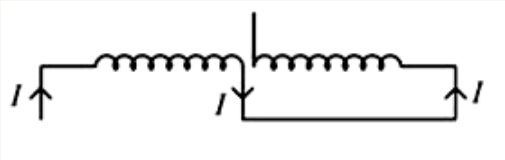
Answer: C

Solution:

$R = \frac{mv}{qB}$
$\frac{R_\alpha}{R_P} = \frac{M_\alpha}{M_P} \times \frac{q_P}{q_\alpha}$
$\frac{R_\alpha}{R_P} = \frac{4}{1} \times \frac{1}{2} = 2$

Question83

A bar magnet having a magnetic moment of $2.0 \times 10^5 JT^{-1}$, is placed along the direction of uniform magnetic field of magnitude $B = 14 \times 10^{-5}$ T. The work done in rotating the magnet slowly through 60° from the direction of field is :



[26-Jun-2022-Shift-2]

Options:

A. 14J

B. 8.4J

C. 4J

D. 1.4J

Answer: A

Solution:

$U = -M \cdot B$
So $U_f - U_i = -MB(1 - \cos\theta)$
$= -14J$
So $W = -\Delta U = 14J$

Question84

A metal surface is illuminated by a radiation of wavelength 4500 Å. The ejected photo-electron enters a constant magnetic field of 2 mT making an angle of 90° with the magnetic field. If it starts revolving in a circular path of radius 2 mm, the work function of the metal is

approximately :
[26-Jun-2022-Shift-2]

Options:

A. 1.36 eV

B. 1.69 eV

C. 2.78 eV

D. 2.23 eV

Answer: A

Solution:

$\frac{hc}{\lambda} - \varphi = K\,E$ ..... . (i)

$R = \frac{mv}{Bq} = \frac{\sqrt{2m(K\,E)}}{Bq}$..... . (ii)

Putting the values,

$\varphi \simeq 1.36\,eV$

Question85

Two 10 cm long, straight wires, each carrying a current of 5A are kept parallel to each other. If each wire experienced a force of 10^{-5}N, then separation between the wires is _____ cm.
[26-Jun-2022-Shift-2]

Answer: 5

Solution:

$\frac{dF}{dl} = \frac{\mu_0 i_1 i_2}{2\pi d}$

So $\frac{2\times10^{-7}\times5\times5}{d} = \frac{10^{-5}}{10\times10^{-2}}$

$d = \frac{2\times10^{-7}\times5\times5}{10^{-4}}$

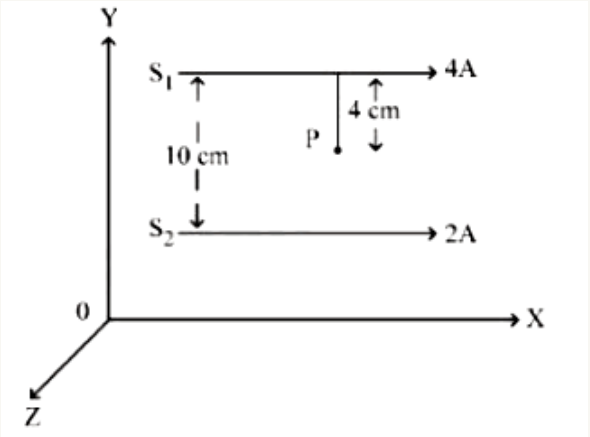
$= 50\,mm$

$= 5\,cm$

Question86

Two long parallel conductors S_1 and S_2 are separated by a distance 10 cm and carrying currents of 4A and 2A respectively. The conductors are placed along X -axis in $X-Y$ plane. There is a point P located between the conductors (as shown in figure).
A charge particle of 3π coulomb is passing through the point P with velocity $\vec{v} = \left(2\hat{i}+3\hat{j}\right)m/s$; where $\hat{i}$ and $\hat{j}$ represents unit vector along x&y axis respectively.

The force acting on the charge particle is $4\pi\times10^{-5}\left(-x\hat{i}+2\hat{j}\right)N$. The value of x is :



[27-Jun-2022-Shift-2]

Options:

A. 2

B. 1

C. 3

D. −3

Answer: C

Solution:

$$\text{Field at P is } = \left(\frac{\mu_0 \times i_1}{2\pi r_1} - \frac{\mu_0 i_2}{2\pi r_2}\right)\left(-\hat{k}\right)$$

$$= -\left(\frac{\mu_0 4}{2\pi \times 0.04} - \frac{\mu_0 \times 2}{2\pi \times 0.06}\right)\hat{k} = -\frac{\mu_0 \times 200}{6\pi}\hat{k}$$

So, force $\vec{F} = q\vec{v} \times \vec{B}$

$$= 3\pi\left(2\hat{i} + 3\hat{j}\right) \times \left(-\left(\frac{\mu_0 \times 200}{6\pi}\hat{k}\right)\right)$$

$$= 3\pi\left(\frac{200\mu_0}{3\pi}\hat{j} - \frac{100\mu_0}{\pi}\hat{i}\right)$$

$$= 200\mu_0\hat{j} - 300\mu_0\hat{i}$$

$$= 4\pi \times 10^{-5}\left(2\hat{j} - 3\hat{i}\right)$$

So, x = 3

Question87

A deuteron and a proton moving with equal kinetic energy enter into a uniform magnetic field at right angle to the field. If r_d and r_p are the radii of their circular paths respectively, then the ratio $\frac{r_d}{r_p}$ will be $\sqrt{x}$: 1 where x is_____
[27-Jun-2022-Shift-2]

Answer: 2

Solution:

$$R = \frac{\sqrt{2mK}}{qB}$$

$$\text{So, } \frac{r_d}{r_p} = \frac{\sqrt{m_d}/q_d}{\sqrt{m_p}/q_p}$$

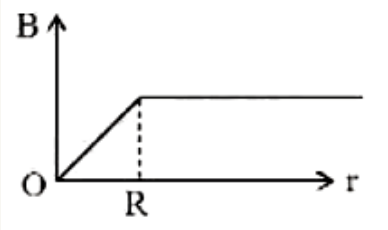
$= \sqrt{2}$
So x = 2

Question88

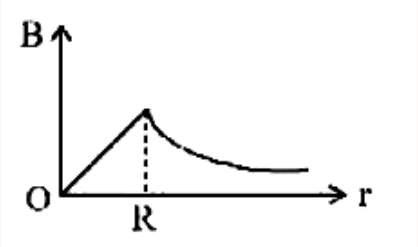
An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface.
Choose the correct representation of magnetic field (B) as a function of radial distance (r) from the axis of cylinder.
[28-Jun-2022-Shift-1]

Options:

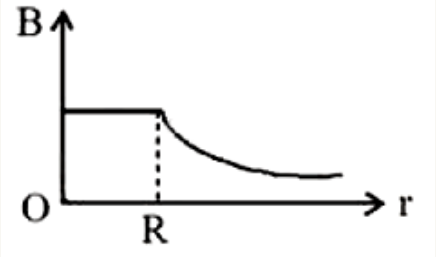
A.



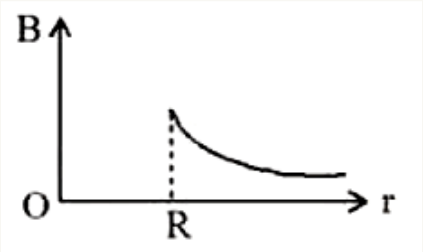
B.



C.



D.



Answer: D

Solution:

Inside a hollow cylindrical conductor with uniform current distribution net magnetic field is zero in hollow space.

But outside the cylindrical conductor $B \propto \frac{1}{r}$

⇒ Graph in option D would be a correct one

Question89

A singly ionized magnesium atom (A = 24) ion is accelerated to kinetic energy 5 keV, and is projected perpendicularly into a magnetic field B of the magnitude 0.5T. The radius of path formed will be _____cm.
[28-Jun-2022-Shift-1]

Answer: 10

Solution:

$$R = \frac{mv}{qB}$$

$$R = \frac{\sqrt{2mK\,E}}{qB}$$

$$= \frac{\sqrt{2 \times 24 \times 1.67 \times 10^{-27} \times 5 \times 1.6 \times 10^{-16}}}{1.6 \times 10^{-19} \times 0.5}$$

$$= 10.009\,\text{cm} = 10\,\text{cm}$$

Question90

Two parallel, long wires are kept 0.20m apart in vacuum, each carrying current of xA in the same direction. If the force of attraction per meter of each wire is 2×10^{-6}N, then the value of x is approximately :
[28-Jun-2022-Shift-2]

Options:

A. 1

B. 2.4

C. 1.4

D. 2

Answer: C

Solution:

$$\frac{dF}{dl} = 2 \times 10^{-6} \text{N/m} = \frac{\mu_0 i_1 i_2}{2\pi d}$$

$$2 \times 10^{-6} = \frac{2 \times 10^{-7} \times x^2}{0.2}$$

$$x = \sqrt{2} \simeq 1.4$$

Question91

A coil is placed in a time varying magnetic field. If the number of turns in the coil were to be halved and the radius of wire doubled, the electrical power dissipated due to the current induced in the coil would be :
(Assume the coil to be short circuited.)
[28-Jun-2022-Shift-2]

Options:

A. Halved

B. Quadrupled

C. The same

D. Doubled

Answer: B

Solution:

As number of turns are halved so length of wire is halved, and radius is doubled, then area will be 4 times the previous one if previous resistance is R then new resistance is $\frac{R}{8}$ and if previous emf is E then new emf will be $\frac{E}{2}$ so

$P_i = \frac{E^2}{R}$

$P_f = \frac{(E/2)^2}{R/8} = \frac{2E^2}{R} = 2P_i$

Question92

A charge particle moves along circular path in a uniform magnetic field in a cyclotron. The kinetic energy of the charge particle increases to 4 times its initial value. What will be the ratio of new radius to the original radius of circular path of the charge particle : [29-Jun-2022-Shift-1]

Options:

A. 1 : 1

B. 1 : 2

C. 2 : 1

D. 1 : 4

Answer: C

Solution:

$R = \frac{mv}{Bq} = \frac{\sqrt{2mK}}{Bq}$
$\Rightarrow R \propto \sqrt{K}$
$\Rightarrow$ ratio $= 2 : 1$

Question93

Two long current carrying conductors are placed parallel to each other at a distance of 8 cm between them. The magnitude of magnetic field produced at mid-point between the two conductors due to current flowing in them is 300μT . The equal current flowing in the two conductors is:
[29-Jun-2022-Shift-2]

Options:

A. 30A in the same direction.

B. 30A in the opposite direction.

C. 60A in the opposite direction.

D. 300A in the opposite direction.

Answer: B

Solution:

As $B_{net} \neq 0$ that is the wires are carrying current in opposite direction.
$\frac{\mu_0 I \times 2}{2\pi(4 \times 10^{-2})} = 30 \times 10^{-6} T$
$\Rightarrow I = \frac{30 \times 10^{-6}}{10^{-6}} A = 30A$ in opposite direction.

Question94

Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is 6 : 5

and their respective masses ratio is 9 : 4. Then, the ratio of their charges will be :
[25-Jul-2022-Shift-1]

Options:

A. 8 : 5

B. 5 : 4

C. 5 : 3

D. 8 : 7

Answer: B

Solution:

We know that $R = \frac{mv}{Bq} = \sqrt{\frac{2mK}{Bq}}$

$\Rightarrow$ Ratio of radii $= \frac{R_1}{R_2} = \sqrt{\frac{m_1}{m_2}\frac{q_2}{q_1}}$

$\Rightarrow \frac{6}{5} = \sqrt{\frac{9}{4}\frac{q_2}{q_1}}$

$\Rightarrow \frac{q_1}{q_2} = \frac{3}{2} \times \frac{5}{6} = \frac{5}{4}$

Question95

The electric current in a circular coil of 2 turns produces a magnetic induction B_1 at its centre. The coil is unwound and in rewound into a circular coil of 5 tuns and the same current produces a magnetic induction B_2 at its centre. The ratio of $\frac{B_2}{B_1}$ is
[25-Jul-2022-Shift-2]

Options:

A. $\frac{5}{2}$

B. $\frac{25}{4}$

C. $\frac{5}{4}$

D. $\frac{25}{2}$

Answer: B

Solution:

$B = \frac{n\mu_0 I}{2R}$

$B_1 = \frac{2\mu_0 I}{2R_1}$

$B_2 = \frac{5\mu_0 I}{2R_2}$

$R_2 = \frac{2R_1}{5}$

$\Rightarrow \frac{B_2}{B_1} = \frac{5}{2} \times \frac{R_1}{R_2} = \frac{25}{4}$

Question96

A charge particle is moving in a uniform magnetic field $\left(2\hat{i} + 3\hat{j}\right)$ T. If it has an acceleration of $\left(\alpha\hat{i} - 4\hat{j}\right) m/s^2$, then the value of α will be : [26-Jul-2022-Shift-1]

Options:

A. 3

B. 6

C. 12

D. 2

Answer: B

Solution:

As magnetic force is perpendicular to magnetic field
So, $\vec{F} \cdot \vec{B}$ must be 0
So, $2\alpha - 12 = 0$
$\alpha = 6$

Question97

B_X and B_Y are the magnetic fields at the centre of two coils X and Y respectively each carrying equal current. If coil X has 200 turns and 20 cm radius and coil Y has 400 turns and 20 cm radius, the ratio of B_X and B_Y is :
[26-Jul-2022-Shift-1]

Options:

A. 1 : 1

B. 1 : 2

C. 2 : 1

D. 4 : 1

Answer: B

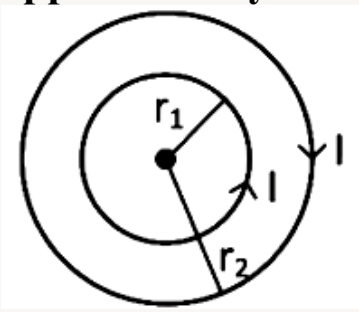
Solution:

$$B = \frac{\mu_0 N I}{2R}$$
$$\frac{B_X}{B_Y} = \frac{N_x R_y}{N_y R_x}$$
$$= \frac{200 \times 20}{400 \times 20} = \frac{1}{2}$$

Question98

Two concentric circular loops of radii $r_1 = 30$ cm and $r_2 = 50$ cm are placed in X − Y plane as shown in the figure. A current I = 7A is

flowing through them in the direction as shown in figure. The net magnetic moment of this system of two circular loops is approximately :



[26-Jul-2022-Shift-2]

Options:

A. $\frac{7}{2}\hat{k}Am^2$

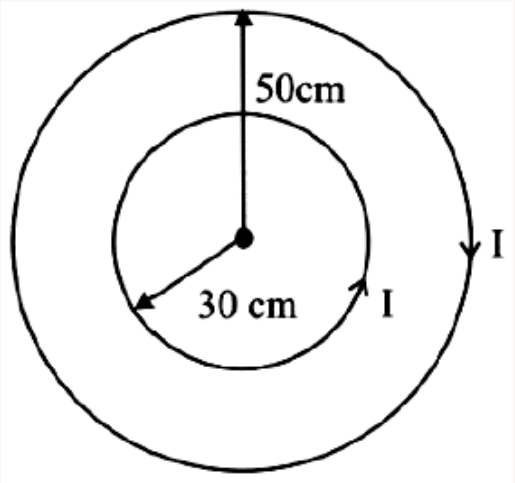
B. $-\frac{7}{2}\hat{k}Am^2$

C. $7\hat{k}Am^2$

D. $-7\hat{k}Am^2$

Answer: B

Solution:



Magnetic moment

$\vec{M} = -i\pi(0.5)^2\hat{k} + i\pi(0.3)^2\hat{k}$

$\vec{M} = -7\times\frac{22}{7}\left(\frac{25}{100} - \frac{9}{100}\right)\hat{k}$

$= -22\left(\frac{16}{100}\right)\hat{k}$

$\vec{M} = -3.52\hat{k}Am^2$

$= -\frac{7}{2}\hat{k}Am^2$

Question99

A cyclotron is used to accelerate protons. If the operating magnetic field is 1.0T and the radius of the cyclotron 'dees' is 60 cm, the kinetic energy of the accelerated protons in MeV will be :
[use $m_p = 1.6 \times 10^{-27}$ kg, $e = 1.6 \times 10^{-19}$C]
[27-Jul-2022-Shift-2]

Options:

A. 12

B. 18

C. 16

D. 32

Answer: B

Solution:

Kinetic energy of electron in cyclotron

$= \left[\frac{q^2B^2r_0^2}{2m}\right]$

$= 18\,MeV$

Question100

The current sensitivity of a galvanometer can be increased by :
(A) decreasing the number of turns
(B) increasing the magnetic freld
(C) decreasing the area of the coil

(D) decreasing the torsional constant of the spring
Choose the most appropriate answer from the options given below : [28-Jul-2022-Shift-1]

Options:

A. (B) and (C) only

B. (C) and (D) only

C. (A) and (C) only

D. (B) and (D) only

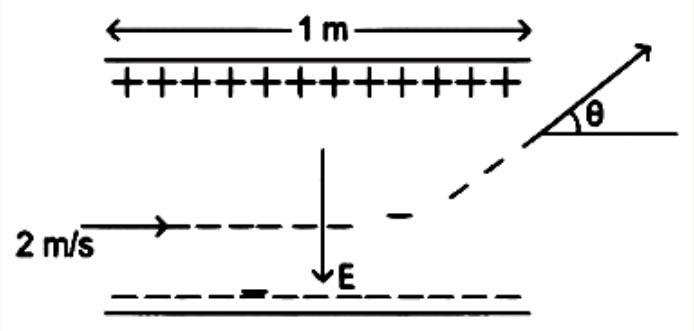
Answer: D

Solution:

$NiAB = k\theta$
$\Rightarrow \frac{\theta}{i} = \frac{NAB}{k}$
$\Rightarrow$ Sensitivity increases if $B\uparrow$ and $k\downarrow$

Question101

A uniform electric field E = (8m/e)V/m is created between two parallel plates of length 1m as shown in figure, (where m = mass of electron and e = charge of electron). An electron enters the field symmetrically between the plates with a speed of 2m/ s. The angle of the deviation (θ) of the path of the electron as it comes out of the field will be_____.



[28-Jul-2022-Shift-2]

Options:

A. $\tan^{-1}(4)$

B. $\tan^{-1}(2)$

C. $\tan^{-1}\left(\frac{1}{3}\right)$

D. $\tan^{-1}(3)$

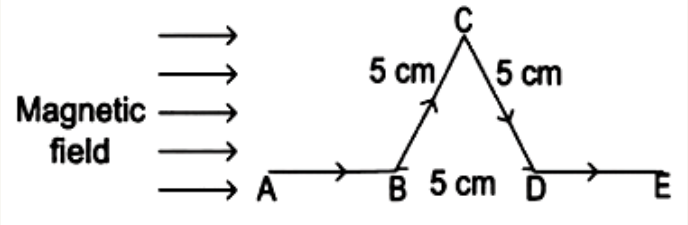
Answer: B

Solution:

$E = \frac{8m}{e} V/m$
$l = 1m$
$v_x = 2m/s$
$a_y = -8m/s^2$
$t = \frac{l}{v_x} = \frac{1}{2} s$
$\Rightarrow |v_y| = 4m/s$
$\Rightarrow$ angle of deviation $= \theta$
$\tan\theta = \frac{v_y}{v_x}$
$\theta = \tan^{-1}\left(\frac{4}{2}\right) = \tan^{-1}(2)$

Question102

A triangular shaped wire carrying 10A current is placed in a uniform magnetic field of 0.5T, as shown in figure. The magnetic force on segment CD is
(Given BC = CD = BD = 5 cm.



[28-Jul-2022-Shift-2]

Options:

A. 0.126N

B. 0.312N

C. 0.216N

D. 0.245N

Answer: C

Solution:

$$\vec{F} = \vec{il} \times \vec{B}$$
$$= il\,B\sin 60^\circ$$
$$= 10\times \frac{5}{100}\times 0.5\times \frac{\sqrt{3}}{2}$$
$$= 0.2165N$$

Question103

A closely wounded circular coil of radius 5 cm produces a magnetic field of 37.68×10^{-4}T at its center. The current through the coil is _____ A.
[Given, number of turns in the coil is 100 and $\pi = 3.14$]
[29-Jul-2022-Shift-1]

Answer: 3

Solution:

$$B = \frac{\mu_0 nI}{2R}$$
$$37.68 \times 10^{-4} = \frac{4\pi \times 10^{-7} 100I}{2 \times 5 \times 10^{-2}}$$
$$I = \frac{300A}{100}$$
$$= 3A$$

Question104

Magnetic fields at two points on the axis of a circular coil at a distance of 0.05m and 0.2m from the centre are in the ratio 8 : 1. The radius of coil is
[25 Feb 2021 Shift 1]

Options:

A. 0.2m

B. 0.1m

C. 0.15m

D. 1.0m

Answer: B

Solution:

Given, $d_1 = 0.05\text{m}, d_2 = 0.2\text{m}$

and $B_1 : B_2 = 8 : 1$

Let radius of coil be r.

As we know that, magnetic field at a point on axis of a coil,

$$B = \frac{\mu_0}{2} \cdot \frac{I r^2}{(d_1^2 + r^2)^{3/2}}$$

$$\therefore \quad \frac{B_1}{B_2} = \frac{r^2 \cdot (d_2^2 + r^2)^{3/2}}{(d_1^2 + r^2)^{3/2} \cdot r^2} = \frac{(d_2^2 + r^2)^{3/2}}{(d_1^2 + r^2)^{3/2}}$$

$$\frac{8}{1} = \left(\frac{d_2^2 + r^2}{d_1^2 + r^2} \right)^{3/2} \Rightarrow \frac{2^3}{1} = \left(\frac{d_2^2 + r^2}{d_1^2 + r^2} \right)^{3/2}$$

$$\frac{4}{1} = \frac{d_2^2 + r^2}{d_1^2 + r^2}$$

$$\Rightarrow 4d_1^2 + 4r^2 = d_2^2 + r^2$$

$$\Rightarrow 4(0.05)^2 + 4r^2 = (0.2)^2 + r^2 \Rightarrow 3r^2 = 0.03$$

$$\Rightarrow r = 0.1\text{m}$$

Question105

A proton, a deuteron and an alpha-particle are moving with same momentum in a uniform magnetic field. The ratio of magnetic forces acting on them is ..... and their speed is ..... in the ratio.
[25 Feb 2021 Shift 1]

Options:

A. 1 : 2 : 4 and 2 : 1 : 1

B. 2 : 1 : 1 and 4 : 2 : 1

C. 4 : 2 : 1 and 2 : 1 : 1

D. 1 : 2 : 4 and 1 : 1 : 2

Answer: B

Solution:

Let F_p, F_d and F_α be the forces and v_p, v_d and v_α be the velocities of proton, deuteron and α−particle, respectively.
Since, $F = Bqv$
On dividing and multiplying F by m, we get

$$F = Bqv\frac{m}{m}$$

$$\Rightarrow F = B\frac{q}{m}p \quad (\because p = mv)$$

$$\Rightarrow F \propto q/m \quad (\because p \text{ and } B \text{ are same})$$

$$\therefore F_p : F_d : F_\alpha = \frac{+q}{m} : \frac{+q}{2m} : \frac{+2q}{4m}$$

$$= 1 : 1/2 : 1/2 = 2 : 1 : 1$$

and $= Bqv$

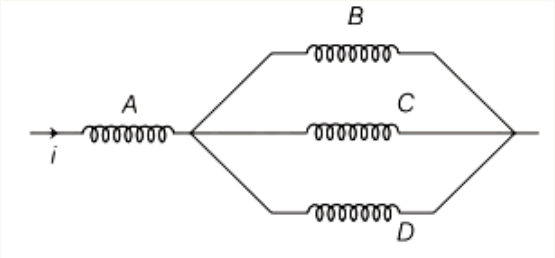
$$\Rightarrow \propto F/q$$

$$\therefore v_p : v_d : v_\alpha = \frac{F_p}{q_p} : \frac{F_d}{q_d} : \frac{F_\alpha}{q_\alpha} = \frac{2x}{q} : \frac{x}{q} : \frac{x}{2q}$$

$$= 2 : 1 : 1/2 = 4 : 2 : 1$$

Question106

Four identical long solenoids A, B, C and D are connected to each other as shown in the figure. If the magnetic field at the centre of A is 3T , the field at the centre of C would be (Assume that, the magnetic field is confined with in the volume of respective solenoid)



[18 Mar 2021 Shift 1]

Options:

A. 12T

B. 6T

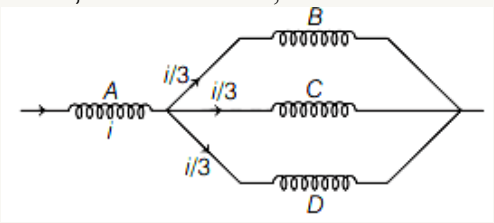
C. 9T

D. 1T

Answer: D

Solution:

Given, magnetic field at the centre at A is 3T .
Since, the solenoid B, C and D are connected in parallel arrangement.



We know that,
The magnetic field in the circuit is directly proportional to the
$B \propto i$

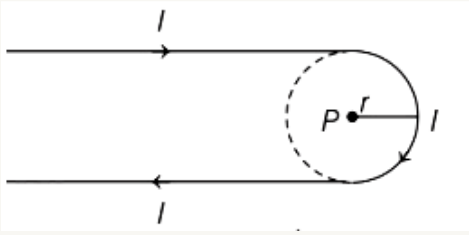
$$\Rightarrow \frac{B_A}{B_C} = \frac{I_A}{I_C} \Rightarrow \frac{3T}{B_C} = \frac{i}{i/3}$$

$$\Rightarrow B_C = 1T$$

So, the field at the centre at C = 1T .

Question107

A hairpin like shape as shown in figure is made by bending a long current carrying wire. What is the magnitude of a magnetic field at point P which lies on the centre of the semicircle?



[17 Mar 2021 Shift 2]

Options:

A. $\frac{\mu_0 I}{4\pi r}(2-\pi)$

B. $\frac{\mu_0 I}{4\pi r}(2+\pi)$

C. $\frac{\mu_0 I}{2\pi r}(2+\pi)$

D. $\frac{\mu_0 I}{2\pi r}(2-\pi)$

Answer: B

Solution:

For the straight wire, the expression of the magnetic field,

$B_1 = B_3 = \frac{\mu_0 I}{4\pi r}$

For the semicircular wire, the expression of the magnetic field,

$B_2 = \frac{\mu_0 I}{4r}$

Hence, the total magnetic field at the point P, $B = B_1 + B_2 + B_3$

Substituting the values in the above equation, we get

$B = \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4r} + \frac{\mu_0 I}{4\pi r} \Rightarrow B = \frac{\mu_0 I}{4\pi r}(1+1+\pi)$

$B = \frac{\mu_0 I}{4\pi r}(2+\pi)$

Question108

A proton and an α-particle, having kinetic energies K_p and K_α, respectively, enter into a magnetic field at right angles.
The ratio of the radii of trajectory of proton to that of α-particle is

2 : 1. The ratio of $K_p : K_\alpha$ is [18 Mar 2021 Shift 2]

Options:

A. 1 : 8

B. 8 : 1

C. 1 : 4

D. 4 : 1

Answer: D

Solution:

Given, the charge on the proton, $q_p = +e$
The charge on the alpha particles, $q_a = +2e$
The expression of the radius of the orbit,

$$r = \frac{mv}{qB}$$

$$\Rightarrow r = \frac{p}{qB} \quad (\because p = mv)$$

$$\because \frac{r_p}{r_\alpha} = \frac{p_p}{q_p} \times \frac{q_\alpha}{p_\alpha}$$

$$\Rightarrow \frac{2}{1} = \frac{+2e}{+e} \times \frac{P_p}{p_\alpha}$$

$$\left[\because \frac{r_p}{r_\alpha} = \frac{2}{1}\right]$$

$$\Rightarrow \frac{p_p}{p_a} = 1$$

Since, kinetic energy and momentum are related as

$$K = \frac{p^2}{2m}$$

$$\therefore \frac{K_p}{K_\alpha} = \frac{p_p{}^2 m_\alpha}{m_p p_\alpha{}^2} \Rightarrow \frac{K_p}{K_\alpha} = \frac{m_\alpha}{m_p} = \frac{4m_p}{m_p}$$

$$\Rightarrow \frac{K_p}{K_\alpha} = \frac{4}{1}$$

Question109

A charge Q is moving $d\vec{l}$ distance in the magnetic field B. Find the value of work done by B.
[16 Mar 2021 Shift 2]

Options:

A. 1

B. Infinite

C. Zero

D. -1

Answer: C

Solution:

According to question, a charge Q is moving d distance in magnetic field B.
As we know that force acting on a point charge by magnetic field is always perpendicular to the velocity of particle v.
$\therefore \vec{F} = Q(\vec{v} \times \vec{B})$.....(i)
The above, statement means that $\vec{F}$ is perpendicular to v.
$\because$ Work done, $W = \vec{F} \cdot d\vec{s}$
$= Q(\vec{v} \times \vec{B}) \cdot d\vec{l}$ [using Eq. (i)]
$= Q(\vec{v} \times \vec{B}) dl \cos\theta$
$= Q(\vec{v} \times \vec{B}) dl \cos 90^\circ$ [$\because \theta = 90^\circ \vec{F} \perp v$ or $d\vec{l}$]
$= 0$
$\therefore$ The work done by the magnetic field will be zero.

Question110

A loop of flexible wire of irregular shape carrying current is placed in an external magnetic field.
Identify the effect of the field on the wire.
[18 Mar 2021 Shift 1]

Options:

A. Loop assumes circular shape with its plane normal to the field.

B. Loop assumes circular shape with its plane parallel to the field.

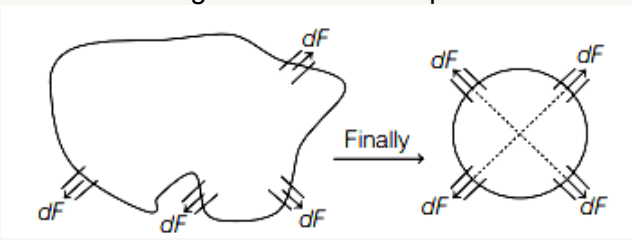
C. Wire gets stretched to become straight.

D. Shape of the loop remains unchanged.

Answer: A

Solution:

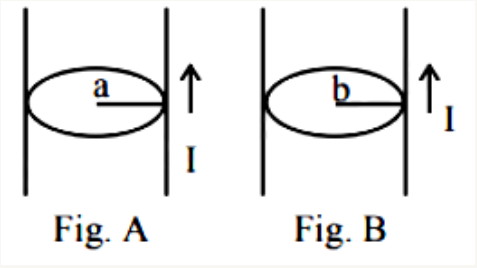
A loop of wire will act like magnetic dipole. In equilibrium, direction of dipole will be parallel to direction of magnetic field and its plane remains normal to the field.



Now on each element of wire force due to magnetic field will be perpendicular to length of element and outward in direction. So, loop will become circular.
Hence, the correct option is (a).

Question111

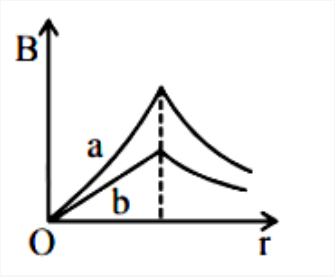
Figure A and B shown two long straight wires of circular cross-section (a and b with a < b), carrying current I which is uniformly distributed across the cross-section. The magnitude of magnetic field B varies with radius r and can be represented as :



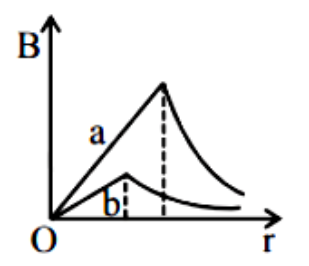
[27 Jul 2021 Shift 2]

Options:

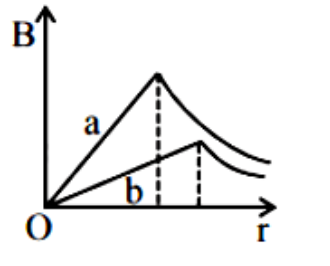
A.



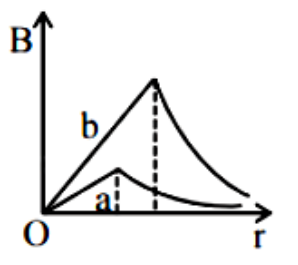
B.



C.



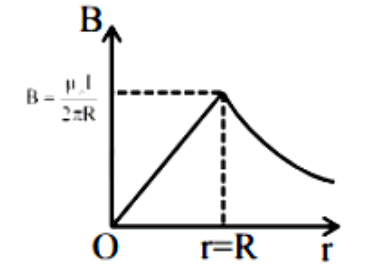
D.



Answer: C

Solution:

Graph for wire of radius R :



As $b > a$

$B_a > B_b$

$B_a = \frac{\mu_0 i}{2\pi a}$

$B_b = \frac{\mu_0 i}{2\pi b}$

Question112

Match List I with List II.

List-I	List-II
(a) Capacitance, C	(i) $M^1L^1T^{-3}A^{-1}$
(b) Permittivity of free space, ε_0	(ii) $M^{-1}L^{-3}T^4A^2$
(c) Permeability of free space, μ_0	(iii) $M^{-1}L^{-2}T^4A^2$
(d) Electric field, E	(iv) $M^1L^1T^{-2}A^{-2}$

Choose the correct answer from the options given below [27 Jul 2021 Shift 2]

Options:

A. (a) → (iii), (b) → (ii), (c) → (iv), (d) → (i)

B. (a) → (iii), (b) → (iv), (c) → (ii), (d) → (i)

C. (a) → (iv), (b) → (ii), (c) → (iii), (d) → (i)

D. (a) → (iv), (b) → (iii), (c) → (ii), (d) → (i)

Answer: A

Solution:

$q = CV$

$$[C] = \left[\frac{q}{V}\right] = \frac{(A \times T)^2}{M L^2 T^{-2}}$$

$$= M^{-1}L^{-2}T^4A^2$$

$$[E] = \left[\frac{F}{q}\right] = \frac{M LT^{-2}}{AT}$$

$$= M LT^{-3}A^{-1}$$

$$F = \frac{q_1 q_2}{4\pi \in_o r^2}$$

$$[\in_0] = M^{-1}L^{-3}T^4A^2$$

Speed of light $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

$\mu_0 = \frac{1}{\epsilon_0 c^2}$

$[\mu_0] = \frac{1}{[M^{-1}L^{-3}T^4A^2][LT^{-1}]^2}$

$= [M^1L^1T^{-2}A^{-2}]$

Question113

Two ions having same mass have charges in the ratio 1 : 2. They are projected normally in a uniform magnetic field with their speeds in the ratio 2 : 3. The ratio of the radii of their circular trajectories is : [25 Jul 2021 Shift 2]

Options:

A. 1 : 4

B. 4 : 3

C. 3 : 1

D. 2 : 3

Answer: B

Solution:

$R = \frac{mv}{qB} \Rightarrow \frac{R_1}{R_2} = \frac{\frac{mv_1}{q_1B}}{\frac{mv_2}{q_2B}} = \frac{v_1}{q_1} \times \frac{q_2}{v_2} = \frac{q_2}{q_1} \times \frac{v_1}{v_2}$

$= \frac{2}{1} \times \left(\frac{2}{3}\right) = \frac{4}{3}$

Question114

A deuteron and an alpha particle having equal kinetic energy enter perpendicular into a magnetic field. Let r_d and r_α be their respective

radii of circular path. The value of $\frac{r_d}{r_\alpha}$ is equal to :
[20 Jul 2021 Shift 1]

Options:

A. $\frac{1}{\sqrt{2}}$

B. $\sqrt{2}$

C. 1

D. 2

Answer: B

Solution:

$r = \frac{mv}{qB} = \frac{\sqrt{2mk}}{qB}$

$\frac{r_d}{r_\alpha} = \sqrt{\frac{m_d}{m_\alpha}\frac{q_\alpha}{q_d}} = \sqrt{\frac{2}{4}}\left(\frac{2}{1}\right) = \sqrt{2}$

Hence option (2).

Question115

A coil having N turns is wound tightly in the form of a spiral with inner and outer radii a and b, respectively. Find the magnetic field at centre, when a current / passes through coil
[31 Aug 2021 Shift 1]

Options:

A. $\frac{\mu_0 IN}{2(b-a)}\ln\left(\frac{b}{a}\right)$

B. $\frac{\mu_0 I}{8}\ln\left[\frac{a+b}{a-b}\right]$

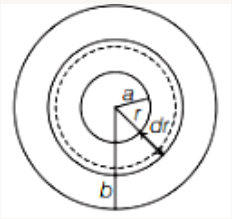
C. $\frac{\mu_0 I}{4(a-b)}\ln\left[\frac{1}{a}-\frac{1}{b}\right]$

D. $\frac{\mu_0 I}{8}\ln\left(\frac{a-b}{a+b}\right)$

Answer: A

Solution:

Given, number of turns = N
Inner radius = a
Outer radius = b
Current = I
Consider an element of thickness dr at a distance r from the centre.



Now, number of turns in dr element, $dN = \frac{N}{(b-a)}dr$
Magnetic field due to this element at the centre,

$$dB = \frac{\mu_0}{2}\frac{dN \cdot I}{r}$$

$$\therefore dB = \frac{1}{2}\frac{\mu_0 I}{r}\frac{N}{(b-a)}dr$$

∴ Total magnetic field at the centre,

$$B = \int_a^b dB = \int_a^b \frac{\mu_0 IN}{2(b-a)}\frac{dr}{r}$$

$$= \frac{N}{2}\frac{\mu_0 I}{(b-a)}\ln r|_a^b$$

$$= \frac{N}{2}\frac{\mu_0 I}{(b-a)}\ln\left(\frac{b}{a}\right)$$

Question116

A current of 1.5A is flowing through a triangle, of side 9 cm each. The magnetic field at the centroid of the triangle is (Assume that, the current is flowing in the clockwise direction.) [31 Aug 2021 Shift 2]

Options:

A. 3×10^{-7}T, outside the plane of triangle

B. $2\sqrt{3} \times 10^{-7}$T, outside the plane of triangle

C. $2\sqrt{3} \times 10^{-5}$T, inside the plane of triangle

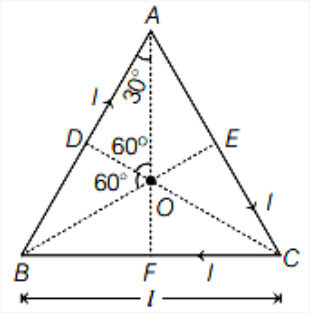
D. 3×10^{-5}T, inside the plane of triangle

Answer: D

Solution:

Given, electric current, I = 1.5 A
Side of triangle, $l = 9\,\text{cm} = 9 \times 10^{-2}\text{m}$
Let OD = OE = OF = r

$AD = \frac{l}{2}$



In Δ AOD, $\frac{OD}{AD} = \tan 30°$

$OD = AD \tan 30°$

$r = OD = \frac{l}{2} \times \frac{1}{\sqrt{3}} = \frac{l}{2\sqrt{3}}$

Magnetic field due to AB at O,

$B_1 = \frac{\mu 0^I}{4\pi r}[\sin 60° + \sin 60°] = \frac{\mu 0^I}{4\pi r} \times \frac{2\sqrt{3}}{2}$

$\Rightarrow B_1 = \frac{\mu_0 d\sqrt{3}}{4\pi r}$ (perpendicular inward)

Now, total magnetic field at O due to all sides,

$B = 3B_1 = \frac{\mu_0}{4\pi r} 3\sqrt{3}$

By substituting the values, we get

$B = \frac{10^{-7} \times 1.5 \times 3\sqrt{3} \times 2\sqrt{3}}{9 \times 10^{-2}} = 3 \times 10^{-5}$T (inside the plane of triangle)

Question117

A co-axial cable consists of an inner wire of radius a surrounded by an outer shell of inner and outer radii b and c, respectively. The inner wire carries an electric current i_0, which is distributed uniformly across cross-sectional area. The outer shell carries an

equal current in opposite direction and distributed uniformly. What will be the ratio of the magnetic field at a distance x from the axis when (i) x < a and (ii) a < x < b ?
[27 Aug 2021 Shift 2]

Options:

A. $\frac{x^2}{a^2}$

B. $\frac{a^2}{x^2}$

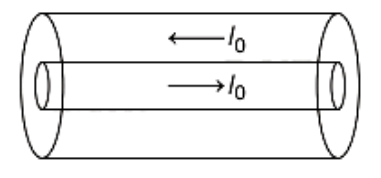
C. $\frac{x^2}{b^2-a^2}$

D. $\frac{b^2-a^2}{x^2}$

Answer: A

Solution:

Given, radius of inner wire of co-axial cable $= a$
Inner radius of outer shell $= b$
Outer radius of outer shell $= c$
Let, magnetic field at distance $x < a = B_1$



and magnetic field at distance
$a < x < b = B_2$
By using Ampere circuital law,
$\oint B \cdot dl = n\mu_0 I_{enc}$
where, $dl =$ perimeter of small circular element,
$n =$ number of turns,
$\mu_0 =$ free space permeability
and $I =$ current.
$\therefore \; B_1 \cdot 2\pi x = n\mu_0 I_{enc}$

$$\Rightarrow \; B_1 = \frac{\mu_0}{2\pi x} \frac{I_0}{\pi a^2} \cdot \pi x^2$$

$$= \frac{\mu_0}{2\pi} \frac{I_0 x}{a^2}$$

Now, for $B_2 (a < x < b)$

$B_2 \cdot 2\pi x = \mu_0 I_0$

$B_2 = \frac{1}{2\pi}\frac{\mu_0 I_0}{x}$

∴ From Eqs. (i) and (ii) we will get

$\frac{B_1}{B_2} = \frac{\frac{\mu_0 I_0 x}{2\pi a^2}}{\frac{\mu_0 I_0}{2\pi x}} = \frac{x^2}{a^2}$

Question118

The fractional change in the magnetic field intensity at a distance r from centre on the axis of current carrying coil of radius a to the magnetic field intensity at the centre of the same coil is (Take, r < a) [26 Aug 2021 Shift 1]

Options:

A. $\frac{3}{2}\frac{a^2}{r^2}$

B. $\frac{2}{3}\frac{a^2}{r^2}$

C. $\frac{2}{3}\frac{r^2}{a^2}$

D. $\frac{3}{2}\frac{r^2}{a^2}$

Answer: D

Solution:

The magnetic field intensity at a distance r from centre on the axis of a coil of radius a carrying current i is given by

$B_{axis} = \frac{\mu_0}{4\pi}\frac{2\pi i a^2}{(a^2+r^2)^{\frac{3}{2}}}$ ...(i)

The magnetic field intensity at the centre of the coil is given by

$B_{centre} = \frac{\mu_0}{4\pi}\frac{2\pi i}{a}$ ...(ii)

According to question, the fractional change in the magnetic fieldintensity is given by

$\frac{\Delta B}{B} = 1 - \frac{B_{axis}}{B_{centre}}$ ...(iii)

Substituting the values of B_{axis} and B_{centre} from Eqs. (i) and (ii) respectively in Eq. (iii), we get

$$\frac{\Delta B}{B} = 1 - \frac{\frac{\mu_0}{4\pi}\frac{2\pi i a^2}{(a^2+r^2)^{\frac{3}{2}}}}{\frac{\mu_0}{4\pi}\frac{2\pi i}{a}}$$

$$= 1 - \frac{a^3}{(a^2+r^2)^{\frac{3}{2}}}$$

$$= 1 - \frac{a^3}{a^3\left(1+\frac{r^2}{a^2}\right)^{\frac{3}{2}}}$$

$$= 1 - \left(1+\frac{r^2}{a^2}\right)^{-\frac{3}{2}}$$

Given that, r < a, applying Binomial expansion,

$(1-x)^{-n} = 1 - x + \frac{2x^2}{2!} - \frac{3\times 2x^3}{3!} \ldots (-1)^n x^n,$

$= (1+nx)$

we get $\frac{\Delta B}{B} = 1 - \left(1 - \frac{3}{2}\frac{r^2}{a^2}\right)$

$\frac{\Delta B}{B} = 1 - 1 + \frac{3}{2}\frac{r^2}{a^2}$

$\frac{\Delta B}{B} = \frac{3}{2}\frac{r^2}{a^2}$, which is the required expression.

Question119

Two ions of masses 4 amu and 16 amu have charges +2e and +3e, respectively. These ions pass through the region of constant perpendicular magnetic field. The kinetic energy of both ions is same. Then,
[27 Aug 2021 Shift 1]

Options:

A. lighter ion will be deflected less than heavier ion

B. lighter ion will be deflected more than heavier ion

C. both ions will be deflected equally

D. no ion will be deflected

Answer: B

Solution:

Given, mass of 1 st ion, $m_1 = 4\,amu$

Mass of 2 nd ion, $m_2 = 16\,amu$

Charge on 1 st ion, $q_1 = +2e$

Charge on 2 nd ion, $q_2 = +3e$

Kinetic energy of both ion is same,

$KE_1 = KE_2$

The radius of path traced by a charged particle in magnetic field is given as

$$r = \frac{\sqrt{2\,mKE}}{qB}$$

$$\Rightarrow r \propto \frac{\sqrt{m}}{q}$$

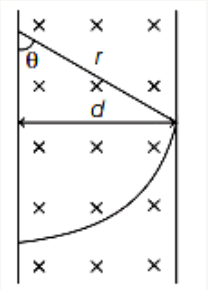
$$\Rightarrow \frac{r_1}{r_2} = \sqrt{\frac{m_1}{m_2}} \times \left(\frac{q_2}{q_1}\right)$$

$$\Rightarrow \frac{r_1}{r_2} = \sqrt{\frac{4}{16}} \times \left(\frac{+3e}{2e}\right)$$

$$= \frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$$

$$\Rightarrow r_1 = \frac{3}{4} r_2$$

$$\Rightarrow r_1 < r_2$$



Now, the deflection of ion in region with constant magnetic field as shown in figure is denoted by θ.

From diagram,

$$\sin\theta = \frac{d}{r}$$

$\theta \propto \frac{1}{r}$ [∵ d = constant]

As, $r_1 < r_2$ from above expression $\theta'_1 > \theta_2$.

Thus, deflection of lighter ion will be more than the deflection of heavier ion.

Question120

If the maximum value of accelerating potential provided by a radio frequency oscillator is 12 kV . The number of revolution made by a proton in a cyclotron to achieve one sixth of the speed of light is..... .

[Given, $m_p = 1.67 \times 10^{-27}$ kg, $e = 1.6 \times 10^{-19}$C, $c = 3 \times 10^8$m/s]

[26 Aug 2021 Shift 2]

Answer: 543

Solution:

Given, maximum value of accelerating potential,
V = 12 kV = 12000 V,

Speed achieved by a proton cyclotron, $v = \frac{c}{6}$

where, c is the speed of light $= 3 \times 10^8 ms^{-1}$

Mass of proton, $m_p = 1.67 \times 10^{-27} kg$

Charge of proton, $q_p = 1.6 \times 10^{-19} C$

According to the question, assume a proton makes n revolutions in the cyclotron to achieve the speed v.

The energy absorbed by proton due to the accelerating potential provided by the radio oscillator $= n \times 2 \times q_p \times V$

This energy absorbed by proton will be equal to the kinetic energy gain by the proton i.e.,

$2nq_pV = \frac{m_p v^2}{2}$

Substituting the given values in above expression, we get

$n \times 2 \times 1.6 \times 10^{-19} \times 12000$

$= \frac{1}{2} \times 1.67 \times 10^{-27} \times \left(\frac{3 \times 10^8}{6}\right)^2$

$n \times 3.84 \times 10^{-15} = 2.0875 \times 10^{-12}$

n = 543.62

As, n is number of revolutions, so it should be an integer.

∴ n = 543

Question121

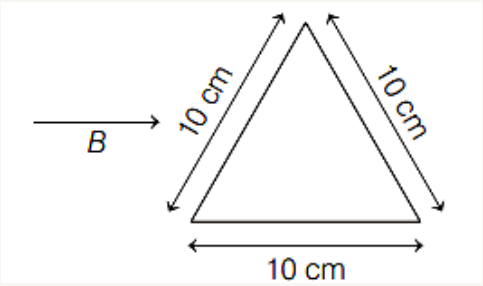
A coil in the shape of an equilateral triangle of s ide 10 cm lies in a vertical plane between the pole pieces of permanent magnet

producing a horizontal magnetic field 20 mT. The torque acting on the coil when a current of 0.2A is passed through it and its plane becomes parallel to the magnetic field will be $\sqrt{x} \times 10^{-5}$ Nm. The value of x is..... .
[26 Aug 2021 Shift 2]

Answer: 3

Solution:

Side of equilateral triangular coil, a = 10 cm = 0.1 m
Horizontal magnetic field, $B = 20\,mT = 20 \times 10^{-3} T$
Current in coil, i = 0.2 A
Torque acting on coil, $\tau = \sqrt{x} \times 10^{-5} N - m$
Torque acting on coil can be calculated as
τ = MBsin θ



Here, M is pole strength and B is magnetic field. Magnetic pole strength, M = iA

$\tau = i\left(\frac{\sqrt{3}}{4}a^2\right) B \sin\theta$ [$\because A = \frac{\sqrt{3}}{4}a^2$. (for equilateral triangle)]

$= 0.2 \times \frac{\sqrt{3}}{4} \times (0.1)^2 \times 20 \times 10^{-3} \times \sin 90°$ [$\because A \perp B$]

$= \sqrt{3} \times 10^{-5} N - m$

After comparing obtained value with the given value, x = 3.

Question122

A uniform conducting wire of length is 24a, and resistance R is wound up as a current carrying coil in the shape of an equilateral triangle of side a and then in the form of a square of side a. The coil is connected to a voltage source V_0. The ratio of magnetic moment of the coils in case of equilateral triangle to that for square is 1 : $\sqrt{y}$.

The value of where y is ..... .
[27 Aug 2021 Shift 1]

Answer: 3

Solution:

Length of conducting wire, l = 24 a
Resistance of wire is R.
Ratio of magnetic moment of coil for equilateral triangle to square is $1:\sqrt{y}$.
Side of equilateral triangle is a and also side of square is a.
Number of turns in equilateral triangle will be

$n_T = \frac{24a}{3a} = 8$ [$\because$ Length of one turn is 3a]

Number of turns in square will be

$n_S = \frac{24a}{4a} = 6$ [$\because$ Length of one turn is 4a]

Magnetic moment of coil is given as
M = nIA
The ratio of magnetic moment for triangular coil to square coil is given as

$\frac{M_T}{M_S} = \frac{n_T I A_T}{n_S I A_S}$ ($\because$ I is same in both coils)

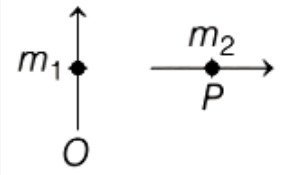
Substituting the values in above expression, we get

$$\frac{1}{\sqrt{y}} = \frac{\left(8 \times \frac{\sqrt{3}}{4}a^2\right)}{6 \times a^2} = \frac{8\sqrt{3}}{24} = \frac{1}{\sqrt{3}}$$

$\Rightarrow \sqrt{y} = \sqrt{3}$
$\Rightarrow$ y = 3
Hence, the value of y is 3.

Question123

Two short magnetic dipoles m_1 and m_2 each having magnetic moment of 1 Am^2 are placed at point O and P, respectively. The distance between OP is 1 m. The torque experienced by the magnetic dipole m_2 due to the presence of $m_1$is ..... $\times 10^{-7}$ Nm.

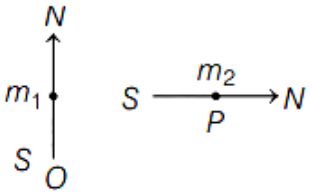


[26 Aug 2021 Shift 1]

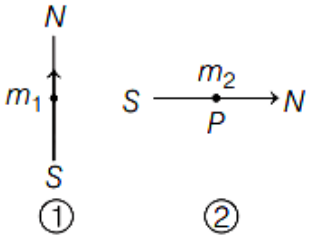
Answer: 1

Solution:

In the given figure,



The direction of magnetic moment is from South to North.



When second magnet is kept in the magnetic field produced by first magnet, the torque acting on second magnet is given by

$\tau = M_2 \times B_1$

$|\tau| = M_2 B_1 \sin\theta$ ($\therefore \theta = 90°$)

$|\tau| = M_2 B_1$

Point P lie on equatorial position, so

$$B_1 = \frac{\mu_0}{4\pi}\frac{M_1}{r^3}$$

$$|\tau| = \left(\frac{\mu_0}{4\pi}\frac{M_1}{r^3}\right)M_2$$

$M_1 = M_2 = 1A - m^2$, $r = 1m$ (given)

$$|\tau| = \frac{10^{-7} \times 1 \times 1}{(1)^3}$$

$\Rightarrow |\tau| = 10^{-7}\,Nm$

$\Rightarrow |\tau| = 1 \times 10^{-7}\,Nm$

Question124

Consider a galvanometer shunted with 5Ω resistance and 2% of current passes through it. What is the resistance of the given galvanometer ?
[31 Aug 2021 Shift 1]

Options:

A. 300Ω

B. 344Ω

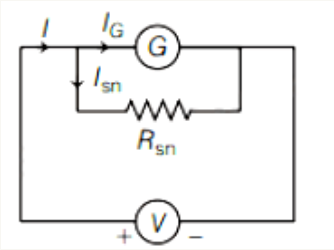
C. 245Ω

D. 226Ω

Answer: C

Solution:

Given,
Shunt resistance, $R_{sn} = 5\Omega$
Let galvanometer resistance = G
Total, current = I
Voltage across galvanometer $(V_G) =$ Voltage across shunt resistance R_{sn}



$I_G \times G = I_{sn} \times R_{sn}$
$0.021 \times G = 0.981 \times 5$
$\Rightarrow G = \frac{98 \times 5}{2} = 49 \times 5 = 245\Omega$

Question125

For full scale deflection of total 50 divisions, 50 mV voltage is required in galvanometer.
The resistance of galvanometer if its current sensitivity is 2 div/mA

will be
[27 Aug 2021 Shift 2]

Options:

A. 1Ω

B. 5Ω

C. 4Ω

D. 2Ω

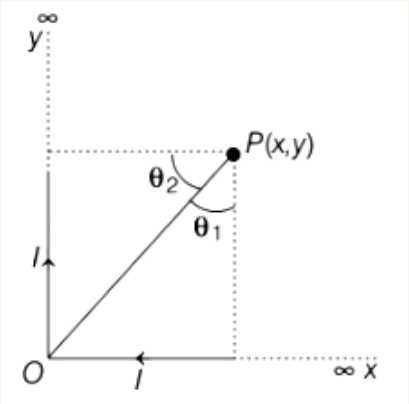
Answer: D

Solution:

Given, number of divisions, $n = 50$
Voltage, $V = 50mV = 50 \times 10^{-3}V$
Current sensitivity, $k = \frac{2d\,iv}{mA} = \frac{2d\,iv}{10^{-3}A}$
$\therefore$ Current, $I = \frac{n}{k} = \frac{50 \times 10^{-3}}{2} = 25 \times 10^{-3}A$
By using Ohm's law,
Resistance $(R) = \frac{Potential\ (V)}{Current\ (I)}$
$= \frac{50 \times 10^{-3}}{25 \times 10^{-3}} = 2\Omega$

Question126

There are two infinitely long straight current carrying conductors and they are held at right angles to each other so that their common ends meet at the origin as shown in the figure given below. The ratio of current in both conductors is 1 : 1. The magnetic field at point P is



[1 Sep 2021 Shift 2]

Options:

A. $\frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2}+(x+y)\right]$

B. $\frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2}-(x+y)\right]$

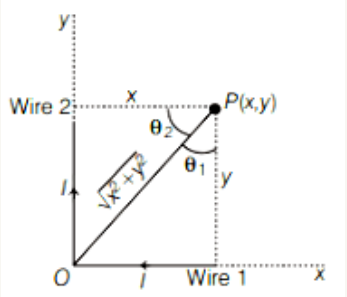
C. $\frac{\mu_0 Ixy}{4\pi}\left[\sqrt{x^2+y^2}-(x+y)\right]$

D. $\frac{\mu_0 Ixy}{4\pi}\left[\sqrt{x^2+y^2}+(x+y)\right]$

Answer: A

Solution:

The diagram is given below,



The magnetic field due to the wire 1,

$B_{\text{wire 1}} = \frac{\mu_0 I}{4\pi y}(\sin 90° + \sin\theta_1)$

$\frac{\mu_0 I}{4\pi y}\left(1+\frac{x}{\sqrt{x^2+y^2}}\right)$

The magnetic field due to the wire 2,

$\frac{\mu_0 I}{4\pi y}(\sin 90° + \sin\theta_2)$

$$B_{wire\,2} = \frac{\mu_0 I}{4\pi x}\left(1 + \frac{y}{\sqrt{x^2+y^2}}\right)$$

Thus, the total magnetic field,

$$B = B_1 + B_2$$

$$= \frac{\mu_0 I}{4\pi}\left(\frac{1}{y} + \frac{x}{y\sqrt{x^2+y^2}} + \frac{1}{x} + \frac{y}{x\sqrt{x^2+y^2}}\right)$$

$$= \frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2} + (x+y)\right]$$

Question127

Proton with kinetic energy of 1M eV moves from south to north. It gets an acceleration of $10^{12} m/s^2$ by an applied magnetic field (west to east). The value of magnetic field:
(Rest mass of proton is 1.6×10^{-27}kg)
[8 Jan 2020, I]

Options:

A. 0.71mT

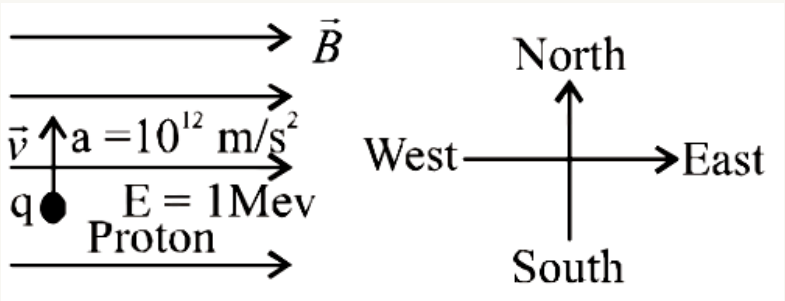
B. 7.1mT

C. 0.071mT

D. 71mT

Answer: A

Solution:



As we know, magnetic force $F = qvB = ma$

$\therefore \vec{a} = \left(\frac{qvB}{m}\right)$ perpendicular to velocity.

$\therefore$ Also $v = \sqrt{\frac{2KE}{m}} = \sqrt{\frac{2 \times e \times 10^6}{m}}$

$\therefore a = \frac{qvB}{m} = \frac{eB}{m}\sqrt{\frac{2 \times e \times 10^6}{m}}$

$\therefore 10^{12} = \left(\frac{1.6 \times 10^{-19}}{1.67 \times 10^{-27}}\right)^{\frac{3}{2}} . \sqrt{2} \times 10^3 B$

$\therefore B \simeq \frac{1}{\sqrt{2}} \times 10^{-3} T = 0.71 mT$ (approx)

Question128

A long, straight wire of radius a carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to the wire at distance $\frac{a}{3}$ and 2a respectively from the axis of the wire is:
[9 Jan 2020, I]

Options:

A. $\frac{2}{3}$

B. 2

C. $\frac{1}{2}$

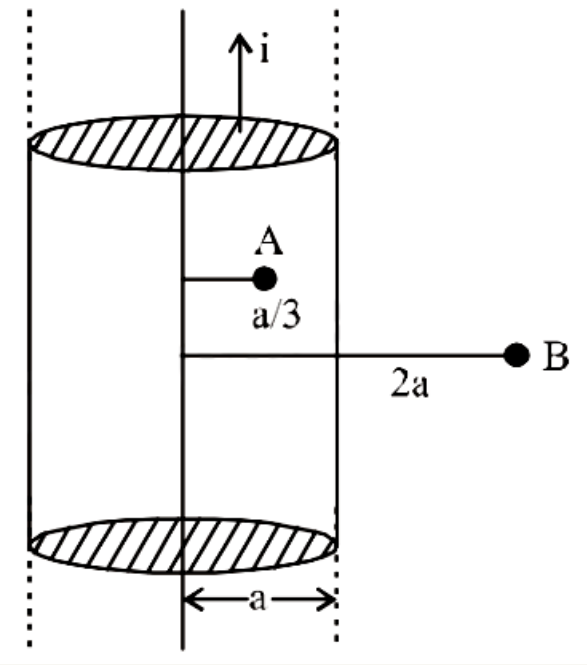
D. $\frac{3}{2}$

Answer: A

Solution:

Let a be the radius of the wire
Magnetic field at point A (inside)

$$B_A = \frac{\mu_0 ir}{2\pi a^2} = \frac{\mu_0 i\frac{a}{3}}{2\pi a^2} = \frac{\mu_0 i}{\pi a^2}\frac{a}{6} = \frac{\mu_0 i}{6\pi a}$$



Magnetic field at point B (outside)

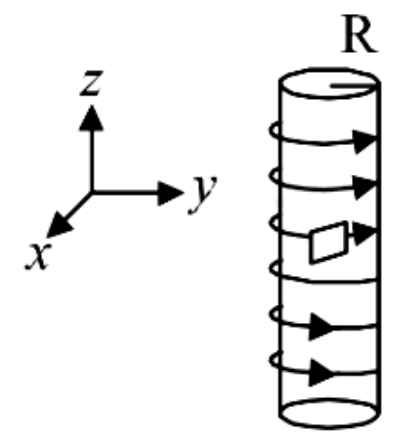
$$B_B = \frac{\mu_0 i}{2\pi(2a)}$$

$$\therefore \frac{B_A}{B_B} = \frac{\frac{\mu_0 i}{6\pi a}}{\frac{\mu_0 i}{2\pi(2a)}} = \frac{4}{6} = \frac{2}{3}$$

Question129

An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns/length and carries a current I. The electron gun shoots an electron along the radius of the solenoid with speed v. If the electron does not hit the surface of the solenoid, maximum possible value of v is (all symbols have their standard

meaning):



[9 Jan 2020, II]

Options:

A. $\frac{e\mu_0 nI\,R}{m}$

B. $\frac{e\mu_0 nI\,R}{2m}$

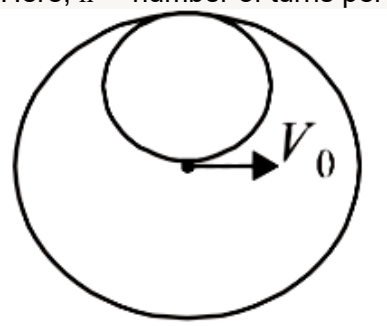
C. $\frac{e\mu_0 nI\,R}{4m}$

D. $\frac{2e\mu_0 nI\,R}{m}$

Answer: B

Solution:

Magnetic field inside the solenoid is given by
$B = \mu_0 nI$ .....(i)
Here, $n =$ number of turns per unit length



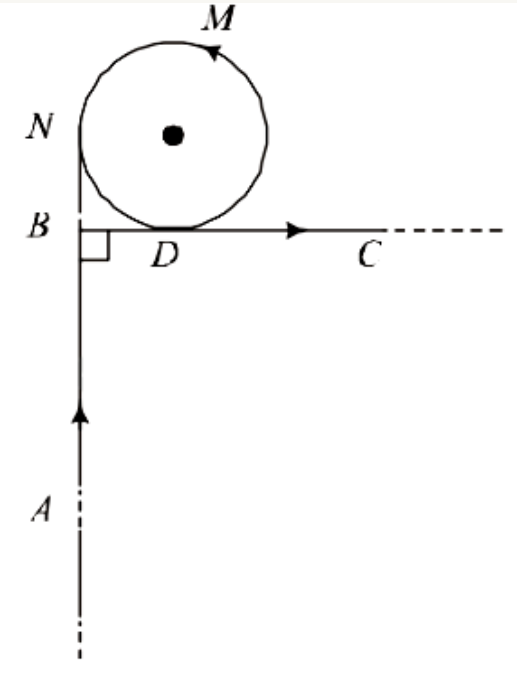
The path of charge particle is circular. The maximum possible radius of electron $= \frac{R}{2}$

$\therefore \frac{mV_{max}}{qB} = \frac{R}{2}$

$\Rightarrow V_{max} = \frac{qBR}{2m} = \frac{eR\mu_0 nI}{2m}$ (using (i))

Question130

A very long wire ABDMNDC is shown in figure carrying current I. AB and BC parts are straight, long and at right angle. At D wire forms a circular turn DMND of radius R.
AB, BC parts are tangential to circular turn at N and D. Magnetic field at the centre of circle is:



[8 Jan 2020, II]

Options:

A. $\frac{\mu_0 I}{2\pi R}\left(\pi + \frac{1}{\sqrt{2}}\right)$

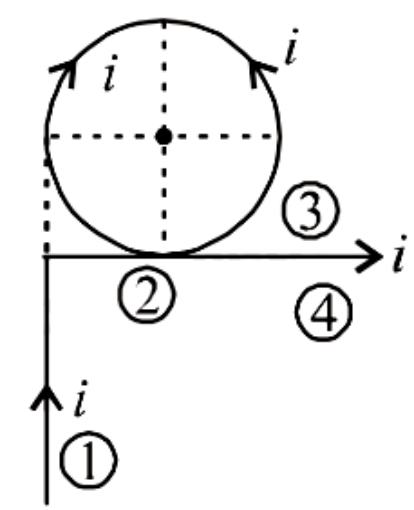
B. $\frac{\mu_0 I}{2\pi R}\left(\pi - \frac{1}{\sqrt{2}}\right)$

C. $\frac{\mu_0 I}{2\pi R}(\pi + 1)$

D. $\frac{\mu_0 I}{2R}$

Answer: A

Solution:



$B_0 = B_1 + B_2 + B_3 + B_4$

$= \frac{\mu_0 I}{4\pi R}[\sin 90° - \sin 45°] + \frac{\mu_0 I}{2R} + \frac{\mu_0 I}{4\pi R}$

$[\sin 45° + \sin 90°]$

$= -\frac{\mu_0 I}{4\pi R}\left(1 - \frac{1}{\sqrt{2}}\right) + \frac{\mu_0 I}{2R} + \frac{\mu_0 I}{4\pi R}\left(1 + \frac{1}{\sqrt{2}}\right)$

$\vec{B}_0 \odot = \frac{\mu_0 I}{2\pi R}\left(\pi + \frac{1}{\sqrt{2}}\right) \odot$

Question131

A small circular loop of conducting wire has radius a and carries current I. It is placed in a uniform magnetic field B perpendicular to its plane such that when rotated slightly about its diameter and released, it starts performing simple harmonic motion of time period T. If the mass of the loop is m then :
[9 Jan 2020, II]

Options:

A. $T = \sqrt{\frac{2m}{IB}}$

B. $T = \sqrt{\frac{\pi m}{2IB}}$

C. $T = \sqrt{\frac{2\pi m}{I B}}$

D. $T = \sqrt{\frac{\pi m}{I B}}$

Answer: C

Solution:

Torque on circular loop, $\tau = M B \sin\theta$
where, M = magnetic moment
B = magnetic field
Now, using $\tau = I\alpha$
$\therefore \tau = M B \sin\theta = I\alpha$

$\Rightarrow \pi R^2 I B\theta = \frac{mR^2\alpha}{2}$

$\left(\because m = I A \text{ and moment of inertia of circular loop, } I = \frac{mR^2}{2}\right)$

$\Rightarrow \pi R^2 I B\theta = \frac{mR^2}{2}\omega\theta$

$\Rightarrow \omega = \sqrt{\frac{2\pi I B}{m}} \Rightarrow \frac{2\pi}{T} = \sqrt{\frac{2\pi I B}{m}}$

$\Rightarrow T = \sqrt{\frac{2\pi m}{I B}}$

Question132

An electron is moving along +x direction with a velocity of 6 times 10^6 ms^{-1}. It enters a region of uniform electric field of 300V / cm pointing along +y direction. The magnitude and direction of the magnetic field set up in this region such that the electron keeps moving along the x direction will be:
[Sep. 06, 2020 (I)]

Options:

A. 3×10^{-4}T , along +z direction

B. 5×10^{-3}T , along −z direction

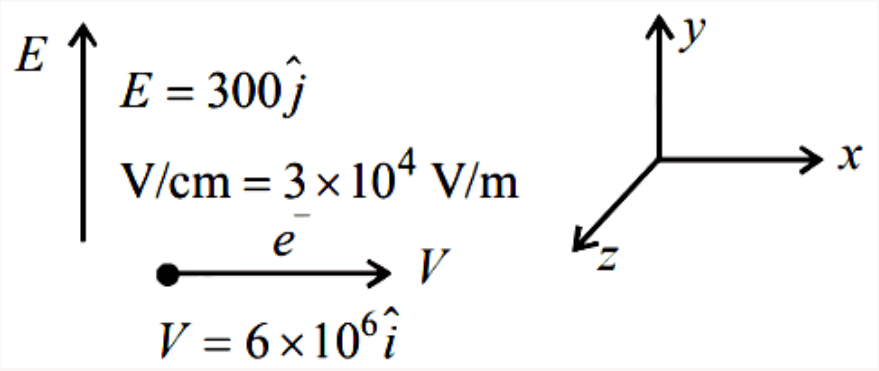
C. 5×10^{-3}T , along +z direction

D. 3×10^{-4}T , along $-z$ direction

Answer: C

Solution:

$\vec{E} = 300\hat{j}\,V/cm = 3 \times 10^4 V/m$
$\vec{V} = 6 \times 10^6 \hat{i}$



$\vec{B}$ must be in +z axis.
$q\vec{E} + q\vec{V} \times \vec{B} = 0$
$E = VB$
$\therefore B = \frac{E}{V} = \frac{3 \times 10^4}{6 \times 10^6} = 5 \times 10^{-3}$T

Question133

A particle of charge q and mass m is moving with a velocity $-v\hat{i}(v \neq 0)$ towards a large screen placed in the Y-Z plane at a distance d . If there is a magnetic field $\vec{B} = B_0\hat{k}$, the minimum value of v for which the particle will not hit the screen is:
[Sep. 06, 2020 (I)]

Options:

A. $\frac{qdB_0}{3m}$

B. $\frac{2qdB_0}{m}$

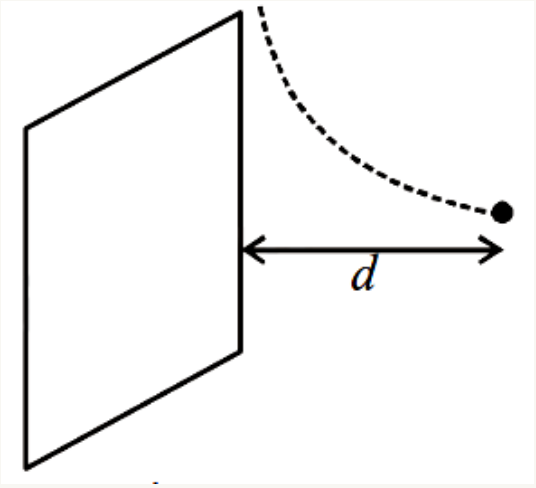
C. $\frac{qdB_0}{m}$

D. $\frac{qd\,B_0}{2m}$

Answer: C

Solution:

In uniform magnetic field particle moves in a circular path, if the radius of the circular path is 'r', particle will not hit the screen.



$r = \frac{mv}{qB_0}\left[\because \frac{mv^2}{r} = qvB_0\right]$

Hence, minimum value of v for which the particle will not hit the screen.

$v = \frac{qB_0d}{m}$

Question134

A charged particle carrying charge 1μC is moving with velocity $\left(2\hat{i}+3\hat{j}+4\hat{k}\right)ms^{-1}$. If an external magnetic field of $\left(5\hat{i}+3\hat{j}-6\hat{k}\right)\times10^{-3}T$ exists in the region where the particle is moving then the force on the particle is $\vec{F}\times10^{-9}N$. The vector $\vec{F}$ is: [Sep. 03, 2020 (I)]

Options:

A. $-0.30\hat{i}+0.32\hat{j}-0.09\hat{k}$

B. $-30\hat{i}+32\hat{j}-9\hat{k}$

C. $-300\hat{i}+320\hat{j}-90\hat{k}$

D. $-3.0\hat{i}+3.2\hat{j}-0.9\hat{k}$

Answer: A

Solution:

[Given: $q=1\mu C=1\times10^{-6}C$
$\vec{V}=\left(2\hat{i}+3\hat{j}+4\hat{k}\right)m/s$ and
$\vec{B}=\left(5\hat{i}+3\hat{j}-6\hat{k}\right)\times10^{-3}T$]

$$\vec{F}=q\left(\vec{V}\times\vec{B}\right)=10^{-6}\times10^{-3}\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&4\\5&3&-6\end{vmatrix}$$

$$=\left(-30\hat{i}+32\hat{j}-9\hat{k}\right)\times10^{-9}N$$
$$\therefore\vec{F}=\left(-30\hat{i}+32\hat{j}-9\hat{k}\right)$$

Question135

A beam of protons with speed $4\times10^{5}ms^{-1}$ enters a uniform magnetic field of 0.3T at an angle of 60° to the magnetic field. The pitch of the resulting helical path of protons is close to : (Mass of the proton $=1.67\times10^{-27}$kg, charge of the proton $=1.69\times10^{-19}C$)
[Sep. 02, 2020 (I)]

Options:

A. 2cm

B. 5cm

C. 12cm

D. 4cm

Answer: D

Solution:

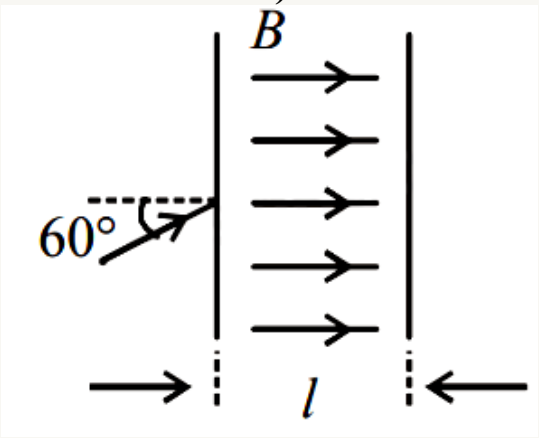
Pitch $= (v\cos\theta)T$ and $T = \frac{2\pi m}{qB}$

$\therefore$ Pitch $= (V\cos\theta)\frac{2\pi m}{qB}$

$= (4\times10^5\cos60°)\frac{2\pi}{0.3}\left(\frac{1.67\times10^{-27}}{1.69\times10^{-19}}\right) = 4\text{cm}$

Question136

The figure shows a region of length ' l ' with a uniform magnetic field of 0.3T in it and a proton entering the region with velocity $4\times10^5 ms^{-1}$ making an angle 60° with the field. If the proton completes 10 revolution by the time it cross the region shown, ' l ' is close to (mass of proton $= 1.67\times10^{-27}$kg, charge of the proton $= 1.6\times10^{-19}$C)



[Sep. 02, 2020 (II)]

Options:

A. 0.11m

B. 0.88m

C. 0.44m

D. 0.22m

Answer: C

Solution:

Time period of one revolution of proton, $T = \frac{2\pi m}{qB}$

Here, m = mass of proton
q = charge of proton
B = magnetic field.
Linear distance travelled in one revolution,
$p = T(v\cos\theta)$(Here ,v = velocity of proton)
$\therefore$ Length of region, $l = 10 \times (v\cos\theta)T$

$\Rightarrow l = 10 \times v\cos 60° \times \frac{2\pi m}{qB}$

$\Rightarrow l = \frac{20\pi mv}{qB} = \frac{20 \times 3.14 \times 1.67 \times 10^{-27} \times 4 \times 10^5}{1.6 \times 10^{-19} \times 0.3}$

$\Rightarrow l = 0.44m$

Question137

A charged particle going around in a circle can be considered to be a current loop. A particle of mass m carrying charge q is moving in a plane wit speed v under the influence of magnetic field $\vec{B}$. The magnetic moment of this moving particle :
[Sep. 06, 2020 (II)]

Options:

A. $\frac{mv^2\vec{B}}{2B^2}$

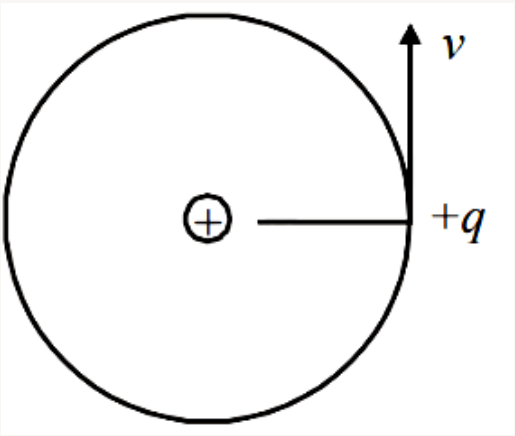
B. $-\frac{mv^2\vec{B}}{2\pi B^2}$

C. $-\frac{mv^2\vec{B}}{B^2}$

D. $-\frac{mv^2\vec{B}}{2B^2}$

Answer: D

Solution:



Length of the circular path, $l = 2\pi r$

Current, $i = \frac{q}{T} = \frac{qv}{2\pi r}$

Magnetic moment M = Current $\times$ Area

$= i \times \pi r^2 = \frac{qv}{2\pi r} \times \pi r^2$

$M = \frac{1}{2} q \,.\, v \,.\, r$

Radius of circular path in magnetic field, $r = \frac{mv}{qB}$

$\therefore M = \frac{1}{2} qv \times \frac{mv}{qB} \Rightarrow M = \frac{mv^2}{2B}$

Direction of $\vec{M}$ is opposite of $\vec{B}$ therefore

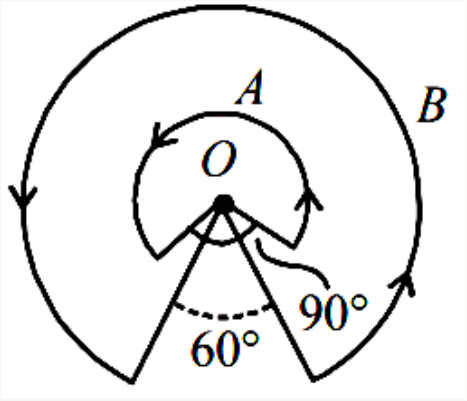
$\vec{M} = \frac{-mv^2 \vec{B}}{2B^2}$

(By multiplying both numerator and denominator by B).

Question138

A wire A, bent in the shape of an arc of a circle, carrying a current of 2 A and having radius 2 cm and another wire B, also bent in the shape of arc of a circle, carrying a current of 3 A and having radius of 4 cm, are placed as shown in the figure. The ratio of the magnetic

fields due to the wires A and B at the common centre O is :



[Sep. 04, 2020 (I)]

Options:

A. 4 : 6

B. 6 : 4

C. 2 : 5

D. 6 : 5

Answer: D

Solution:

Given: $I_A = 2A, R_A = 2cm, \theta_A = 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$

$I_B = 3A, R_B = 4cm, \theta_B = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

Using, magnetic field, $B = \frac{\mu_0 I \theta}{4\pi R}$

$$\frac{B_A}{B_B} = \frac{I_A}{I_B} \times \frac{\theta_A R_B}{\theta_B R_A} = \frac{2 \times \frac{3\pi}{2} \times 4}{3 \times \frac{5\pi}{3} \times 2} = \frac{6}{5}$$

Question139

Magnitude of magnetic field (in SI units) at the centre of a hexagonal shape coil of side 10cm, 50 turns and carrying current I (Ampere) in units of $\frac{\mu_0 I}{\pi}$ is :
[Sep. 03,2020 (I)]

Options:

A. $250\sqrt{3}$

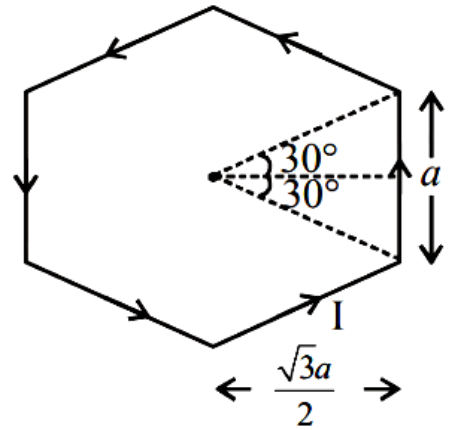
B. $50\sqrt{3}$

C. $500\sqrt{3}$

D. $5\sqrt{3}$

Answer: C

Solution:



Magnetic field due to one side of hexagon $B = \frac{\mu_0 I}{4\pi \frac{\sqrt{3}a}{2}}(\sin 30° + \sin 30°)$

$$\Rightarrow B = \frac{\mu_0 I}{2\sqrt{3}a}\left(\frac{1}{2}+\frac{1}{2}\right) = \frac{\mu_0 I}{2\sqrt{3}a\pi}$$

Now, magnetic field due to one hexagon coil

$$B = 6 \times \frac{\mu_0 I}{2\sqrt{3}a\pi}$$

Again magnetic field at the centre of hexagonal shape coil of 50 turns,

$$B = 50 \times 6 \times \frac{\mu_0 I}{2\sqrt{3}a\pi} \left[\because a = \frac{10}{100} = 0.1\text{m}\right]$$

or, $B = \frac{150\mu_0 I}{\sqrt{3} \times 0.1 \times \pi} = 500\sqrt{3}\frac{\mu_0 I}{\pi}$

Question140

A square loop of side 2a and carrying current I is kept is xz plane with its centre at origin. A long wire carrying the same current I is placed parallel to z-axis and passing through point (0, b, 0), (b > > a). The magnitude of torque on the loop about z-axis will be : [Sep. 06, 2020 (II)]

Options:

A. $\frac{2\mu_0 I^2 a^2}{\pi b}$

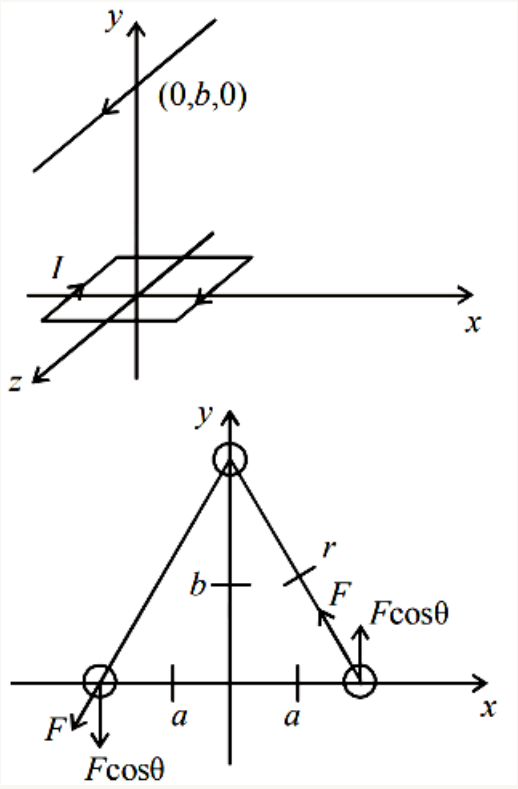
B. $\frac{2\mu_0 I^2 a^2 b}{\pi(a^2+b^2)}$

C. $\frac{\mu_0 I^2 a^2 b}{2\pi(a^2+b^2)}$

D. $\frac{\mu_0 I^2 a^2}{2\pi b}$

Answer: B

Solution:



$r = \sqrt{b^2 + a^2}$

Force, $F = BI\,2a = \frac{\mu_0 I}{2\pi r} I \times 2a$

Force, $F = \frac{\mu_0 I^2 a}{\pi\sqrt{b^2 + a^2}}$

Torque, $\tau = F_1 \times$ Perpendicular distance $= F\cos\theta \times 2a$

$= \frac{\mu_0 I^2 a}{\pi\sqrt{b^2 + a^2}} \times \frac{b}{\sqrt{b^2 + a^2}} \times 2a$

$\Rightarrow \tau = \frac{2\mu_0 I^2 a^2 b}{\pi(a^2 + b^2)}$

If $b >> a$ then $\tau = \frac{2\mu_0 I^2 a^2}{\pi b}$

Question141

A square loop of side 2a, and carrying current I , is kept in XZ plane with its centre at origin. A long wire carrying the same current I is placed parallel to the z -axis and passing through the point (0, b, 0), (b>> a). The magnitude of the torque on the loop about z -

axis is given by :
[Sep. 05,2020 (I)]

Options:

A. $\frac{\mu_0 I^2 a^2}{2\pi b}$

B. $\frac{\mu_0 I^2 a^3}{2\pi b^2}$

C. $\frac{2\mu_0 I^2 a^2}{\pi b}$

D. $\frac{2\mu_0 I^2 a^3}{\pi b^2}$

Answer: C

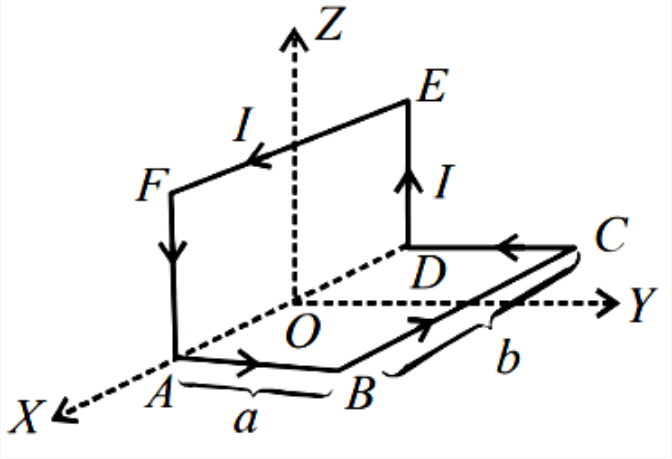
Solution:

Torque on the loop,
$\vec{\tau} = \vec{M} \times \vec{B} = M B \sin\theta = M B \sin 90°$
Magnetic field, $B = \frac{\mu_0 I}{2\pi d}$
$\therefore \tau = I_1 (2a)^2 \left(\frac{\mu_0 I_2}{2\pi d}\right) \sin 90°$
$= \frac{2\mu_0 I_1 I_2}{\pi d} \times a^2 = \frac{2\mu_0 I^2 a^2}{\pi d}$

Question142

A wire carrying current I is bent in the shape ABCDEFA as shown, where rectangle ABCDA and ADEFA are perpendicular to each other. If the sides of the rectangles are of lengths a and b, then the magnitude and direction of magnetic moment of the loop ABCDEFA

is :



[Sep. 02, 2020 (II)]

Options:

A. abI , along $\left(\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}}\right)$

B. $\sqrt{2}$abI , along $\left(\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}}\right)$

C. $\sqrt{2}$abI , along $\left(\frac{\hat{j}}{\sqrt{5}}+\frac{2\hat{k}}{\sqrt{5}}\right)$

D. abI , along $\left(\frac{\hat{j}}{\sqrt{5}}+\frac{2\hat{k}}{\sqrt{5}}\right)$

Answer: B

Solution:

Magnetic moment of loop ABCD,
$M_1 =$ area of loop $\times$ current
$\vec{M}_1 = (abI)\left(\hat{j}\right)$ (Here ,ab = area of rectangle)
Magnetic moment of loop DE F A,
$\vec{M}_2 = (abI)\left(\hat{i}\right)$
Net magnetic moment,

$\vec{M} = \vec{M}_1 + \vec{M}_2 \Rightarrow \vec{M} = abI\left(\hat{i} + \hat{j}\right)$

$\Rightarrow \left|\vec{M}\right| = \sqrt{2}abI\left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}\right)$

Question143

A galvanometer of resistance G is converted into a voltmeter of ragne $0-1V$ by connecting a resistance R_1 in series with it. The additional resistance R_1 in series with it. The additional resistance that should be connected in series with R_1 to increase the range of the voltmeter to $0-2V$ will be:
[Sep. 05, 2020 (I)]

Options:

A. G

B. R_1

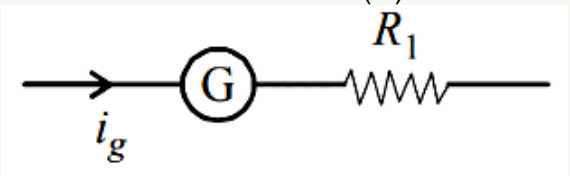
C. $R_1 - G$

D. $R_1 + G$

Answer: D

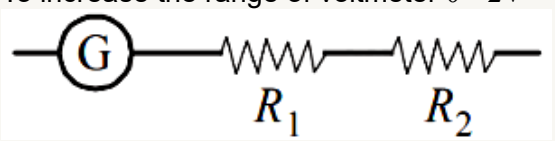
Solution:

Galvanometer of resistance (G) converted into a voltmeter of range $0-1V$



$V = 1 = i_g(G + R_1)$ .....(i)

To increase the range of voltmeter $0-2V$



$2 = i_g(R_1 + R_2 + G)$ .....(ii)

Dividing eq. (i) by (ii),

$$\Rightarrow \frac{1}{2} = \frac{G + R_1}{G + R_1 + R_2}$$
$$\Rightarrow G + R_1 + R_2 = 2G + 2R_1$$
$$\therefore R_2 = G + R_1$$

Question144

A galvanometer is used in laboratory for detecting the null point in electrical experiments. If, on passing a current of 6mA it produces a deflection of 2°, its figure of merit is close to :
[Sep. 05, 2020 (II)]

Options:

A. $333°A/div.$

B. $6 \times 10^{-3} A/div$

C. $666°A/div.$

D. $3 \times 10^{-3} A/div$

Answer: D

Solution:

Given,
Current passing through galvanometer, $I = 6mA$
Deflection, $\theta = 2°$
Figure of merit of galvanometer
$$= \frac{I}{\theta} = \frac{6 \times 10^{-3}}{2} = 3 \times 10^{-3} A/div$$

Question145

A galvanometer coil has 500 turns and each turn has an average area of $3 \times 10^{-4} m^2$. If a torque of 1.5N m is required to keep this coil parallel to a magnetic field when a current of 0.5A is flowing

through it, the strength of the field (in T) is _____ .
[NA Sep. 03, 2020 (II)]

Answer: 20

Solution:

Given.
Area of galvanometer coil, $A = 3 \times 10^{-4} m^2$
Number of turns in the coil, $N = 500$
Current in the coil. $I = 0.5A$
Torque $\tau = \left|\overrightarrow{M} \times \overrightarrow{B}\right| = N\,iAB\sin(90°) = NiAB$
$\Rightarrow B = \frac{\tau}{N\,iA} = \frac{1.5}{500 \times 0.5 \times 3 \times 10^{-4}} = 20T$

Question146

A proton and an α -particle (with their masses in the ratio of 1: 4 and charges in the ratio 1: 2) are accelerated from rest through a potential difference V . If a uniform magnetic field (B) is set up perpendicular to their velocities, the ratio of the radii $r_p : r_\alpha$ of the circular paths descrfibed by them will be:
[12 Jan 2019, I]

Options:

A. $1 : \sqrt{2}$

B. 1: 2

C. 1: 3

D. $1 : \sqrt{3}$

Answer: A

Solution:

Radius of the circular path will be $r = \frac{mv}{qB}$

$\Rightarrow r = \frac{\sqrt{2mK\,E}}{qB}$ $(\because p = mv = \sqrt{2mK\,E})$

$\because K\,E = q\Delta V$

$\therefore r = \frac{\sqrt{2mq\Delta V}}{qB} \Rightarrow r \propto \sqrt{\frac{m}{q}}$

$\therefore \frac{r_p}{r_\alpha} = \frac{1}{\sqrt{2}}$

Question147

In an experiment, electrons are accelerated, from rest, by applying a voltage of 500V . Calculate the radius of the path if a magnetic field 100mT is then applied.

[Charge of the electron $= 1.6 \times 10^{-19}$C

Mass of the electron $= 9.1 \times 10^{-31}$kg]

[11 Jan 2019, I]

Options:

A. 7.5×10^{-3}m

B. 7.5×10^{-2}m

C. 7.5m

D. 7.5×10^{-4}m

Answer: D

Solution:

Radius of the path (r) is given by $r = \frac{mv}{qB}$

$r = \frac{\sqrt{2mk}}{eB}(\because p = mv = \sqrt{2mk})$

$= \frac{\sqrt{2mV}}{eB}(\because k = eV)$

$$r = \frac{\sqrt{\frac{2m}{e}V}}{B} = \frac{\sqrt{\frac{2\times 9.1\times 10^{-31}}{1.6\times 10^{-19}}(500)}}{100\times 10^{-3}}$$

$$r = \frac{\sqrt{\frac{9.1}{0.16}\times 10^{-10}}}{10^{-1}} = \frac{3}{.4}\times 10^{-4}$$

$$= 7.5\times 10^{-4} m$$

Question148

The region between y = 0 and y = d contains a magnetic field $\vec{B} = B\hat{z}$. A particle of mass m and charge q enters the region with a velocity $\vec{v} = v\hat{i}$. if $d = \frac{mv}{2qB}$, the acceleration of the charged particle at the point of its emergence at the other side is :
[11 Jan 2019, II]

Options:

A. $\frac{qvB}{m}\left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right)$

B. $\frac{qvB}{m}\left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}\right)$

C. $\frac{qvB}{m}\left(\frac{-\hat{j}+\hat{i}}{\sqrt{2}}\right)$

D. (BONUS)

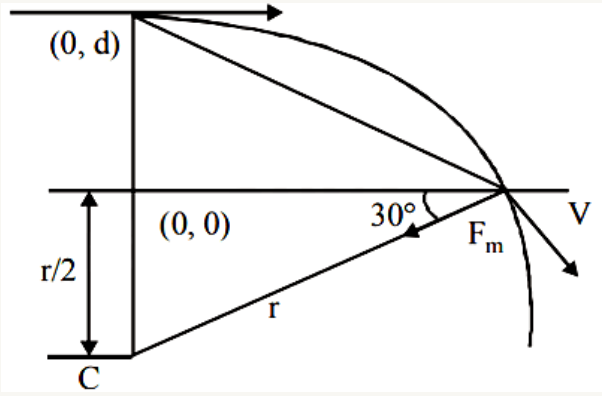
E.

Answer: D

Solution:

Assuming particle enters from $(0, d)$

$r = \frac{mv}{qB}, d = \frac{r}{2}$

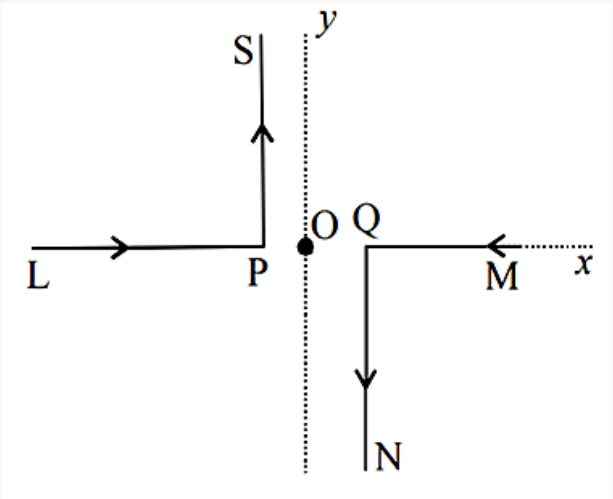


$a = \frac{qVB}{m}\left[\frac{-\sqrt{3}i - j}{2}\right]$

this option is not given in the all above four choices

Question149

As shown in the figure, two infinitely long, identical wires are bent by 90° and placed in such a way that the segments LP and QM are along the x -axis, while segments PS and QN are parallel to the y -axis. If OP = OQ = 4cm, and the magnitude of the magnetic field at O is 10^{-4}T , and the two wires carry equal currents (see figure), the magnitude of the current in each wire and the direction of the magnetic field at O will be ($\mu_0 = 4\pi \times 10^{-7}$N A^{-2}):



[12 Jan 2019, I]

Options:

A. 20 A, perpendicular out of the page

B. 40 A, perpendicular out of the page

C. 20 A, perpendicular into the page

D. 40 A, perpendicular into the page

Answer: C

Solution:

Let I be the current in each wire. (directed inwards)
Magnetic field at ' O ' due to LP and QM will be zero.
i.e., $B_0 = B_{PS} + B_{QN}$

$\therefore$ Net magnetic field $B_0 = \frac{\mu_0 i}{4\pi d} + \frac{\mu_0 i}{4\pi d}$

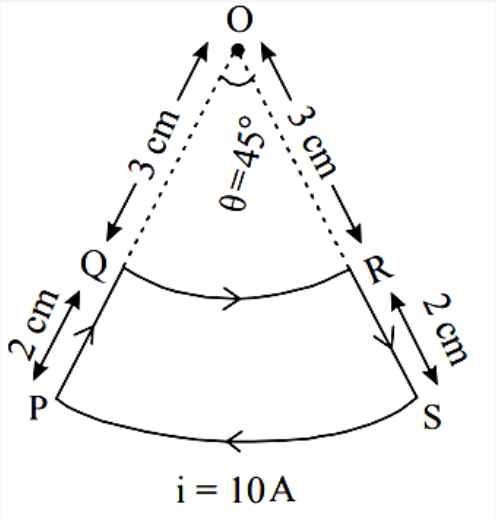
or $10^{-4} = \frac{\mu_0 i}{2\pi d} + \frac{2 \times 10^{-7} \times i}{4 \times 10^{-2}}$

$\therefore i = 20A$ and the direction of magnetic field is perpendicular into the plane

---

Question150

A current loop, having two circular arcs joined by two radial lines is shown in the figure. It carries a current of 10 A. The magnetic field at point O will be close to:



[9 Jan. 2019 I]

Options:

A. $1.0 \times 10^{-7}\mathrm{T}$

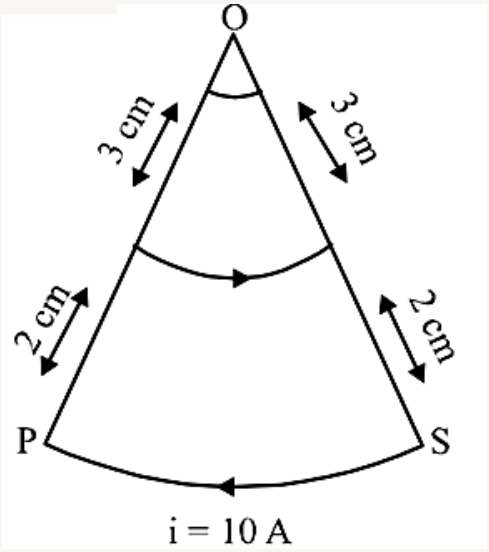
B. $1.5 \times 10^{-7}\mathrm{T}$

C. $1.5 \times 10^{-5}\mathrm{T}$

D. $1.0 \times 10^{-5}\mathrm{T}$

Answer: D

Solution:



There will be no magnetic field at O due to wire PQ and RS
Magnetic field at ' O ' due to arc QR

$$= \frac{\mu_0}{4\pi} \frac{\left(\frac{\pi}{4}\right) . \mathrm{I}}{\mathrm{r}_1}$$

Magnetic field at ' O ' due to are PS

$$= \frac{\mu_0}{4\pi} \frac{\left(\frac{\pi}{4}\right) . \mathrm{I}}{\mathrm{r}_2}$$

$\therefore$ Net magnetic field at ' O'

$$\mathrm{B} = \frac{\mu_0}{4\pi}(\pi/4) \times 10 \left[\frac{1}{(3 \times 10^{-2})} - \frac{1}{(5 \times 10^{-2})} \right]$$

$$\Rightarrow \left| \overrightarrow{\mathrm{B}} \right| = \frac{\pi}{3} \times 10^{-5}\mathrm{T} \approx 1 \times 10^{-5}\mathrm{T}$$

Question151

One of the two identical conducting wires of length L is bent in the form of a circular loop and the other one into a circular coil of N identical turns. If the same current is passed in both, the ratio of the magnetic field at the central of the loop (B_1) to that at the centre of the coil (B_C), i.e. ,$\frac{B_L}{B_C}$ will be:
[9 Jan 2019, II]

Options:

A. N

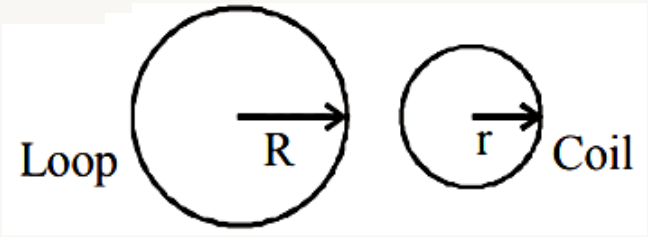
B. $\frac{1}{N}$

C. N^2

D. $\frac{1}{N^2}$

Answer: D

Solution:



$L = 2\pi R \quad L = N \times 2\pi r$

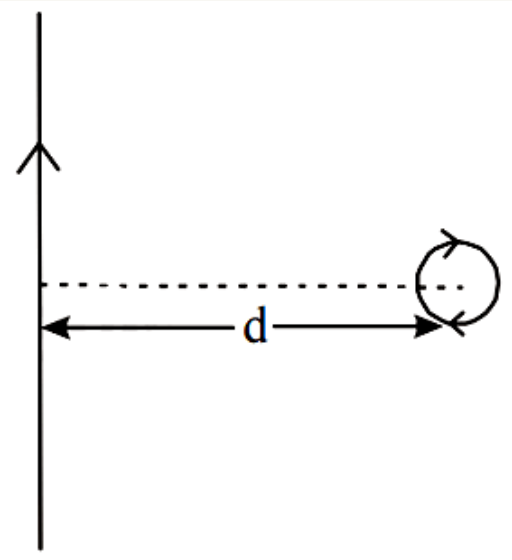
$R = N\,r \Rightarrow r = \frac{R}{N}$

$B_{Loop} = \frac{\mu_0 i}{2R} \quad B_{coil} = \frac{\mu_0 N\,i}{2r} = \frac{\mu_0 N\,i}{2\left(\frac{R}{N}\right)} = \frac{\mu_0 N^2 i}{2R}$

$\therefore \frac{B_L}{B_C} = \frac{1}{N^2}$

Question152

An infinitely long current carrying wire and a small current carrying loop are in the plane of the paper as shown. The radius of the loop is a and distance of its centre from the wire is d (d > > a). If the loop applies a force F on the wire then:



[9 Jan. 2019 I]

Options:

A. $F = 0$

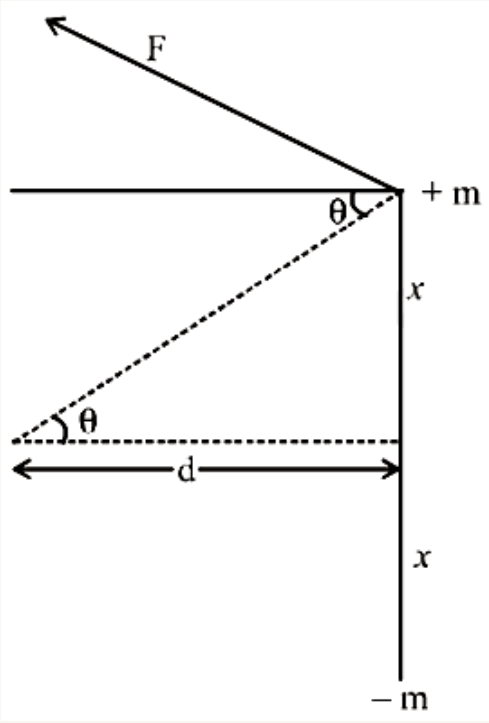
B. $F \propto \left(\frac{a}{d}\right)$

C. $F \propto \left(\frac{a^2}{d^3}\right)$

D. $F \propto \left(\frac{a}{d}\right)^2$

Answer: D

Solution:



Force on one pole,

$$F = m \times \frac{\mu_0 I}{2\pi\sqrt{d^2 + x^2}}$$

Total force, $F_{total} = 2F \sin\theta$

$$= 2 \times \frac{\mu_0 I m}{2\pi\sqrt{d^2 + a^2}} \times \frac{x}{\sqrt{d^2 + a^2}}$$

$$= \frac{\mu_0 I m x}{\pi(d^2 + a^2)}$$

Magnetic moment, $M = I\pi a^2 = m \times 2$

or, Total force, $F_{total} = \frac{\mu_0 I a^2}{2(d^2 + a^2)}$

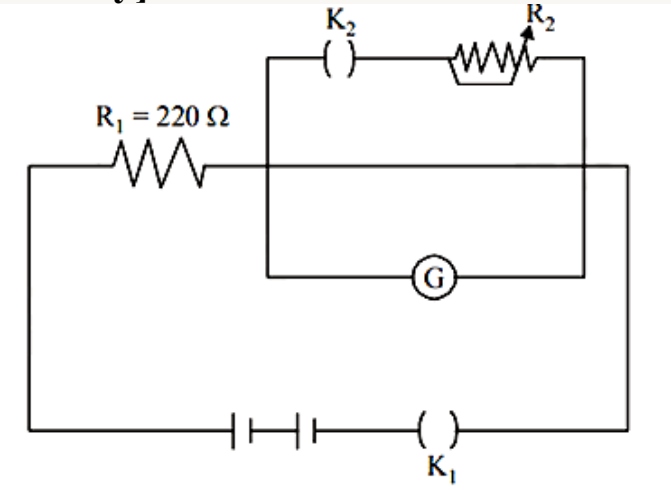
$$= \frac{\mu_0 I a^2}{2d^2} [\because d >> a]$$

Clearly $F_{total} \propto \frac{a^2}{d^2}$

Question153

The galvanometer deflection, when key K_1 is closed but K_2 is open, equals $theta_0$ (see figure). On closing K_2 also and adjusting R_2 to 5Ω, the deflection in galvanometer becomes $\frac{\theta_0}{5}$. The resistance of the galvanometer is, then, given by [Neglect the internal resistance of

battery]:



[12 Jan 2019, I]

Options:

A. 5Ω

B. 22Ω

C. 25Ω

D. 12Ω

Answer: B

Solution:

When key K_1 is closed and key K_2 is open

$$i_g = \frac{E}{220 + R_g} = C\theta_0 \text{ .....(i)}$$

When both the keys are closed

$$i_g = \left(\frac{E}{220 + \frac{5R_g}{5 + R_g}} \right) \times \frac{5}{(R_g + 5)} = \frac{C\theta_0}{5}$$

$$\Rightarrow \frac{5E}{225R_g + 1100} = \frac{C\theta_0}{5} \text{ .....(ii)}$$

$$\frac{E}{220 + R_g} = C\theta_0 \text{ .....(i)}$$

Dividing (i) by (ii), we get

$$\Rightarrow \frac{225R_g + 1100}{1100 + 5R_g} = 5$$

$$\Rightarrow 5500 + 25R_g = 225R_g + 1100$$

$200R_g = 4400$
$R_g = 22\Omega$

Question154

A galvanometer, whose resistance is 50 ohm, has 25 divisions in it. When a current of 4×10^{-4} A passes through it, its needle (pointer) deflects by one division. To use this galvanometer as a voltmeter of range 2.5V , it should be connected to a resistance of :
[12 Jan 2019, II]

Options:

A. 250 ohm

B. 200 ohm

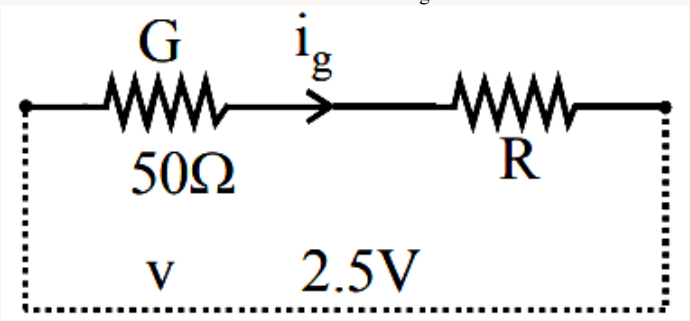
C. 6200 ohm

D. 6250 ohm

Answer: B

Solution:

Galvanometer has 25 divisions $I_g = 4 \times 10^{-4} \times 25 = 10^{-2}$A



$V = I_g R_{net}$
$v = I_g(G + R)$
$2.5 = (50 + R)10^{-2} \therefore R = 200\Omega$

Question155

A galvanometer having a resistance of 20 Ω and 30 division on both sides has figure of merit 0.005 ampere/division. The resistance that should be connected in series such that it can be used as a voltmeter upto 15 volt, is:
[11 Jan 2019, II]

Options:

A. 100 Ω

B. 120 Ω

C. 80 Ω

D. 125 Ω

Answer: C

Solution:

Deflection current
$= I g_{max} = n \times k = 0.005 \times 30$
Where, n = Number of divisions $= 30$ and k = 0.005amp/division
$= 15 \times 10^{-2} = 0.15$
$v = I_g[20 + R]$
$15 = 0.15[20 + R]$
$100 = 20 + R$
$R = 80\Omega$

Question156

A galvanometer having a coil resistance 100 Ω gives a full scale deflection when a current of 1 mA is passed through it. What is the value of the resistance which can convert this galvanometer into a voltmeter giving full scale deflection for a potential difference of 10 V?
[8 Jan 2019, II]

Options:

A. 10 kΩ

B. 8.9 kΩ

C. 7.9 kΩ

D. 9.9 kΩ

Answer: D

Solution:

Given,
Resistance of galvanometer, $G = 100\Omega$ Current, $i_g = 1mA$
A galvanometer can be converted into voltmeter by connecting a large resistance R in series with it.
Total resistance of the combination $= G + R$
According to Ohm's law, $V = i_g(G + R)$
$\therefore 10 = 1 \times 10^{-3}(100 + R_0)$
$\Rightarrow 10000 - 100 = 9900\Omega = R_0$
$\Rightarrow R_0 = 9.9k\Omega$

Question157

A proton, an electron, and a Helium nucleus, have the same energy. They are in circular orbits in a plane due to magnetic field perpendicular to the plane. Let r_p, r_e and r_{He} be their respective radii, then,
[10 April 2019, I]

Options:

A. $r_e > r_p = r_{He}$

B. $r_e < r_p = r_{He}$

C. $r_e < r_p < r_{He}$

D. $r_e > r_p > r_{He}$

Answer: B

Solution:

As $mvr = qvB \Rightarrow r = \frac{mv}{qB} = \frac{\sqrt{2mK\ .E}}{qB}$

$\left[\text{As} : \frac{1}{2}mv^2 = K\ .E\ .\right.$

$\Rightarrow m^2v^2 = 2mK\ .E$

$\left.\Rightarrow mv = \sqrt{2mK\ .E\ .}\right]$

For proton, electron and α -particle,

$m_{He} = 4m_p$ and $m_p > > m_e$

Also $a_{He} = 2q_p$ and $q_p = q_e$

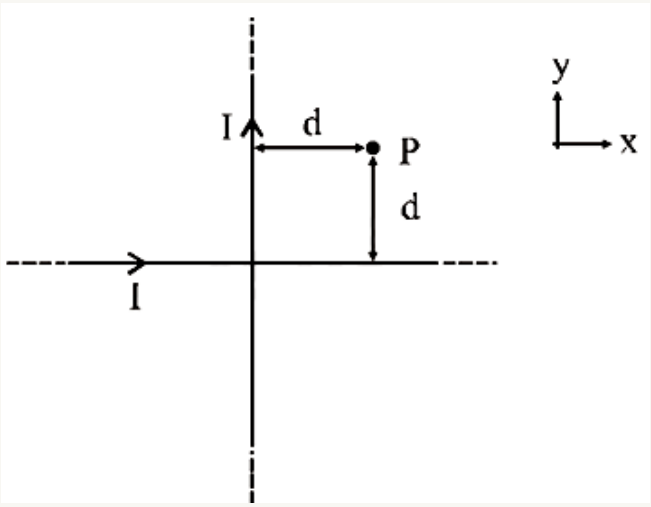
$\therefore$ As KE of all the particles is same then,

$r\ \alpha \frac{\sqrt{m}}{q}$

$\therefore r_{He} = r_p > r_e$

Question158

Two very long, straight, and insulated wires are kept at 90° angle from each other in xy -plane as shown in the figure.



These wires carry currents of equal magnitude I, whose directions are shown in the figure. The net magnetic field at point P will be: [12 April 2019, I]

Options:

A. Zero

B. $-\frac{\mu_0 I}{2\pi d}\left(\hat{x}+\hat{y}\right)$

C. $\frac{+\mu_0 I}{\pi d}\left(\hat{z}\right)$

D. $\frac{\mu_0 I}{2\pi d}\left(\hat{x}+\hat{y}\right)$

Answer: A

Solution:

$\vec{B} = \vec{B}_1 + \vec{B}_2$

$= \frac{\mu_0}{2\pi} \cdot \left(\frac{i^o}{d} \cdot \hat{k} + \frac{i^o}{d}\left(-\hat{k}\right) \right) = 0$

Question159

A thin ring of 10cm radius carries a uniformly distributed charge. The ring rotates at a constant angular speed of 40 À rad s^{-1} about its axis, perpendicular to its plane. If the magnetic field at its centre is 3.8×10^{-9}T , then the charge carried by the ring is close to ($\mu_0 = 4\pi \times 10^{-7} N/A^2$).
[12 April 2019, I]

Options:

A. 2×10^{-6}C

B. 3×10^{-5}C

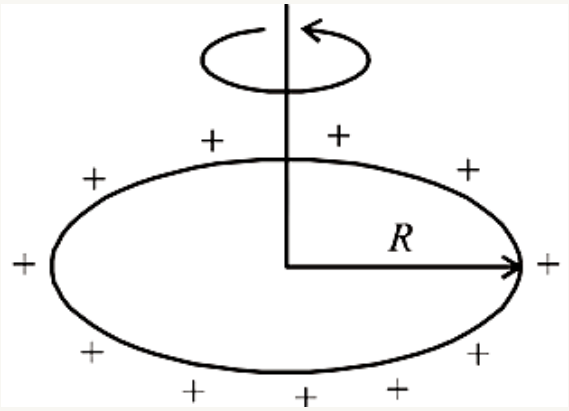
C. 4×10^{-5}C

D. 7×10^{-6}C

Answer: B

Solution:

If q is the charge on the ring, then



$i = \frac{q}{T} = \frac{q\omega}{2\pi}$

Magnetic field,

$B = \frac{\mu_0 i}{2R}$

$= \frac{\mu_0\left(\frac{q\omega}{2\pi}\right)}{2R}$

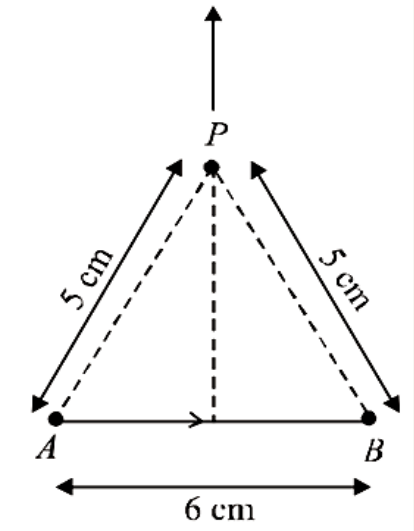
or $3.8 \times 10^{-9} = \left(\frac{\mu_0}{4\pi}\right)\frac{q\omega}{R} = (10^{-7})\frac{q \times 40\pi}{0.10}$

$\therefore q = 3 \times 10^{-5}$C

Question160

Find the magnetic field at point P due to a straight line segment AB of length 6cm carrying a current of 5A. (See figure)

$(\mu_o = 4\pi \times 10^{-7} N - A^{-2})$



[12 April 2019, II]

Options:

A. 2.0×10^{-5}T

B. 1.5×10^{-5}T

C. 3.0×10^{-5}T

D. 2.5×10^{-5}T

Answer: B

Solution:

$B = \frac{\mu_0}{4\pi}, \frac{i}{r}(\sin\alpha + \sin\beta)$

Here $r = \sqrt{5^2 - 3^2} = 4$cm

$\alpha = \beta = 37°$

$\therefore B = 10^{-7} \times \frac{5}{4} 2 \sin 37° = 1.5 \times 10^{-5}$T

Question161

The magnitude of the magnetic field at the center of an equilateral triangular loop of side 1m which is carrying a current of 10 A is :
[Take $\mu_0 = 4\pi \times 10^{-7} N\,A^{-2}$]
[10 April 2019, II]

Options:

A. 18μT

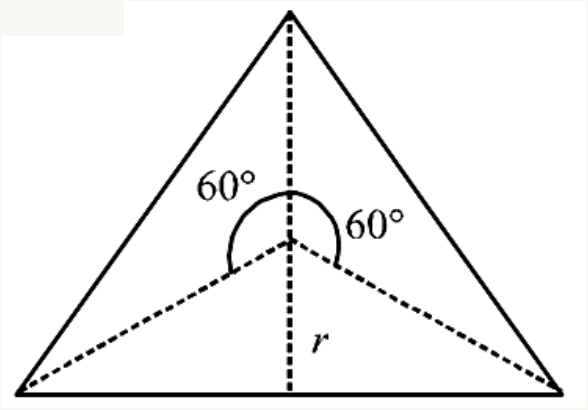
B. 9μT

C. 3μT

D. 1μT

Answer: A

Solution:



$r = \left(\frac{1}{3}\right)(a \sin 60)$

$r = \frac{a}{3} \times \frac{\sqrt{3}}{2} = \left(\frac{a}{2\sqrt{3}}\right)$

$B_0 = 3\left[\frac{\mu_0 I}{4\pi r}(\sin 60° + \sin 60°)\right]$

$= \frac{3\mu_0 I}{4\pi\left(\frac{a}{2\sqrt{3}}\right)} \times (2)\left(\frac{\sqrt{3}}{2}\right) = \frac{9}{2}\left(\frac{\mu_0 I}{\pi a}\right)$

$= \frac{9 \times 2 \times 10^{-7} \times 10}{1} = 18\mu T$

Question162

A square loop is carrying a steady current I and the magnitude of its magnetic dipole moment is m. If this square loop is changed to a circular loop and it carries the same current, the magnitude of the magnetic dipole moment of circular loop will be :
[10 April 2019, II]

Options:

A. $\frac{m}{\pi}$

B. $\frac{3m}{\pi}$

C. $\frac{2m}{\pi}$

D. $\frac{4m}{\pi}$

Answer: D

Solution:

Let a be the area of the square and r be the radius of circular loop.

$2\pi r = 4a \Rightarrow r = \left(\frac{2a}{\pi}\right)$

For square $M = (I)a^2$

For circular loop $M_1 = (I)\pi r^2$

$M_1 = (I)(\pi)\left(\frac{4a^2}{\pi^2}\right)$

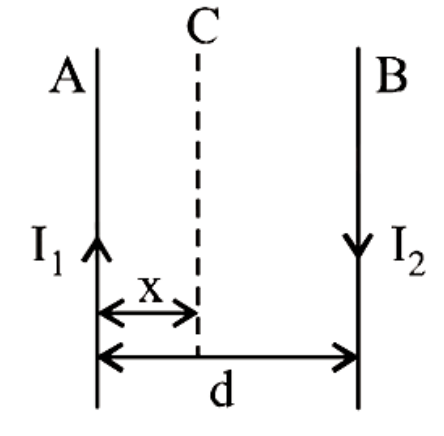
$M_1 = \frac{4I\,a^2}{\pi}$

$M_1 = \frac{4M}{\pi}(\because M = I\,a^2)$

Question163

Two wires A & B are carrying currents I_1 and I_2 as shown in the figure. The separation between them is d. A third wire C carrying a current I is to be kept parallel to them at a distance x from A such

that the net force acting on it is zero. The possible values of x are :



[10 April 2019, I]

Options:

A. $x = \left(\frac{I_1}{I_1 - I_2}\right) d$ and $x = \frac{I_2}{(I_1 + I_2)} d$

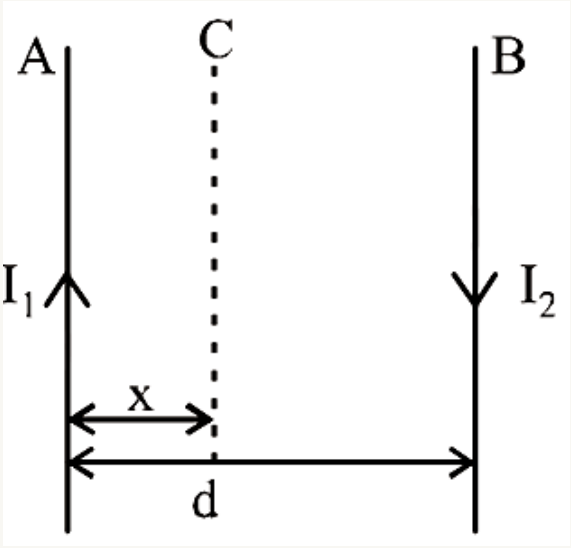
B. $x = \left(\frac{I_2}{(I_1 + I_2)}\right) d$ and $x = \left(\frac{I_2}{(I_1 - I_2)}\right) d$

C. $x = \left(\frac{I_1}{(I_1 + I_2)}\right) d$ and $x = \left(\frac{I_2}{(I_1 - I_2)}\right) d$

D. $x = \pm \frac{I_1 d}{(I_1 - I_2)}$

Answer: D

Solution:



As net force on the third wire C is zero.

$$\Rightarrow \vec{F} = \frac{\mu_0 I_1}{2\pi x} + \frac{\mu_0 I_2}{2\pi(d - x)} = 0$$

$$\frac{\mu_0 I_1}{2\pi x} = \frac{\mu_0 I_2}{2\pi(x - d)}$$

$$I_1 x - I_1 d = I_2 x$$

$$x = \frac{I_1 d}{I_1 - I_2}$$

Two cases may be possible if $I_1 > I_2$ or $I_2 > I_1$

Question164

A rectangular coil (Dimension 5 cm × 2.5 cm) with 100 turns, carrying a current of 3 A in the clock-wise direction, is kept centered at the origin and in the X-Z plane. A magnetic field of 1 T is applied along X-axis. If the coil is tilted through 45° about Z-axis, then the torque on the coil is:
[9 April 2019 I]

Options:

A. 0.38 Nm

B. 0.55 Nm

C. 0.42 Nm

D. 0.27 Nm

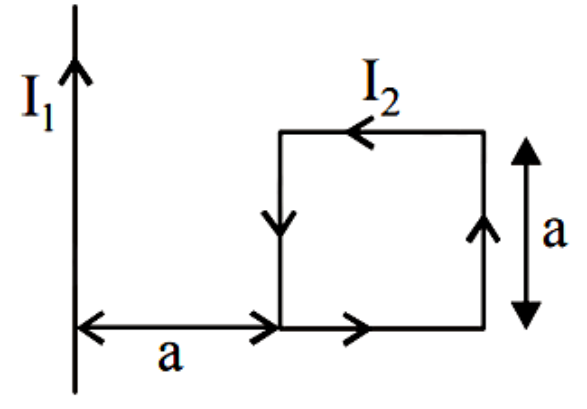
Answer: D

Solution:

$\tau = M B \sin 45° = N (iA) B \sin 45°$
$= 100 \times 3(5 \times 2.5) \times 10^{-4} \times 1 \times \frac{1}{\sqrt{2}}$
$= 0.27 N - m$

Question165

A rigid square of loop of side 'a' and carrying current I_2 is lying on a horizontal surface near a long current I_1 carrying wire in the same plane as shown in figure. The net force on the loop due to the wire will be:



[9 April 2019 I]

Options:

A. Repulsive and equal to $\frac{\mu_O I_1 I_2}{2\pi}$

B. Attractive and equal to $\mu_O I_1 I_2 3\pi$

C. Repulsive and equal to $\frac{\mu_O I_1 I_2}{4\pi}$

D. Zero

Answer: C

Solution:

$$F = \frac{\mu_0}{2\pi}\left(\frac{i_1 i_2}{a} - \frac{i_1 i_2}{2a}\right) \times a = \frac{\mu_0 i_1 i_2}{4\pi}$$

Question166

A circular coil having N turns and radius r carries a current I. It is held in the X Z plane in a magnetic field B. The torque on the coil due to the magnetic field is :
[8 April 2019 I]

Options:

A. $\frac{Br^2I}{\pi N}$

B. $B\pi r^2 I\, N$

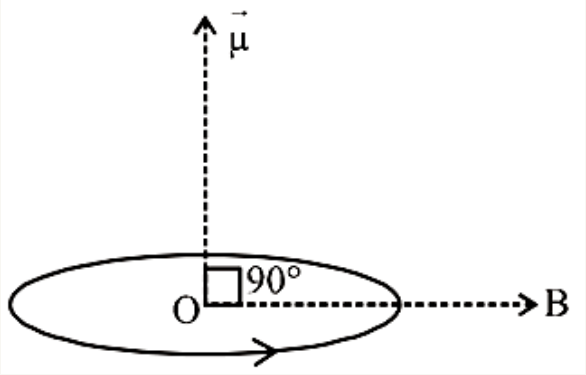
C. $\frac{B\pi r^2 I}{N}$

D. Zero

Answer: B

Solution:

$|\vec{\tau}| = |\vec{\mu} \times \vec{B}|[\mu = N\, I\, A]$
$= N\, I\, A \times B \sin 90°[A = \pi r^2]$
$\Rightarrow \tau = N\, I\, \pi r^2 B$

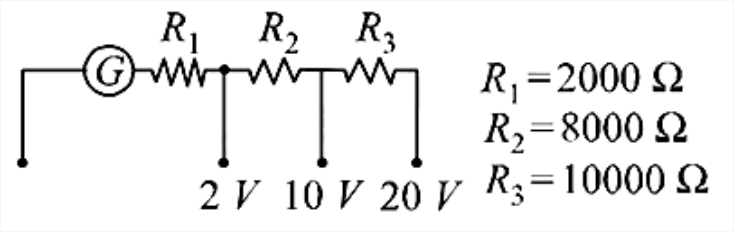


Question167

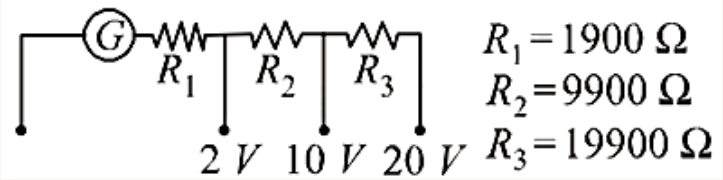
A galvanometer of resistance 100Ω has 50 divisions on its scale and has sensitivity of 20μA/ division. It is to be converted to a voltmeter with three ranges, of 0 − 2V , 0 − 10 V and 0 − 20 V. The appropriate circuit to do so is :
[12 April 2019, I]

Options:

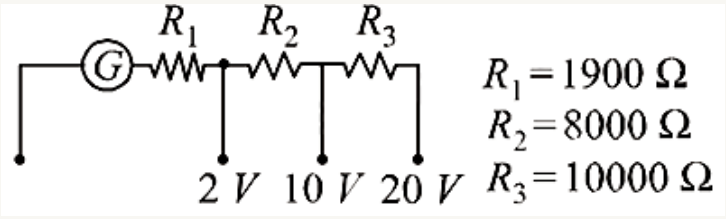
A.



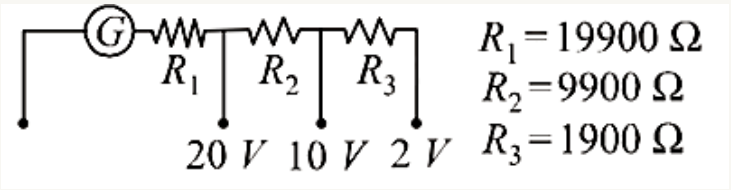
B.



C.



D.



Answer: C

Solution:

$i_g = 20 \times 50 = 1000\mu A = 1mA$

Using, $V = i_g(G + R)$, we have

$2 = 10^{-3}(100 + R_1)$

$R_1 = 1900\Omega$

when, $V = 10$ volt

$10 = 10^{-3}(100 + R_2 + R_1)$

$10000 = (100 + R_2 + 1900)$

$\therefore R_2 = 8000\Omega$

Question168

A moving coil galvanometer, having a resistance G, produces full scale deflection when a current I_g flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to $I_0 (I_0 > I_g)$ by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to V $(V = GI_0)$ by connecting a series resistance R_v to it. Then,
[12 April 2019, II]

Options:

A. $R_A R_V = G^2 \left(\frac{I_0 - I_g}{I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

B. $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

C. $R_A R_V = G^2 \left(\frac{I_g}{I_0 - I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g}\right)^2$

D. $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \frac{I_g}{(I_0 - I_g)}$

Answer: B

Solution:

In an ammeter,

$i_g = i_0 \frac{R_A}{R_A + G}$

and for voltmeter,

$V = i_g(G + R_V) = Gi_0$

On solving above equations, we get

$R_A R_V = G^2$

and $\frac{R_A}{R_V} = \left(\frac{i_g}{i_0 - i_g}\right)^2$

Question169

A moving coil galvanometer allows a full scale current of 10^{-4} A. A series resistance of 2M Ω is required to convert the above galvanometer into a voltmeter of range 0 − 5V . Therefore the value of shunt resistance required to convert the above galvanometer into an ammeter of range 0 − 10mA is :
[10 April 2019, I]

Options:

A. 500Ω

B. 100Ω

C. 200Ω

D. 10Ω

E. (Bonus)

Answer: E

Solution:

$v = i_g(R + G)$

$\Rightarrow 5 = 10^{-4}(2 \times 10^6 + x)$

$x = -195 \times 10^4 \Omega$

Question170

A moving coil galvanometer has resistance 50Ω and it indicates full deflection at 4 mA current. A voltmeter is made using this galvanometer and a 5kΩ resistance. The maximum voltage, that can be measured using this voltmeter, will be close to:
[9 April 2019 I]

Options:

A. 40V

B. 15V

C. 20V

D. 10V

Answer: C

Solution:

$V = i_g(G+R) = 4 \times 10^{-3}(50+5000) = 20V$

Question171

A moving coil galvanometer has a coil with 175 turns and area $1cm^2$. It uses a torsion band of torsion constant $10^{-6}N-m/rad$. The coil is placed in a magnetic field B parallel to its plane. The coil deflects by 1° for a current of 1mA. The value of B (in Tesla) is approximately:
[9 April 2019, II]

Options:

A. 10^{-4}

B. 10^{-2}

C. 10^{-1}

D. 10^{-3}

Answer: D

Solution:

$C\theta = N\,BiA\sin 90°$

or $10^{-6}\left(\frac{\pi}{180}\right) = 175B(10^{-3}) \times 10^{-4}$

$\therefore B = 10^{-3}T$

Question172

The resistance of a galvanometer is 50 ohm and the maximum current which can be passed through it is 0.002 A. What resistance must be connected to it order to convert it into an ammeter of range 0 − 0.5A?
[9 April 2019, II]

Options:

A. 0.5 ohm

B. 0.002 ohm

C. 0.02 ohm

D. 0.2 ohm

Answer: D

Solution:

Using, $i_g = i\frac{S}{S+G}$

$0.002 = 0.5\frac{S}{S+50}$

On solving, we get

$S = \frac{100}{498} \simeq 0.2\Omega$

Question173

A particle having the same charge as of electron moves in a circular path of radius 0.5cm under the influence of a magnetic field of 0.5T . If an electric field of 100V / mmakes it to move in a straight path then the mass of the particle is (Given charge of electron $= 1.6 \times 10^{-19}\text{C}$)
[12 April 2019, I]

Options:

A. $9.1 \times 10^{-31}\text{kg}$

B. $1.6 \times 10^{-27}\text{kg}$

C. $1.6 \times 10^{-19}\text{kg}$

D. $2.0 \times 10^{-24}\text{kg}$

Answer: D

Solution:

As particle is moving along a circular path

$\therefore R = \frac{mv}{qB}$ .....(i)

Path is straight line, then $qE = qvB$

$E = vB \Rightarrow v = \frac{E}{B}$ .....(ii)

From equation (i) and (ii)

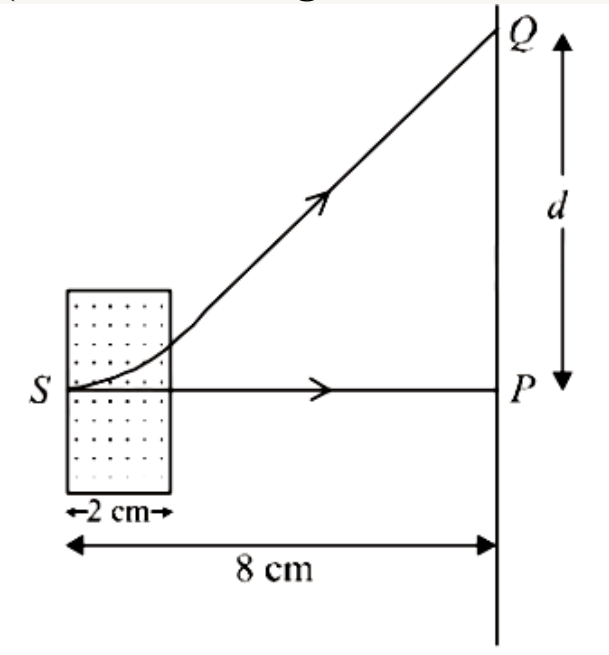
$m = \frac{qB^2R}{E} = \frac{1.6 \times 10^{-19} \times (0.5)^2 \times 0.5 \times 10^{-2}}{100}$

$\therefore m = 2.0 \times 10^{-24}\text{kg}$

Question174

An electron, moving along the x -axis with an initial energy of 100eV , enters a region of magnetic field $\vec{B} = (1.5 \times 10^{-3}\text{T})\hat{k}$ at S (see figure). The field extends between x = 0 and x = 2cm. The electron is detected at the point Q on a screen placed 8cm away from the point S. The distance d between P and Q (on the screen) is:

(Electron's charge $= 1.6 \times 10^{-19}$C, mass of electron $= 9.1 \times 10^{-31}$kg)



[12 April 2019, II]

Options:

A. 11.65cm

B. 12.87cm

C. 1.22cm

D. 2.25cm

Answer: B

Question175

An electron, a proton and an alpha particle having the same kinetic energy are moving in circular orbits of radii r_e r_p, r_α respectively in a uniform magnetic field B. The relation between r_e, r_p, r_α is :
[2018]

Options:

A. $r_e > r_p = r_\alpha$

B. $r_e < r_p = r_\alpha$

C. $r_e < r_p < r_\alpha$

D. $r_e < r_\alpha < r_p$

Answer: B

Solution:

As we know, radius of circular path in magnetic field $r = \frac{\sqrt{2K\,m}}{qB}$

For electron, $r_e = \frac{\sqrt{2K\,m_e}}{eB}$ .....(i)

For proton, $r_p = \frac{\sqrt{2K\,m_p}}{eB}$ .....(ii)

For α particle, $r_\alpha = \frac{\sqrt{2K\,m_a}}{q_\alpha B} = \frac{\sqrt{2K\,4m_p}}{2eB} = \frac{\sqrt{2K\,m_p}}{eB}$ .....(iii)

$\therefore r_e < r_p = r_\alpha \quad (\because m_e < m_p)$

Question176

The dipole moment of a circular loop carrying a current I, is m and the magnetic field at the centre of the loop is B_1. When the dipole moment is doubled by keeping the current current constant, the magnetic field at the centre of the loop is B_2. The ratio $\frac{B_1}{B_2}$ is:

[2018]

Options:

A. 2

B. $\sqrt{3}$

C. $\sqrt{2}$

D. $\frac{1}{\sqrt{2}}$

Answer: C

Solution:

Magnetic field at the centre of loop, $B_1 = \frac{\mu_0 I}{2R}$

Dipole moment of circular loop is $m =$ IA

$m_1 = I \cdot A = I \cdot \pi R^2 \{ R =$ Radius of the loop $\}$

If moment is doubled (keeping current constant)R be- comes $\sqrt{2R}$

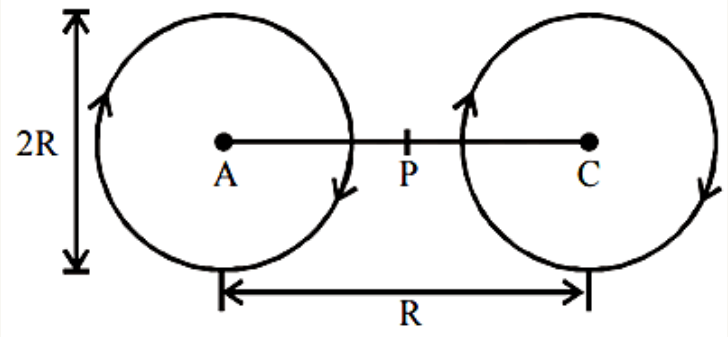
$m_2 = I \cdot \pi(\sqrt{2}R)^2 = 2 \cdot I\pi R^2 = 2m_1$

$B_2 = \frac{\mu_0 I}{2(\sqrt{2}R)}$

$\therefore \frac{B_1}{B_2} = \frac{\frac{\mu_0 I}{2R}}{\frac{\mu_0 I}{2(\sqrt{2}R)}} = \sqrt{2}$

Question177

A Helmholtz coil has pair of loops, each with N turns and radius R. They are placed coaxially at distance R and the same current I flows through the loops in the same direction. The magnitude of magnetic field at P, midway between the centres A and C, is given by (Refer to figure):



[Online April 15, 2018]

Options:

A. $\frac{4N\mu_0 I}{5^{3/2}R}$

B. $\frac{8N\mu_0 I}{5^{3/2}R}$

C. $\frac{4N\mu_0 I}{5^{1/2}R}$

D. $\frac{8N\mu_0 I}{5^{1/2}R}$

Answer: B

Solution:

Point P is situated at the mid-point of the line joining the centres of the circular wires which have same radii (R).

The magnetic fields $(\vec{B})$ at P due to the currents in the wires are in same direction.
Magnitude of magnetic field at point, P

$$B = 2\left\{\frac{\mu_0 N I R^2}{2\left(R^2+\frac{R^2}{4}\right)^{3/2}}\right\} = \frac{\mu_0 N I R^2}{\frac{5^{3/2}}{8}} = \frac{8\mu_0 N I}{5^{3/2}R}$$

Question178

A current of 1A is flowing on the sides of an equilateral triangle of side 4.5×10^{-2}m. The magnetic field at the centre of the triangle will be:
[Online April 15, 2018]

Options:

A. $4 \times 10^{-5} W\,b/m^2$

B. Zero

C. $2 \times 10^{-5} W\,b/m^2$

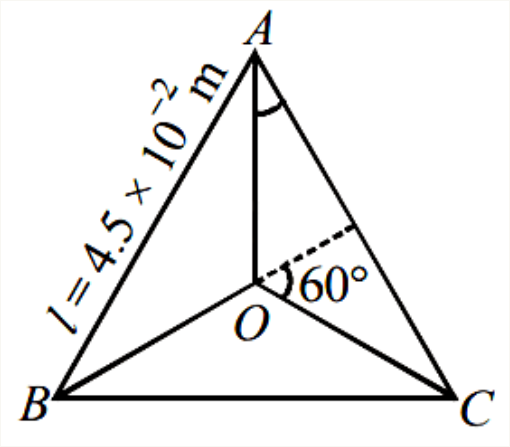
D. $8 \times 10^{-5} W\,b/m^2$

Answer: A

Solution:

Here, side of the triangle, $l = 4.5 \times 10^{-2}$m, current, $I = 1$A magnetic field at the centre of the triangle ' O ' B = ?

From figure, $\tan 60° = \sqrt{3} = \frac{1}{2d}$



$\Rightarrow d = \frac{l}{2\sqrt{3}} = \left(\frac{4.5 \times 10^{-2}}{2\sqrt{3}}\right)$m

Magnetic field, $B = \frac{\mu_0 i}{4\pi d}(\cos\theta_1 + \cos\theta_2)$

Putting value of $mu = 4\pi \times 10^{-7}$ and θ_1 and θ_2 we will get $B = 4 \times 10^{-5}$W b/m^2

Question179

A charge q is spread uniformly over an insulated loop of radius r . If it is rotated with an angular velocity ω with respect to normal axis then the magnetic moment of the loop is [Online April 16, 2018]

Options:

A. $\frac{1}{2}q\omega r^2$

B. $\frac{4}{3}q\omega r^2$

C. $\frac{3}{2}q\omega r^2$

D. $q\omega r^2$

Answer: A

Solution:

Magnetic moment, $\mu = I\,A = \frac{qv}{2\pi r}(\pi r^2)$

or, $\mu = \frac{qr\omega}{2\pi r}(\pi r^2) = \frac{1}{2}qr^2\omega$

Question180

In a circuit for finding the resistance of a galvanometer by half deflection method, a 6 V battery and a high resistance of 11kΩ are used. The figure of merit of the galvanometer 60μA/division. In the absence of shunt resistance, the galvanometer produces a deflection of θ = 9 divisions when current flows in the circuit. The value of the shunt resistance that can cause the deflection of θ/2 , is closest to [Online April 16, 2018]

Options:

A. 55Ω

B. 110Ω

C. 220Ω

D. 550Ω

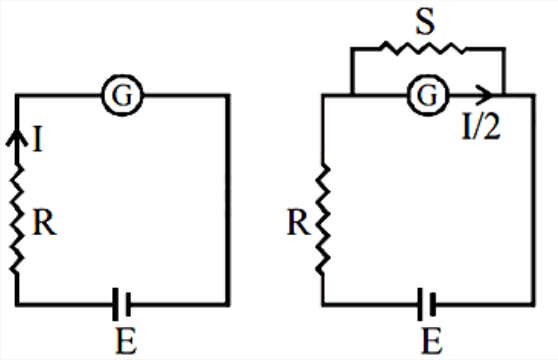
Answer: B

Solution:

Figure of merit of a galvanometer is the correct required to produce a deflection of one division in the galvanometer i.e., figure of merit $= \frac{I}{\theta}$

$I = \frac{\varepsilon}{R+G} \quad G = \frac{1}{9}K\,\Omega$

$\frac{1}{2} = \frac{\varepsilon}{R + \frac{GS}{G+S}} \times \frac{S}{S+G} \Rightarrow \frac{1}{2} = \frac{\varepsilon S}{R(S+G)+GS}$



$$S = \frac{11 \times 10^3 \times \frac{1}{2} \times 10^2 \times 270 \times 10^{-6}}{6 - \left(\frac{6}{2}\right)} = 110\Omega$$

Question181

A galvanometer with its coil resistance 25Ω requires a current of 1mA for its full deflection. In order to construct an ammeter to read up to a current of 2A, the approximate value of the shunt resistance should be
[Online April 16,2018]

Options:

A. $2.5 \times 10^{-2}\Omega$

B. $1.25 \times 10^{-3}\Omega$

C. $2.5 \times 10^{-3}\Omega$

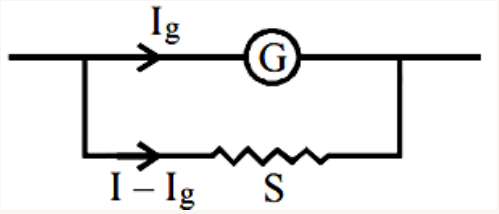
D. $1.25 \times 10^{-2}\Omega$

Answer: D

Solution:

According to question, current through galvanometer,
$I_a = 1mA$
Current through shunt $(I - I_o) = 2A$
Galvanometer resistance $R_g = 25\Omega$

Resistance of shunt, $S = ?$



$I_0R_0 = (I - I_g)S$

$\Rightarrow S = \frac{10^{-3} \times 25}{2}$

$S \simeq 1.25 \times 10^{-2}\Omega$

Question182

A negative test charge is moving near a long straight wire carrying a current. The force acting on the test charge is parallel to the direction of the current. The motion of the charge is : [Online April 9, 2017]

Options:

A. away from the wire

B. towards the wire

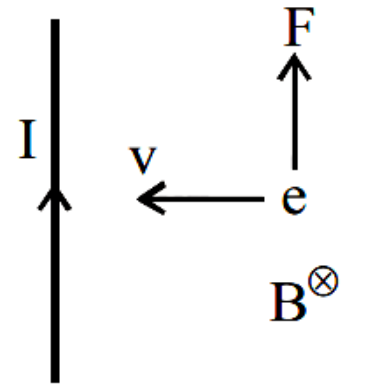
C. parallel to the wire along the current

D. parallel to the wire opposite to the current

Answer: B

Solution:

The force is parallel to the direction of current in magnetic field,
hence $F = q(v \times B)$
According to Fleming's left hand rule,



we have, the direction of motion of charge is towards the wire.

Question183

In a certain region static electric and magnetic fields exist. The magnetic field is given by $\vec{B} = B_0\left(\hat{i} + 2\hat{j} - 4\hat{k}\right)$. If a test charge moving with a velocity $\vec{v} = v_0\left(3\hat{i} - \hat{j} + 2\hat{k}\right)$ experiences no force in that region, then the electric field in the region, in SI units, is :
[Online April 8, 2017]

Options:

A. $\vec{E} = -v_0B_0\left(3\hat{i} - 2\hat{j} - 4\hat{k}\right)$

B. $\vec{E} = -v_0B_0\left(\hat{i} + \hat{j} + 7\hat{k}\right)$

C. $\vec{E} = v_0B_0\left(14\hat{j} + 7\hat{k}\right)$

D. $\vec{E} = -v_0B_0\left(14\hat{j} + 7\hat{k}\right)$

Answer: D

Solution:

According to question, as the test charge experiences no net force in that region i.e., sum of electric force $\left(F_e = q\vec{E}\right)$ and magnetic forces $\left[F_m = q\left(\vec{v} \times \vec{B}\right]\right.$ will be zero.

Hence, $F_e + F_m = 0$

$F_e = -q(\vec{v} \times \vec{B})$

$= -B_0 v_0 [(3\hat{i} - \hat{j} + 2\hat{k}) \times (\hat{i} + 2\hat{j} - 4\hat{k})]$

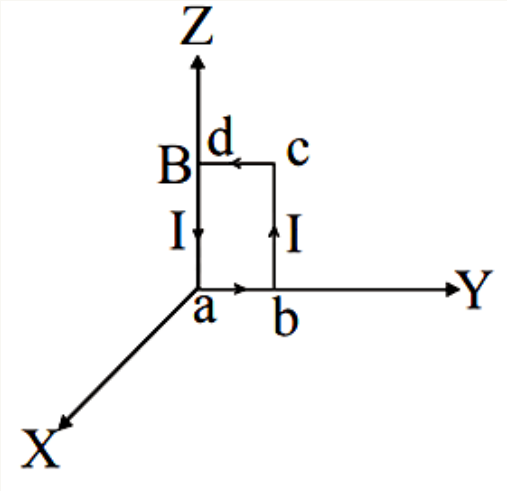
$= -B_0 v_0 (14\hat{j} + 7\hat{k})$

Question184

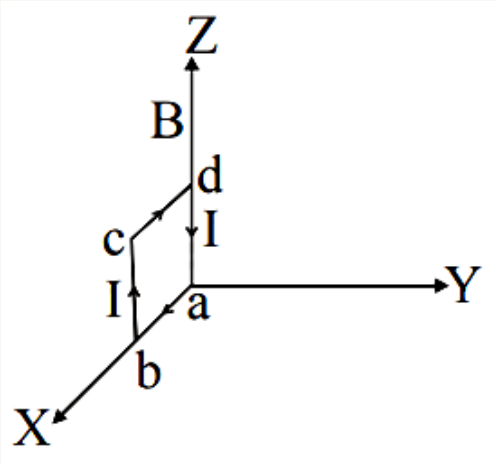
A uniform magnetic field B of 0.3 T is along the positive Z direction. A rectangular loop (abcd) of sides 10 cm × 5 cm carries a current 1 of 12 A. Out of the following different orientations which one corresponds to stable equilibrium ?
[Online April 9, 2017]

Options:

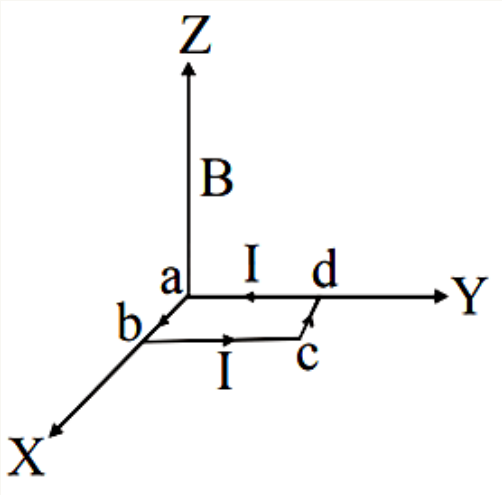
A.



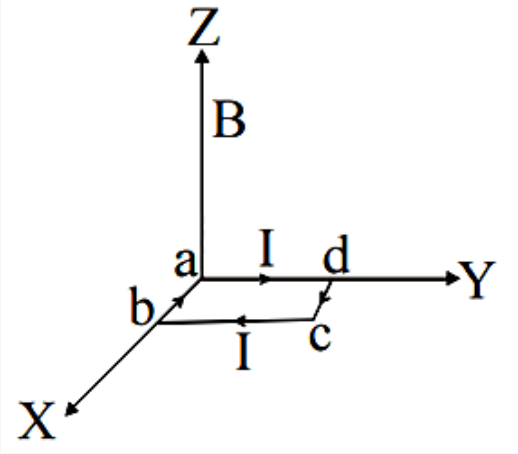
B.



C.



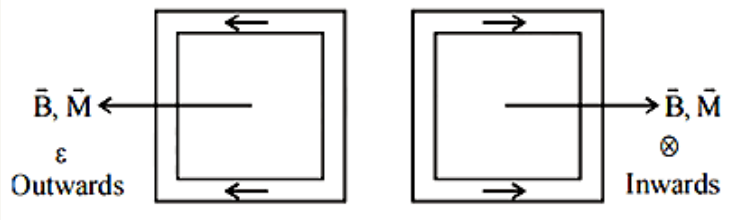
D.



Answer: C

Solution:

Magnetic moment of current carrying rectangular loop of area A is given by $M = NIA$
magnetic moment of current carrying coil is a vector and its direction is given by right hand thumb rule, for rectangular loop, $\bar{B}$ at centre due to current in loop and $\bar{M}$ are always parallel.



Hence, (c) corresponds to stable equilibrium.

Question185

When a current of 5 mA is passed through a galvanometer having a coil of resistance 15Ω, it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into to voltmeter of range $0-10V$ is
[2017]

Options:

A. $2.535 \times 10^3\Omega$

B. $4.005 \times 10^3\Omega$

C. $1.985 \times 10^3\Omega$

D. $2.045 \times 10^3\Omega$

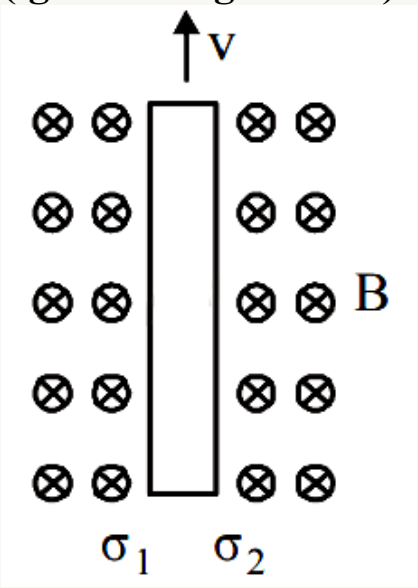
Answer: C

Solution:

Given : Current through the galvanometer,
$i_g = 5 \times 10^{-3}A$
Galvanometer resistance, $G = 15\Omega$
Let resistance R to be put in series with the galvanometer to convert it into a voltmeter.
$V = i_g(R+G)$
$10 = 5 \times 10^{-3}(R+15)$
$\therefore R = 2000 - 15 = 1985 = 1.985 \times 10^3\Omega$

Question186

Consider a thin metallic sheet perpendicular to the plane of the paper moving with speed 'v' in a uniform magnetic field B going into the plane of the paper (See figure). If charge densities σ_1 and σ_2 are induced on the left and right surfaces, respectively, of the sheet then (ignore fringe effects):



[Online April 10, 2016]

Options:

A. $\sigma_1 = \frac{-\epsilon_0 vB}{2}, \sigma_2 = \frac{\epsilon_0 vB}{2}$

B. $\sigma_1 = \epsilon_0 vB, \sigma_2 = -\epsilon_0 vB$

C. $\sigma_1 = \frac{\epsilon_0 vB}{2}, \sigma_2 = \frac{-\epsilon_0 vB}{2}$

D. $\sigma_1 = \sigma_2 = \epsilon_0 vB$

Answer: B

Solution:

$\because F = qE$ and $F = qvB$
$\therefore E = vB$
And Gauss's law in Electrostatics $E = \frac{\sigma}{\varepsilon_0}$

$E = \frac{\sigma}{\varepsilon_0} = vB \Rightarrow \sigma = \varepsilon_0 vB$

$\sigma_1 = -\sigma_2$

Question187

Two identical wires A and B, each of length 'l ', carry the same current I. Wire A is bent into a circle of radius R and wire B is bent to form a square of side a'. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is:

[2016]

Options:

A. $\frac{\pi^2}{16}$

B. $\frac{\pi^2}{8\sqrt{2}}$

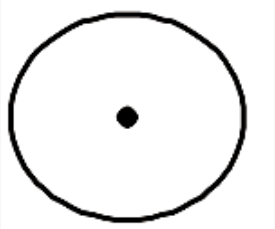
C. $\frac{\pi^2}{8}$

D. $\frac{\pi^2}{16\sqrt{2}}$

Answer: B

Solution:

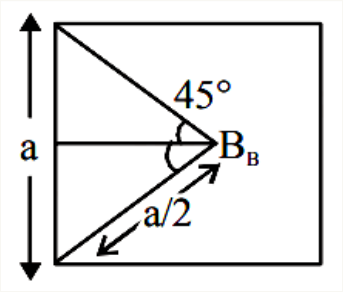
Case (a):



$B_A = \frac{\mu_0 I}{4\pi R} \times 2\pi = \frac{\mu_0 I}{4\pi l/2\pi} \times 2\pi (\because 2\pi R = l)$

$= \frac{\mu_0 I}{4\pi l} \times (2\pi)^2$

Case (b) :



$$B_B = 4 \times \frac{\mu_0}{4\pi}\frac{1}{a/2}[\sin 45° + \sin 45°]$$

$$= 4 \times \frac{\mu_0}{4\pi} \times \frac{I}{1/8} \times \frac{2}{\sqrt{2}} = \frac{\mu_0 I}{4\pi l} \times \frac{64}{\sqrt{2}} = \frac{\mu_0 I}{4\pi l} 32\sqrt{2} \; [4a = 1]$$

$$\Rightarrow \frac{B_A}{B_B} = \frac{\pi^2}{8\sqrt{2}}$$

Question188

A galvanometer having a coil resistance of 100 Ω gives a full scale deflection, when a current of 1 mA is passed through it. The value of the resistance, which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10 A, is :
[2016]

Options:

A. 0.1 Ω

B. 3 Ω

C. 0.01 Ω

D. 2 Ω

Answer: C

Solution:

$$I_g G = (I - I_g)s$$
$$\therefore 10^{-3} \times 100 = (10 - 10^{-3}) \times S$$
$$\therefore S \approx 0.01\Omega$$

Question189

A 50Ω resistance is connected to a battery of 5V . A galvanometer of resistance 100Ω is to be used as an ammeter to measure current through the resistance, for this a resistance r_s is connected to the galvanometer. Which of the following connections should be employed if the measured current is within 1% of the current without the ammeter in the circuit?
[Online April 9,2016]

Options:

A. $r_s = 0.5\Omega$ in series with the galvanometer

B. $r_s = 1\Omega$ in series with galvanometer

C. $r_s = 1\Omega$ in parallel with galvanometer

D. $r_s = 0.5\Omega$ in parallel with the galvanometer.

Answer: D

Solution:

As we know, $I = \frac{V}{R} = \frac{5}{50} = 0.1$

$I' = 0.099$

When Galvanometer is connected

$$R_{eq} = 50 + \frac{100S}{100+S} = \frac{V}{I}$$

$$\Rightarrow \frac{100S}{100+S} = \frac{5}{0.099} - 50$$

$$\Rightarrow \frac{100S}{100+S} = 50.50 - 50 \Rightarrow \frac{100S}{100+S} = 0.5$$

$$\Rightarrow 100S = 50 + 0.55 \Rightarrow 99.5S = 50$$

$$S = \frac{50}{99.05} = 0.5\Omega$$

So, shunt of resistance $= 0.5\Omega$ is connected in parallel with the galvanometer.

Question190

To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle theta. If a shunt of resistance S is needed to get half deflection then G, R and S related by the equation: [Online April 9, 2016]

Options:

A. $S(R+G) = RG$

B. $2S(R+G) = RG$

C. $2G = S$

D. $2S = G$

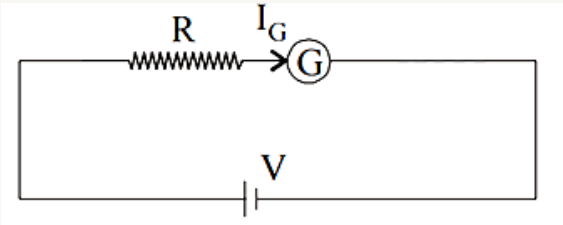
Answer: A

Solution:

According to Ohm's Law, $I = \frac{V}{R}$

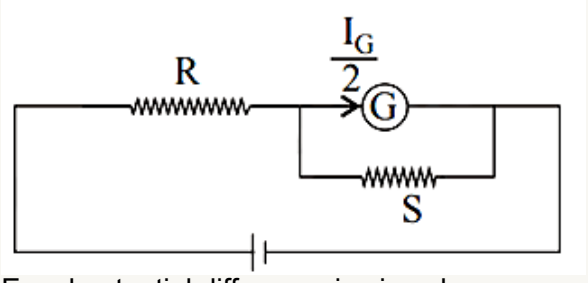
$I_g = \frac{V}{R+G}$

where, I_g -Galvanometer current, G-Galvanometer resistance



When shunt of resistance S is connected parallel to the Galvanometer then $G = \frac{GS}{G+S}$

$\therefore I = \frac{V}{R + \frac{GS}{G+S}}$



Equal potential difference is given by

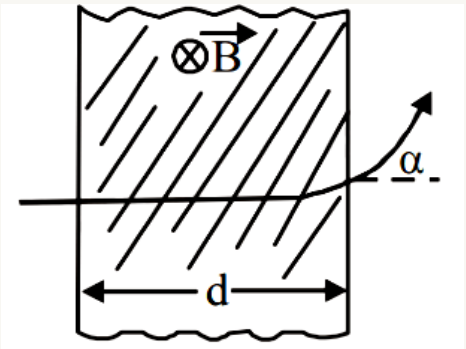
$I'_g G = (I - I'_g)S$

$I'_g(G+S) = I S$

$$\Rightarrow \frac{I_g}{2} = \frac{I S}{G+S}$$
$$\Rightarrow \frac{V}{2(R+G)} = \frac{V}{R+\frac{GS}{G+S}} \times \frac{S}{G+S}$$
$$\Rightarrow \frac{1}{2(R+G)} = \frac{S}{R(G+S)+GS}$$
$$\Rightarrow R(G+S)+GS = 2S(R+G)$$
$$\Rightarrow RG+RS+GS = 2S(R+G)$$
$$\Rightarrow RG = 2S(R+G) - S(R+G)$$
$$\therefore RG = S(R+G)$$

Question191

A proton (mass m) accelerated by a potential difference V flies through a uniform transverse magnetic field B. The field occupies a region of space by width ' d '. If α be the angle of deviation of proton from initial direction of motion (see figure), the value of sin α will be



[Online April 10, 2015]

Options:

A. $qV\sqrt{\frac{Bd}{2m}}$

B. $\frac{B}{2}\sqrt{\frac{qd}{mV}}$

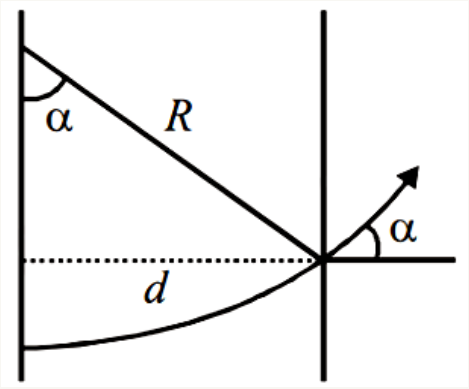
C. $\frac{B}{d}\sqrt{\frac{q}{2mV}}$

D. Bd $\sqrt{\frac{q}{2mV}}$

Answer: D

Solution:

From figure, $\sin\alpha = d/R$



And we know, $\frac{mv^2}{R} = qvB$

$\Rightarrow R = \frac{mv}{qB}$

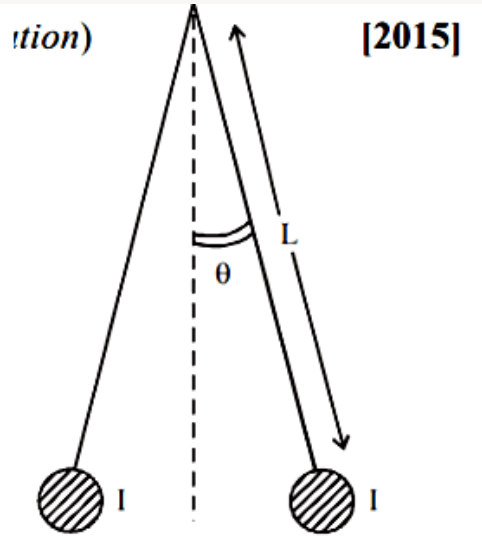
$\therefore \sin\alpha = \frac{d\,qB}{mv}$

$\sin\alpha = Bd\sqrt{\frac{q}{2mV}}\left[\because qV = \frac{1}{2}mv^2\right]$

Question192

Two long current carrying thin wires, both with current I, are held by insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle ' θ ' with the vertical. If wires have mass λ per unit length then the value of I is :

(g = gravitational acceleration)



[2015]

Options:

A. $2\sqrt{\frac{\pi gL}{\mu_0}\tan\theta}$

B. $2\sqrt{\frac{\pi\lambda gL}{\mu_0}\tan\theta}$

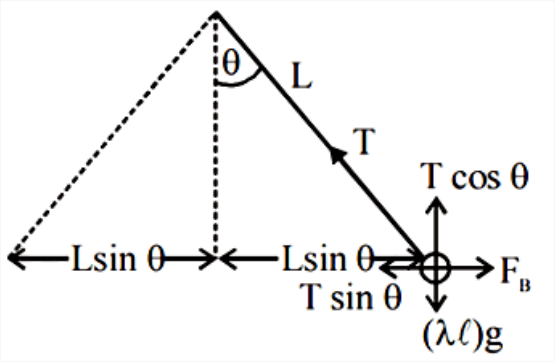
C. $\sin\theta\sqrt{\frac{\pi\lambda gL}{\mu_0\cos\theta}}$

D. $2\sin\theta\sqrt{\frac{\pi\lambda gL}{\mu_0\cos\theta}}$

Answer: D

Solution:

Let us consider ' l ' length of current carrying wire. At equilibrium
$T\cos\theta = \lambda gl$



and $T\sin\theta = \frac{m_0}{2p}\frac{I'I1}{2L\sin q}\left[\because\frac{F_B}{1} = \frac{\mu_0}{4\pi}\frac{2I\times I}{2l\sin\theta}\right]$

Therefore, $I = 2\sin q\sqrt{\frac{pl\,gL}{u_0\cos q}}$

Question193

Two coaxial solenoids of different radius carry current I in the same direction. $\vec{F}_1$ be the magnetic force on the inner solenoid due to the outer one and $\vec{F}_2$ be the magnetic force on the outer solenoid due to the inner one. Then :
[2015]

Options:

A. $\vec{F}_1$ is radially inwards and $\vec{F}_2 = 0$

B. $\vec{F}_1$ is radially outwards and $\vec{F}_2 = 0$

C. $\vec{F}_1 = \vec{F}_2 = 0$

D. $\vec{F}_1$ is radially inwards and $\vec{F}_2$ is radially outwards

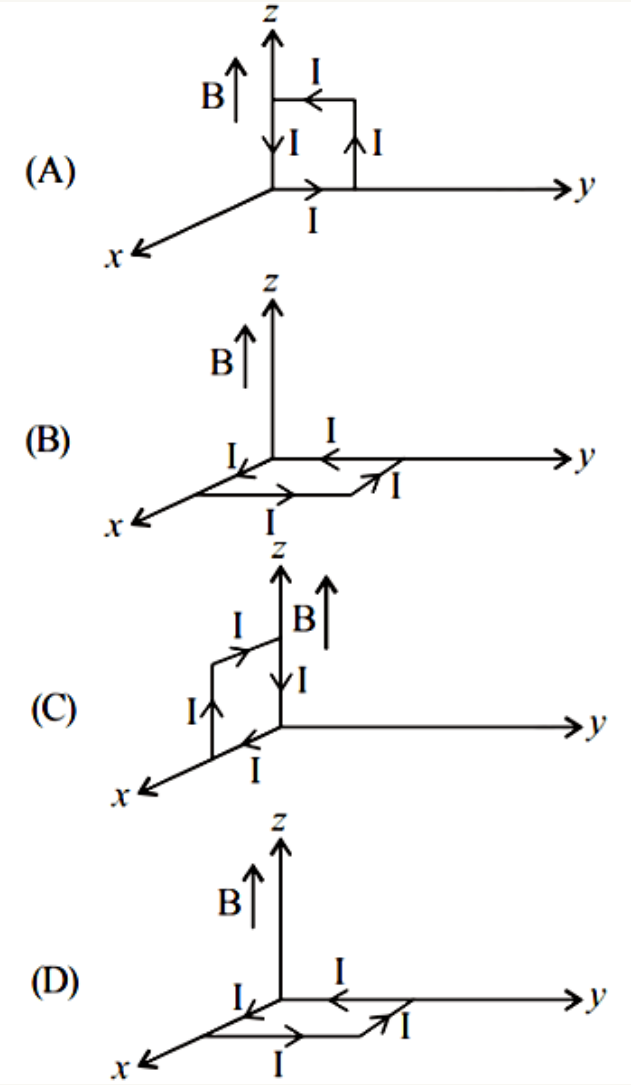
Answer: C

Solution:

$\vec{F}_1 = \vec{F}_2 = 0$
because of action and reaction pair

Question194

A rectangular loop of sides 10 cm and 5 cm carrying a current 1 of 12 A is placed in different orientations as shown in the figures below :



If there is a uniform magnetic field of 0.3T in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium?
[2015]

Options:

A. (B) and (D), respectively

B. (B) and (C), respectively

C. (A) and (B), respectively

D. (A) and (C), respectively

Answer: A

Solution:

For stable equilibrium $\vec{M} \parallel \vec{B}$

For unstable equilibrium $\vec{M} \parallel (-\vec{B})$

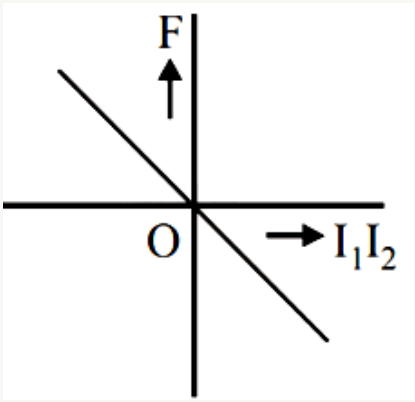
Question195

Two long straight parallel wires, carrying (adjustable) current I_1 and I_2, are kept at a distance d apart. If the force 'F' between the two wires is taken as 'positive' when the wires repel each other and 'negative' when the wires attract each other, the graph showing the dependence of F ′, on the product $I_1 I_2$, would be :

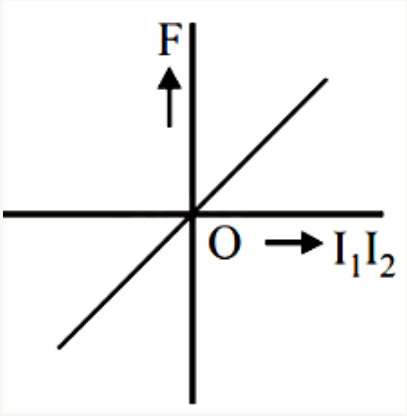
[Online April 11, 2015]

Options:

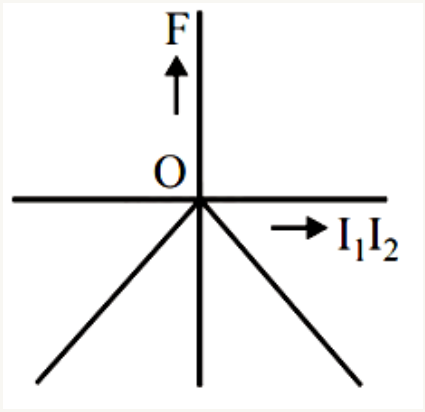
A.



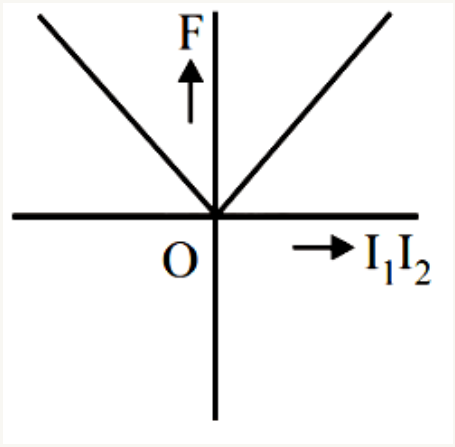
B.



C.



D.



Answer: A

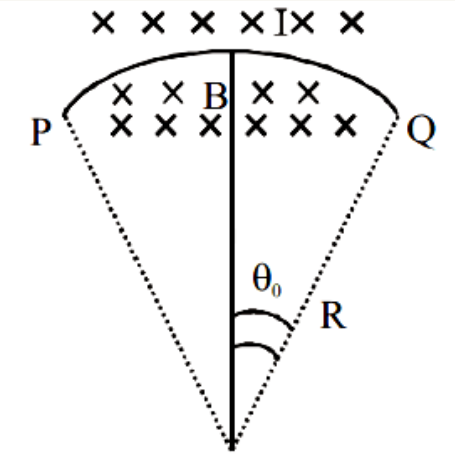
Solution:

I_1I_2 = Positive
(attract)F = Negative
I_1I_2 = Negative

(repell) F = Positive
Hence, option (a) is the correct answer.

Question196

A wire carrying current I is tied between points P and Q and is in the shape of a circular arc of radius R due to a uniform magnetic field B (perpendicular to the plane of the paper, shown by xxx) in the vicinity of the wire. If the wire subtends an angle $2\theta_0$ at the centre of the circle (of which it forms an arc) then the tension in the wire is :



[Online April 11, 2015]

Options:

A. $\frac{I BR}{2 \sin \theta_0}$

B. $\frac{I BR\theta_0}{\sin \theta_0}$

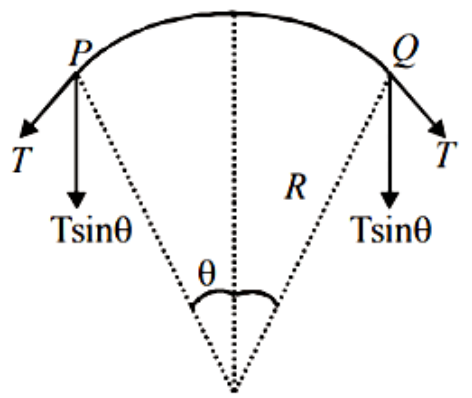
C. IBR

D. $\frac{I BR}{\sin \theta_0}$

Answer: C

Solution:

For small arc length
$2T\ \sin\theta = BIR\,2\theta\ (As\ F\ = BI\,L$ and $L = RZ\,\theta)$
$T\ = BI\,R$



Question197

The AC voltage across a resistance can be measured using a : [Online April 11, 2015]

Options:

A. hot wire voltmeter

B. moving coil galvanometer

C. potential coil galvanometer

D. moving magnet galvanometer

Answer: B

Solution:

To measure AC voltage across a resistance a moving coil galvanometer is used.

Question198

A positive charge 'q' of mass ' m ' is moving along the +x axis. We wish to apply a uniform magnetic field B for time Δt so that the charge reverses its direction crossing the y axis at a distance d.

Then:
[Online April 12, 2014]

Options:

A. $B = \frac{mv}{qd}$ and $\Delta t = \frac{\pi d}{v}$

B. $B = \frac{mv}{2qd}$ and $\Delta t = \frac{\pi d}{2v}$

C. $B = \frac{2mv}{qd}$ and $\Delta t = \frac{\pi d}{2v}$

D. $B = \frac{2mv}{qd}$ and $\Delta t = \frac{\pi d}{v}$

Answer: C

Solution:

The applied magnetic field provides the required centripetal force to the charge particle, so it can move in circular path of radius $\frac{d}{2}$

$\therefore Bqv = \frac{mv^2}{d/2}$

or, $B = \frac{2mv}{qd}$

Time interval for which a uniform magnetic field is applied $\Delta t = \frac{\pi \cdot \frac{d}{2}}{v}$

(particle reverses its direction after time Δ t by covering semi circle).

$\Delta t = \frac{\pi d}{2v}$

Question199

Consider two thin identical conducting wires covered with very thin insulating material. One of the wires is bent into a loop and produces magnetic field B_1, at its centre when a current I passes through it. The ratio $B_1 : B_2$ is:

[Online April 12, 2014]

Options:

A. 1: 1

B. 1: 3

C. 1: 9

D. 9: 1

Answer: B

Solution:

For loop $B = \frac{\mu_0 nI}{2a}$

where, a is the radius of loop. Then, $B_1 = \frac{\mu_0 I}{2a}$

Now, for coil $B = \frac{\mu_0 I}{4\pi} \cdot \frac{2nA}{x^3}$

at the centre x = radius of loop

$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2 \times 3 \times (I/3) \times \pi(a/3)^2}{(a/3)^3} = \frac{\mu_0 \cdot 3I}{2a}$

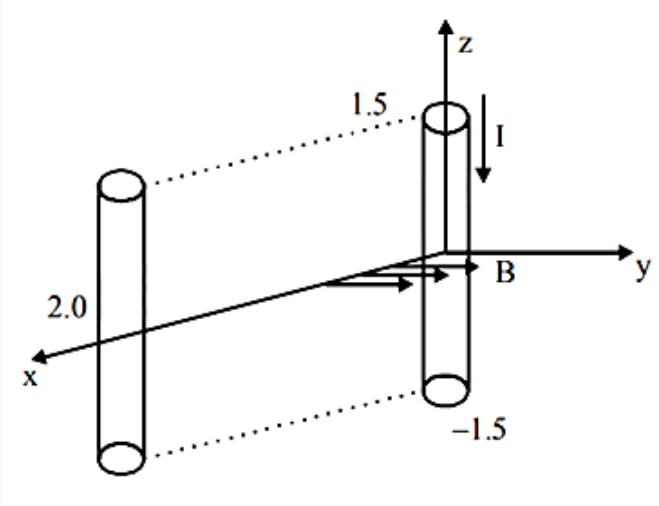
$\therefore \frac{B_1}{B_2} = \frac{\mu_0 I/2a}{\mu_0 \cdot 3I/2a}$

$B_1 : B_2 = 1 : 3$

Question200

A conductor lies along the z -axis at $-1.5 \leq z < 1.5$m and carries a fixed current of 10.0 A in $-\hat{a}_z$ direction (see figure). For a field $\vec{B} = 3.0 \times 10^{-4} e^{-0.2x} \hat{a}_y$T , find the power required to move the conductor at constant speed to x = 2.0m y = 0m in 5×10^{-3}s. Assume

parallel motion along the X-axis.



[2014]

Options:

A. 1.57W

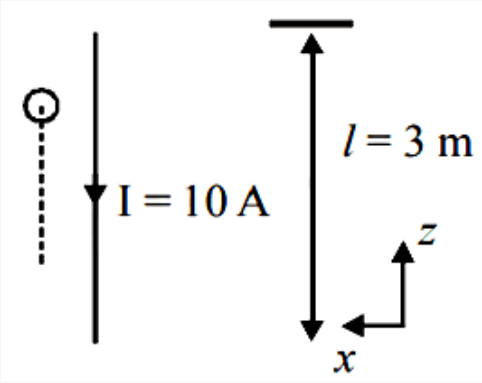
B. 2.97W

C. 14.85W

D. 29.7W

Answer: B

Solution:



Work done in moving the conductor is,

$$W = \int_0^2 F\,dx = \int_0^2 3.0 \times 10^{-4} e^{-0.2x} \times 10 \times 3\,dx$$

$$= 9 \times 10^{-3} \int_0^2 e^{-0.2x} dx$$

$$= \frac{9 \times 10^{-3}}{0.2}[-e^{-0.2 \times 2} + 1]$$

$$= \frac{9 \times 10^{-3}}{0.2} \times [1 - e^{-0.4}]$$

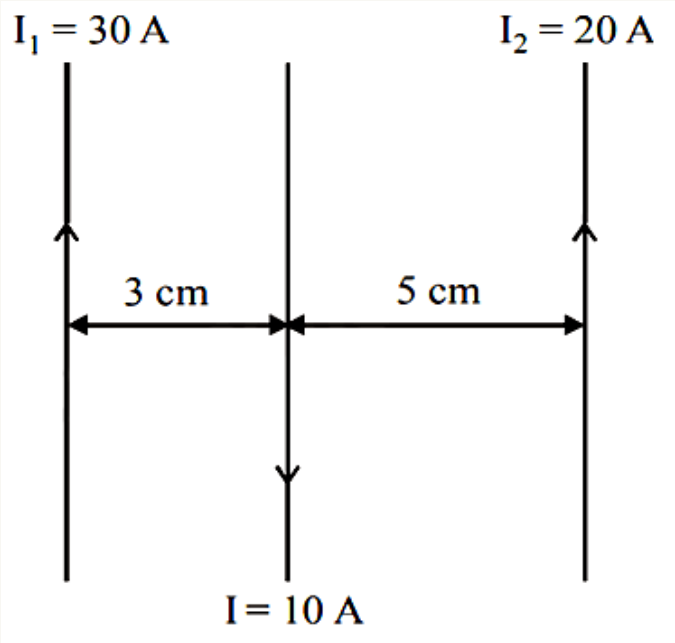
$$= \frac{9 \times 10^{-3} \times (0.33)}{2} = \frac{2.97 \times 10^{-3}}{2}$$

Power required to move the conductor is, $P = \frac{W}{t}$

$$P = \frac{2.97 \times 10^{-3}}{(0.2) \times 5 \times 10^{-3}} = 2.97W$$

Question201

Three straight parallel current carrying conductors are shown in the figure. The force experienced by the middle conductor of length 25cm is:



[Online April 11, 2014]

Options:

A. 3×10^{-4}N toward right

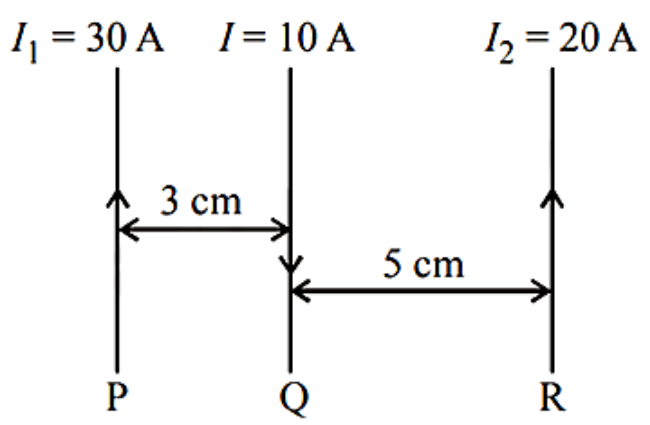
B. 6×10^{-4}N toward right

C. 9×10^{-4}N toward right

D. Zero

Answer: A

Solution:



Also given; length of wire Q
$= 25\text{cm} = 0.25\text{m}$
Force on wire Q due to wire R

$$F_{QR} = 10^{-7} \times \frac{2 \times 20 \times 10}{0.05} \times 0.25$$

$= 20 \times 10^{-5}\text{N}$ (Towards left)
Force on wire O due to wire P

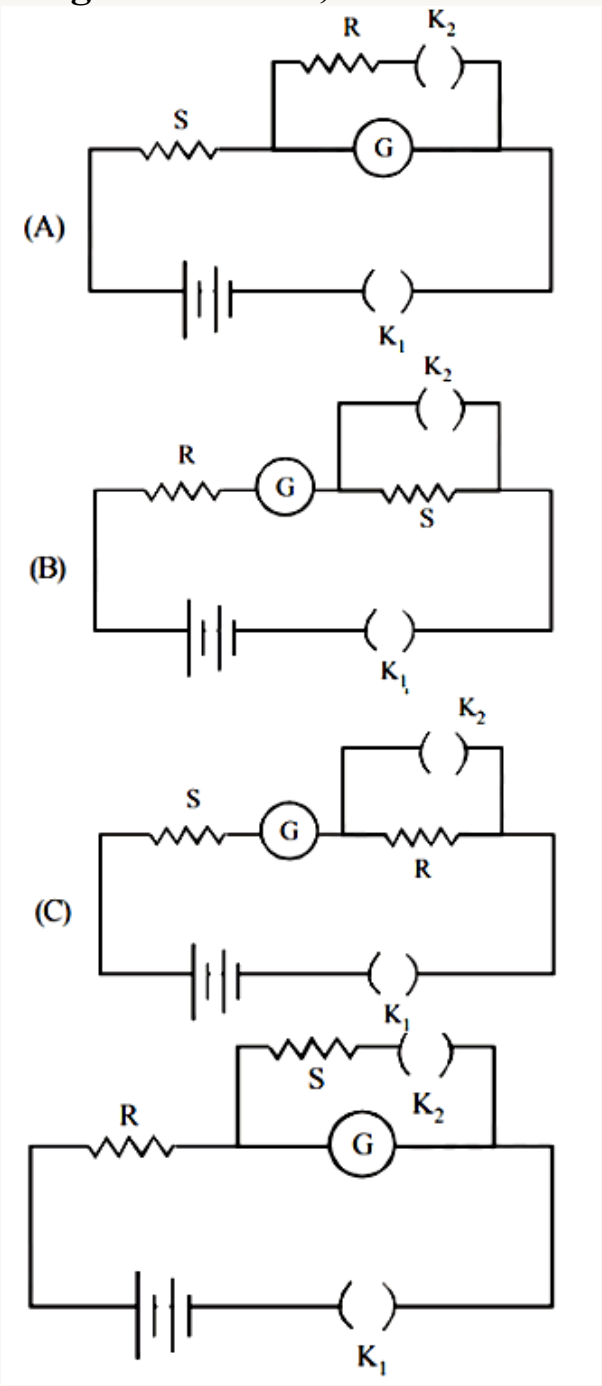
$$F_{QP} = 10^{-7} \times \frac{2 \times 30 \times 10}{0.03} \times 0.25$$

$= 50 \times 10^{-5}\text{N}$ (Towards right)
Hence, $F_{net} = F_{OP} - F_{OR}$
$= 50 \times 10^{-5}\text{N} - 20 \times 10^{-5}\text{N}$
$= 3 \times 10^{-4}\text{N}$ towards right

Question202

In the circuit diagrams (A, B, C and D) shown below, R is a high resistance and S is a resistance of the order of galvanometer resistance G. The correct circuit, corresponding to the half deflection method for finding the resistance and figure of merit of

the galvanometer, is the circuit labelled as:



[Online April 11, 2014]

Options:

A. Circuit A with $G = \frac{RS}{(R-S)}$

B. Circuit B with G = S

C. Circuit C with G = S

D. Circuit D with $G = \frac{RS}{(R-S)}$

Answer: D

Solution:

The correct circuit diagram is D with galvanometer resistance
$G = \frac{RS}{R-S}$

Question203

A particle of charge 16×10^{-16}C moving with velocity $10 ms^{-1}$ along x -axis enters a region where magnetic field of induction $\vec{B}$ is along the y -axis and an electric field of magnitude $10^4 V\ m^{-1}$ is along the negativez -axis. If the charged particle continues moving along x -axis, the magnitude of $\vec{B}$ is:
[Online April 23, 2013]

Options:

A. $16 \times 10^3 W\ bm^{-2}$

B. $2 \times 10^3 W\ bm^{-2}$

C. $1 \times 10^3 W\ bm^{-2}$

D. $4 \times 10^3 W\ bm^{-2}$

Answer: C

Solution:

Since particle is moving undeflected
So, $q_E = qvB$
$\Rightarrow B = \frac{E}{V} = \frac{10^4}{10} = 10^3 wb/m^2$

Question204

A parallel plate capacitor of area 60cm^2 and separation 3 mm is charged initially to 90μC. If the medium between the plate gets slightly conducting and the plate loses the charge initially at the rate of 2.5×10^{-8}C/s, then what is the magnetic field between the plates? [Online April 23,2013]

Options:

A. 2.5×10^{-8}T

B. 2.0×10^{-7}T

C. 1.63×10^{-11}T

D. Zero

Answer: D

Solution:

Magnetic field between the plates in this case is zero.

Question205

A current i is flowing in a straight conductor of length L. The magnetic induction at a point on its axis at a distance $\frac{L}{4}$ from its centre will be :
[Online April 22, 2013]

Options:

A. Zero

B. $\frac{\mu_0 i}{2\pi L}$

C. $\frac{\mu_0 i}{\sqrt{2}L}$

D. $\frac{4\mu_0 i}{\sqrt{5}\pi L}$

Answer: A

Solution:

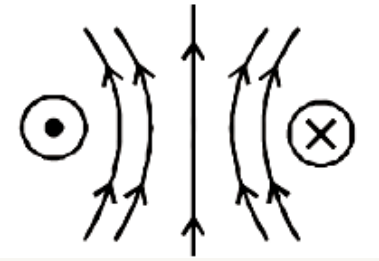
Magnetic field at any point lies on axial position of current carrying conductor $B = 0$

Question206

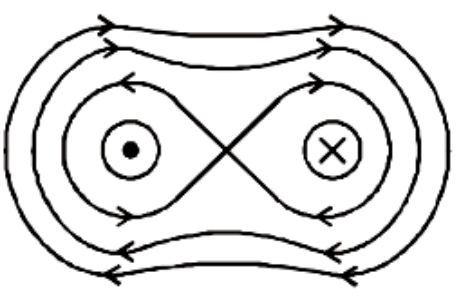
Choose the correct sketch of the magnetic field lines of a circular current loop shown by the dot ◉ and the cross⊗ [Online April 22, 2013]

Options:

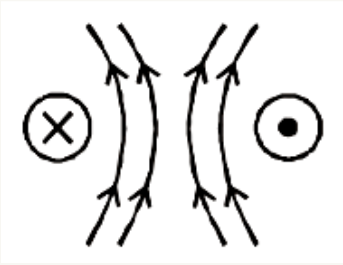
A.



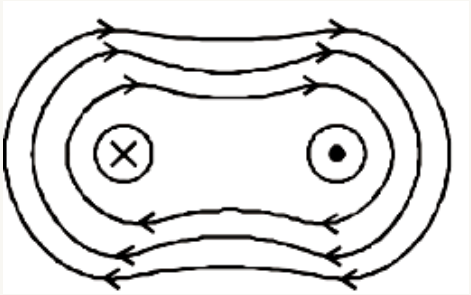
B.



C.



D.



Answer: A

Solution:

If magnetic field is perpendicular and into the plane of the paper, it is represented by cross ⊗ and if the direction of the magnetic field is perpendicular out of the plane of the paper it is represented by dot ◉.

Question207

An electric current is flowing through a circular coil of radius R. The ratio of the magnetic field at the centre of the coil and that at a distance $2\sqrt{2}$R from the centre of the coil and on its axis is : [Online April 9, 2013]

Options:

A. $2\sqrt{2}$

B. 27

C. 36

D. 8

Answer: B

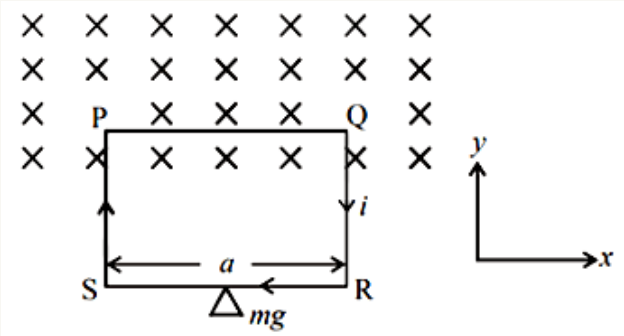
Solution:

Given : Radius $= R$
Distance $x = 2\sqrt{2}R$

$$\frac{B_{centre}}{B_{axis}} = \left(1+\frac{x^2}{R^2}\right)^{3/2} = \left(1+\frac{(2\sqrt{2}R)^2}{R^2}\right)^{3/2}$$
$$= (9)^{3/2} = 27$$

Question208

A rectangular loop of wire, supporting a mass m, hangs with one end in a uniform magnetic field $\vec{B}$ pointing out of the plane of the paper. A clockwise current is set up such that i > mg/Ba, where a is the width of the loop. Then :



[Online April 23, 2013]

Options:

A. The weight rises due to a vertical force caused by the magnetic field and work is done on the system.

B. The weight do not rise due to vertical for caused by the magnetic field and work is done on the system.

C. The weight rises due to a vertical force caused by the magnetic field but no work is done on the system

D. The weight rises due to a vertical force caused by the magnetic field and work is extracted from the magnetic field.

Answer: C

Question209

This questions has Statement I and Statement II. Of the four choices given after the Statements, choose the one that best describes into two Statements.
Statement-I : Higher the range, greater is the resistance of ammeter.
Statement-II : To increase the range of ammeter, additional shunt needs to be used across it.
[2013]

Options:

A. Statement-I is true, Statement-II is true, Statement-II is the correct explanation of Statement-I.

B. Statement-I is true, Statement-II is true, Statement-II is not the correct explanation of Statement-I.

C. Statement-I is true, Statement-II is false.

D. Statement-I is false, Statement-II is true.

Answer: D

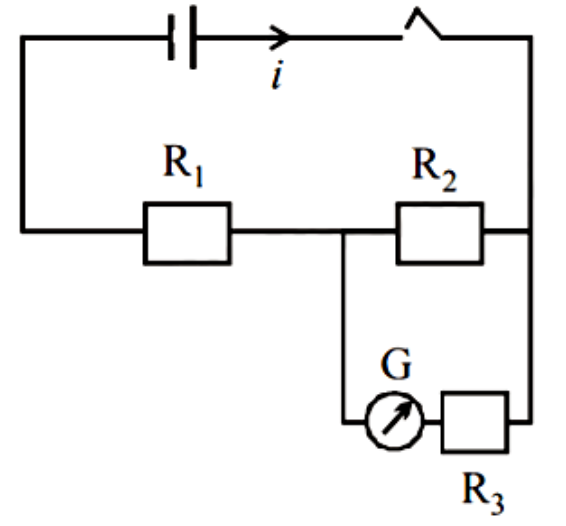
Solution:

Statements I is false and Statement II is true
For ammeter, shunt resistance, $S = \frac{I_g G}{I - I_g}$
Therefore for I to increase, S should decrease, So additional S can be connected across it.

Question210

To find the resistance of a galvanometer by the half deflection method the following circuit is used with resistances

$R_1 = 9970W$, $R_2 = 30W$ and $R_3 = 0$. The deflection in the galvanometer is d . With $R_3 = 107W$ the deflection changed to $\frac{d}{2}$. The galvanometer resistance is approximately:



[Online April 22, 2013]

Options:

A. 107Ω

B. 137Ω

C. 107/2Ω

D. 77Ω

Answer: D

Question211

A shunt of resistance 1Ω is connected across a galvanometer of 120Ω resistance. A current of 5.5 ampere gives full scale deflection in the galvanometer. The current that will give full scale deflection in the absence of the shunt is nearly :
[Online April 9, 2013]

Options:

A. 5.5 ampere

B. 0.5 ampere

C. 0.004 ampere

D. 0.045 ampere

Answer: D

Solution:

The current that will given full scale deflection in the absence of the shunt is nearly equal to the current through the galvanometer when shunt is connected i.e. I_g

As $I_g = \frac{IS}{G+S}$

$= \frac{5.5 \times 1}{120+1} = 0.045$ ampere

Question212

Proton, deuteron and alpha particle of same kinetic energy are moving in circular trajectories in a constant magnetic field. The radii of proton, deuteron and alpha particle are respectively r_p, r_d and r_α. Which one of the following relation is correct? [2012]

Options:

A. $r_\alpha = r_p = r_d$

B. $r_\alpha = r_p < r_d$

C. $r_\alpha > r_d > r_p$

D. $r_\alpha = r_d > r_p$

Answer: B

Solution:

The centripetal force is provided by the magnetic force

$\therefore \frac{mv^2}{R} = qvB \Rightarrow r = \frac{mv}{Bq} \therefore r \propto \frac{\sqrt{m}}{q}$

$\therefore r_p : r_d : r_\alpha = \frac{\sqrt{m_p}}{q_p} : \frac{\sqrt{m_d}}{q_d} : \frac{\sqrt{m_\alpha}}{q_\alpha}$

$= 1 : \sqrt{2} : 1$

Thus we have, $r_\alpha = r_p < r_d$

Question213

This question has Statement 1 and Statement 2 . Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement 1 : A charged particle is moving at right angle to a static magnetic field. During the motion the kinetic energy of the charge remains unchanged.

Statement 2: Static magnetic field exert force on a moving charge in the direction perpendicular to the magnetic field.

[Online May 26, 2012]

Options:

A. Statement 1 is false, Statement 2 is true.

B. Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

C. Statement 1 is true, Statement 2 is false.

D. Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

Answer: D

Solution:

When a charged particle enters the magnetic field in perpendicular direction then it experience a force in perpendicular direction.

i.e. $F = Bqv \sin\theta$

Due to which it moves in a circular path.

Question214

A proton and a deuteron are both accelerated through the same potential difference and enter in a magnetic field perpendicular to the direction of the field. If the deuteron follows a path of radius R, assuming the neutron and proton masses are nearly equal, the radius of the proton's path will be
[Online May 19, 2012]

Options:

A. $\sqrt{2}R$

B. $\frac{R}{\sqrt{2}}$

C. $\frac{R}{2}$

D. R

Answer: B

Solution:

As charge on both proton and deuteron is same i.e. 'e'
Energy acquired by both,
$E = eV$
For Deuteron.
Kinetic energy, $\frac{1}{2}mV^2 = eV$ [V is the potential difference]

$v = \sqrt{\frac{2eV}{m_d}}$

But $m_d = 2m$

Therefore, $v = \sqrt{\frac{2eV}{2m}} = \sqrt{\frac{eV}{m}}$

Radius of path, $R = \frac{mv}{eB}$
Substituting value of 'v' we get

$R = \frac{2m\sqrt{\frac{ev}{m}}}{eB}$

$\frac{R}{2} = \frac{m\sqrt{\frac{ev}{m}}}{eB}$ .....(i)
For proton :

$\frac{1}{2}mV^2 = eV$

$V = \sqrt{\frac{2eV}{m}}$

Radius of path, $R' = \frac{mV}{eB} = \frac{m\sqrt{\frac{2eV}{m}}}{eB}$

$R' = \sqrt{2} \times \frac{R}{2}$ [From eq. (i)]

$R' = \frac{R}{\sqrt{2}}$

Question215

The magnetic force acting on charged particle of charge 2 μC in magnetic field of 2T acting in y-direction, when the particle velocity is $\left(2\hat{i} + 3\hat{j}\right) \times 10^6 ms^{-1}$ is [Online May 12, 2012]

Options:

A. 8N in z-direction

B. 8N in y-direction

C. 4N in y-direction

D. 4N in z-direction

Answer: A

Solution:

$\vec{F} = q\left(\vec{V} \times \vec{B}\right)$

$= 2 \times 10^{-6}\left[\left(2\hat{i} + 3\hat{j}\right) \times 10^6 \times 2\hat{j}\right]$

$= 2 \times 4\hat{k} = 8N$ in Z -direction.

Question216

The velocity of certain ions that pass undeflected through crossed electric field E = 7.7kV / m and magnetic field B = 0.14T is [Online May 7, 2012]

Options:

A. 18km/s

B. 77km/s

C. 55km/s

D. 1078km/s

Answer: C

Solution:

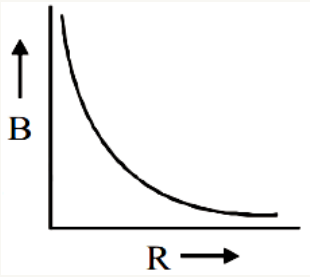
As velocity $v = \frac{E}{B} = \frac{7.7 \times 10^3}{0.14} = 55 km/s$

Question217

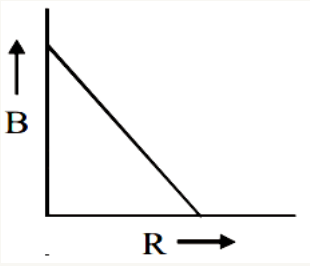
A charge Q is uniformly distributed over the surface of non-conducting disc of radius R. The disc rotates about an axis perpendicular to its plane and passing through its centre with an angular velocity ω. As a result of this rotation a magnetic field of induction B is obtained at the centre of the disc. If we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction at the centre of the disc will be represented by the figure :
[2012]

Options:

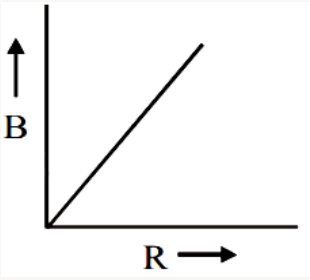
A.



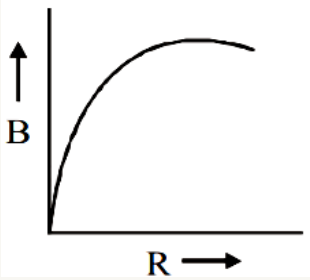
B.



C.



D.



Answer: A

Solution:

The magnetic field due to a disc is given as

$B = \frac{\mu_0 \omega Q}{2\pi R}$ i.e., $B \propto \frac{1}{R}$

Question218

Currents of a 10 ampere and 2 ampere are passed through two parallel thin wires A and B respectively in opposite directions. Wire A is infinitely long and the length of the wire B is 2m. The force acting on the conductor B, which is situated at 10cm distance from A will be [Online May 26, 2012]

Options:

A. 8×10^{-5}N

B. 5×10^{-5}N

C. $8\pi \times 10^{-7}$N

D. $4\pi \times 10^{-7}$N

Answer: A

Solution:

Force acting on conductor B due to conductor A is given by relation

$F = \frac{\mu_0 I_1 I_2 l}{2\pi r}$

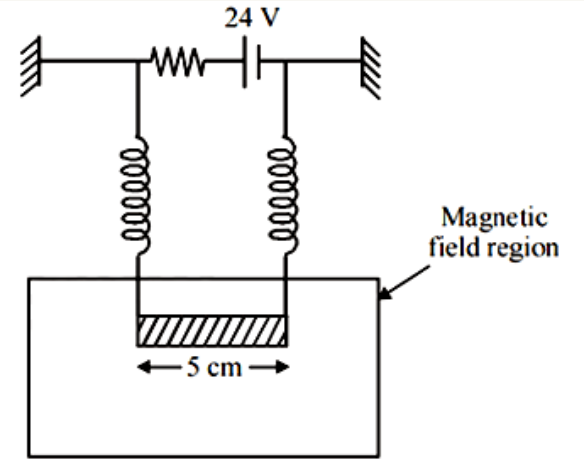
l -length of conductor B

r -distance between two conductors

$\therefore F = \frac{4\pi \times 10^{-7} \times 10 \times 2 \times 2}{2 \times \pi \times 0.1} = 8 \times 10^{-5}$N

Question219

The circuit in figure consists of wires at the top and bottom and identical springs as the left and right sides. The wire at the bottom has a mass of 10g and is 5cm long. The wire is hanging as shown in the figure. The springs stretch 0.5 cm under the weight of the wire and the circuit has a total resistance of 12Ω. When the lower wire is subjected to a static magnetic field, the springs, stretch an additional 0.3cm. The magnetic field is



[Online May 12,2012]

Options:

A. 0.6T and directed out of page

B. 1.2T and directed into the plane of page

C. 0.6T and directed into the plane of page

D. 1.2T and directed out of page

Answer: A

Question220

An electric charge +q moves with velocity $\vec{v} = 3\hat{i} + 4\hat{j} + \hat{k}$ in an electromagnetic field given by $\vec{E} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = i + \hat{j} - 3\hat{k}$ The y - component of the force experienced by + q is :
[2011 RS]

Options:

A. 11q

B. 5q

C. 3q

D. 2q

Answer: A

Solution:

The charge experiences both electric and magnetic force .
Electric force, $F_e = qE$

Magnetic force, $F_m = q(\vec{v} \times \vec{B})$

$\therefore$ Net force, $\vec{F} = q[\vec{E} + \vec{v} \times \vec{B}]$

$$= q\left[3\hat{i} + \hat{j} + 2\hat{k} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 1 \\ 1 & 1 & -3 \end{vmatrix}\right]$$

$$= q[3\hat{i} + \hat{j} + 2\hat{k} + \hat{i}(-12-1) - \hat{j}(-9-1) + \hat{k}(3-4)]$$
$$= q[3\hat{i} + \hat{j} + 2\hat{k} - 13\hat{i} + 10\hat{j} - \hat{k}]$$
$$= q[-10\hat{i} + 11\hat{j} + \hat{k}]$$
$F_y = 11q\hat{j}$
Thus, the y component of the force

Question221

A current I flows in an infinitely long wire with cross section in the form of a semi-circular ring of radius R. The magnitude of the magnetic induction along its axis is:
[2011]

Options:

A. $\frac{\mu_0 I}{2\pi^2 R}$

B. $\frac{\mu_0 I}{2\pi R}$

C. $\frac{\mu_0 I}{4\pi R}$

D. $\frac{\mu_0 I}{\pi^2 R}$

Answer: D

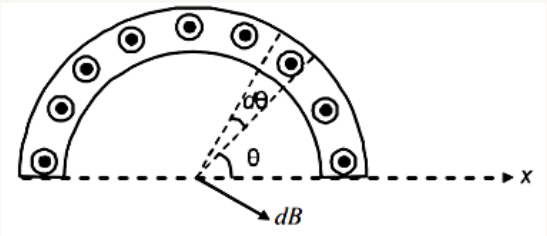
Solution:

Let R be the radius of semicircular ring. Let an elementary length dl is cut for finding magnetic field. So, $dl = Rd\theta$. Current in a small element, $dI = \frac{d\theta}{\pi} I$

Magnetic field due to the element

$$dB = \frac{\mu_0}{4\pi}\frac{2dI}{R} = \frac{\mu_0 I}{2\pi^2 R}$$

The component $dB\cos\theta$, of the field is cancelled by another opposite component. Therefore,



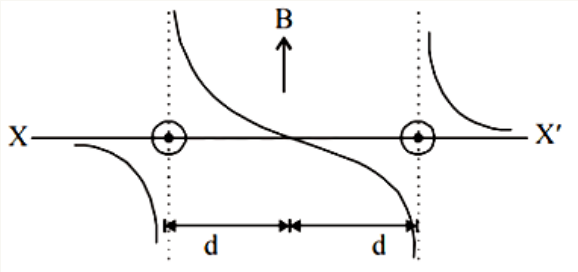
$$B_{net} = \int dB\sin\theta = \frac{\mu_0 I}{2\pi^2 R}\int_0^{\pi}\sin\theta\, d\theta = \frac{\mu_0 I}{\pi^2 R}$$

Question222

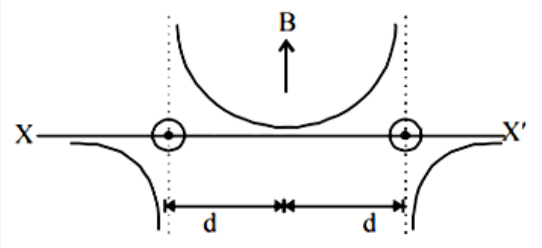
Two long parallel wires are at a distance 2d apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of the magnetic field B along the line XX' is given by [2010]

Options:

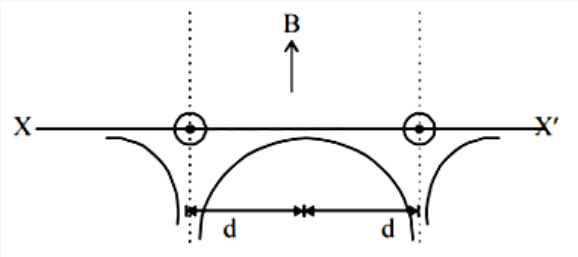
A.



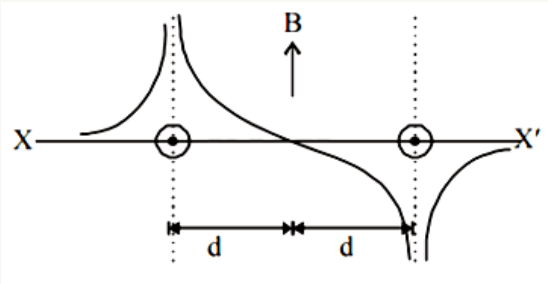
B.



C.



D.



Answer: A

Solution:

The magnetic field varies inversely with the distance for a long conductor. That is, $B \propto \frac{1}{d}$ so, graph in option (a) is the correct one.

Question223

The magnitude of the magnetic field (B) due to the loop ABCD at the origin (O) is : [2009]

Options:

A. $\frac{\mu_o I (b-a)}{24ab}$

B. $\frac{\mu_o I}{4\pi}\left[\frac{b-a}{ab}\right]$

C. $\frac{m_o I}{4p}[2(b-a)+p/3(a+b)]$

D. zero

Answer: A

Solution:

The magnetic field at O due to current in DA is
$B_1 = \frac{\mu_o I}{4\pi a} \times \frac{\pi}{6}$ (directed vertically upwards)
The magnetic field at O due to current in BC is
$B_2 = \frac{\mu_o I}{4\pi b} \times \frac{\pi}{6}$(directed vertically downwards)
The magnetic field due to current AB and CD at O is zero.
Therefore the net magnetic field is $B = B_1 - B_2$ (directed vertically upwards)

$$= \frac{\mu_o I}{4\pi a}\frac{\pi}{6} - \frac{\mu_o I}{4\pi b} \times \frac{\pi}{6}$$
$$= \frac{\mu_O I}{24}\left(\frac{1}{a} - \frac{1}{b}\right) = \frac{\mu_O I}{24ab}(b-a)$$

Question224

Due to the presence of the current I_1 at the origin:
[2009]

Options:

A. The forces on AD and BC are zero.

B. The magnitude of the net force on the loop is givenby $\frac{I_1 I}{4p} m_o[2(b-a)+p/3(a+b]$

C. The magnitude of the net force on the loop is givenby $\frac{\mu_o I I_1}{24ab}(b-a)$

D. The forces on AB and DC are zero.

Answer: D

Solution:

$\vec{F} = I(\vec{l} \times \vec{B})$
The force on AD and BC due to current I_1 is zero. This is because the directions of current element $I\vec{dl}$ and magnetic field $\vec{B}$ are parallel.

Question225

A horizontal overhead powerline is at height of 4m from the ground and carries a current of 100A from east to west. The magnetic field directly below it on the ground is ($\mu_0 = 4\pi \times 10^{-7}$ Tm A^{-1})
[2008]

Options:

A. 2.5×10^{-7} T southward

B. 5×10^{-6} T northward

C. 5×10^{-6} T southward

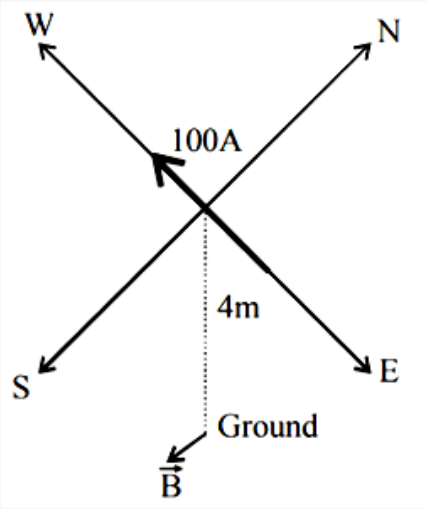
D. 2.5×10^{-7} T northward

Answer: C

Solution:

The magnetic field is

$$B = \frac{\mu_0}{4\pi}\frac{2I}{r} = 10^{-7} \times \frac{2 \times 100}{4} = 5 \times 10^{-6} T$$



Current flows from east to west. Point is below the power line, using right hand thumb rule, the magnetic field is directed towards south.

Question226

A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$ and comes out without any change in magnitude or direction of $\vec{v}$. Then
[2007]

Options:

A. $\vec{v} = \vec{B} \times \vec{E} / E^2$

B. $\vec{v} = \vec{E} \times \vec{B} / B^2$

C. $\vec{v} = \vec{B} \times \vec{E} / B^2$

D. $\vec{v} = \vec{E} \times \vec{B} / E^2$

Answer: B

Solution:

As velocity is not changing, charge particle must go undeflected, then
$qE = qvB$
$\Rightarrow v = \frac{E}{B}$

Also, $\left|\frac{\vec{E} \times \vec{B}}{B^2}\right| = \frac{E B \sin\theta}{B^2}$

$= \frac{E B \sin 90°}{B^2} = \frac{E}{B} = \left|\vec{v}\right| = v$

Question227

A charged particle moves through a magnetic field perpendicular to its direction. Then
[2007]

Options:

A. kinetic energy changes but the momentum is constant

B. the momentum changes but the kinetic energy is constant

C. both momentum and kinetic energy of the particle are not constant

D. both momentum and kinetic energy of the particle are constant

Answer: B

Solution:

When a charged particle enters a magnetic field at a direction perpendicular to the direction of motion, the path of the motion is circular. In circular motion the direction of velocity changes at every point (the magnitude remains constant).
Therefore, the tangential momentum will change at every point. But kinetic energy will remain constant as it is given by $\frac{1}{2}mv^2$ and v^2 is the square of the magnitude of velocity which does not change.

Question228

A long straight wire of radius a carries a steady current i. The current is uniformly distributed across its cross section. The ratio of the magnetic field at a/2 and 2a is
[2007]

Options:

A. 1/2

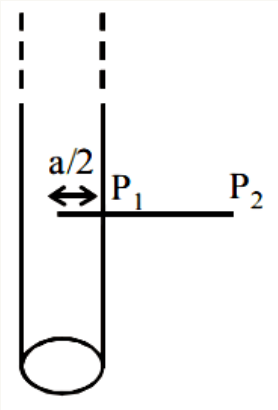
B. 1/4

C. 4

D. 1

Answer: D

Solution:



Since uniform current is flowing through a straight wire, current enclosed in the ampere an path formed at a distance $r_1\left(=\frac{a}{2}\right)$ is

$$i=\left(\frac{\pi r_1^{\,2}}{\pi a^2}\right)\times I,$$

where I is total current
Using Ampere circuital law,

$$\oint B\,.\,\vec{dl}=\mu_0 i$$

$$\Rightarrow B_1=\frac{\mu_0\times \text{current enclosed}}{\text{Path}}$$

$$\Rightarrow B_1=\frac{\mu_0\times\left(\frac{\pi r_1^{\,2}}{\pi a^2}\right)\times I}{2\pi r_1}=\frac{\mu_0\times I\,r_1}{2\pi a^2}$$

Now, magnetic field induction at point

P_2, $B_2 = \frac{\mu_0}{2\pi} \cdot \frac{I}{(2a)} = \frac{\mu_0 I}{4\pi a}$

$\therefore \frac{B_1}{B_2} = \frac{\mu_0 I r_1}{2\pi a^2} \times \frac{4\pi a}{\mu_0 I}$

$\Rightarrow \frac{B_1}{B_2} = \frac{2r_1}{a} = \frac{2 \times \frac{a}{2}}{a} = 1$

Question229

A current I flows along the length of an infinitely long, straight, thin walled pipe. Then [2007]

Options:

A. the magnetic field at all points inside the pipe is the same, but not zero

B. the magnetic field is zero only on the axis of the pipe

C. the magnetic field is different at different points inside the pipe

D. the magnetic field at any point inside the pipe is zero

Answer: D

Solution:

There is no current inside the pipe. From Ampere's circuital law $\oint \vec{B} . \vec{dl} = \mu_0 I$

$\because I = 0$

$\therefore B = 0$

Question230

Two identical conducting wires AOB and COD are placed at right angles to each other. The wire AOB carries an electric current I_1 and COD carries a current I_2. The magnetic field on a point lying at a distance d from O, in a direction perpendicular to the plane of the

wires AOB and COD, will be given by [2007]

Options:

A. $\frac{\mu_0}{2\pi d}(I_1^2 + I_2^2)$

B. $\frac{\mu_0}{2\pi}\left(\frac{I_1 + I_2}{d}\right)^{\frac{1}{2}}$

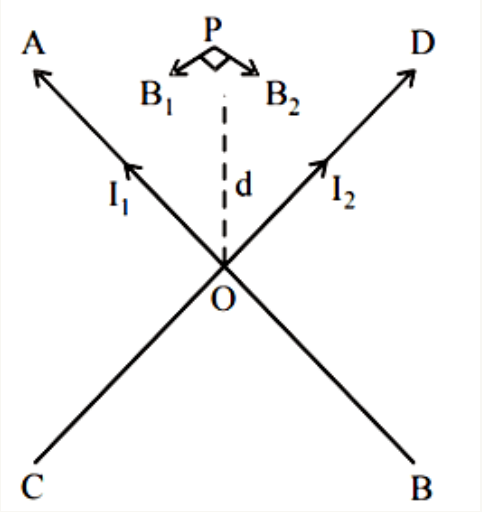
C. $\frac{\mu_0}{2\pi d}(I_1 + I_2)^{\frac{1}{2}}$

D. $\frac{\mu_0}{2\pi d}(I_1 + I_2)$

Answer: C

Solution:

The direction of magnetic field induction due to current through AB and CD at P are indicated as B_1 and B_2. The magnetic fields at a point P, equidistant from AOB and COD will have directions perpendicular to each other, as they are placed normal to each other.



Magnetic field at P due to current through AB,

$B_1 = \frac{\mu_0 I_1}{2\pi d}$

Magnetic field at P due to current through CD

$B_2 = \frac{\mu_0 I_2}{2\pi d}$

∴ Resultant field, $B = \sqrt{B_1^2 + B_2^2}$

$\therefore B = \sqrt{\left(\frac{\mu_0}{2\pi d}\right)^2 (I_1^2 + I_2^2)}$

or, $B = \frac{\mu_0}{2\pi d}(I_1^2 + I_2^2)^{1/2}$

Question231

In a region, steady and uniform electric and magnetic fields are present. These two fields are parallel to each other. A charged particle is released from rest in this region. The path of the particle will be a
[2006]

Options:

A. helix

B. straight line

C. ellipse

D. circle

Answer: B

Solution:

The charged particle will move along the lines of electric field (and magnetic field). Magnetic field will exert no force. The force by electric field will be along the lines of uniform electric field. Hence the particle will move in a straight line.

Question232

A long solenoid has 200 turns per cm and carries a current i. The magnetic field at its centre is 6.28×10^{-2} Weber /m^2. Another long solenoid has 100 turns per cm and it carries a current $\frac{i}{3}$. The value of the magnetic field at its centre is
[2006]

Options:

A. 1.05×10^{-2} Weber/m ^{2}

B. 1.05×10^{-5} Weber /m^2

C. 1.05×10^{-3} Weber /m^2

D. 1.05×10^{-4} Weber /m^2

Answer: A

Solution:

Magnetic field due to long solenoid is given by $B = \mu_0 nI$

In first case $B_1 = \mu_0 n_1 I_1$

In second case, $B_2 = \mu_0 n_2 I_2$

$$\therefore \frac{B_2}{B_1} = \frac{\mu_0 n_2 i_2}{\mu_0 n_1 i_1}$$

$$\Rightarrow \frac{B_2}{6.28 \times 10^{-2}} = \frac{100 \times \frac{i}{3}}{200 \times i}$$

$$\Rightarrow B_2 = \frac{6.28 \times 10^{-2}}{6} = 1.05 \times 10^{-2} \text{W b/m}^2$$

Question233

A charged particle of mass m and charge q travels on a circular path of radius r that is perpendicular to a magnetic field B. The time taken by the particle to complete one revolution is [2005]

Options:

A. $\frac{2\pi q^2 B}{m}$

B. $\frac{2\pi mq}{B}$

C. $\frac{2\pi m}{qB}$

D. $\frac{2\pi qB}{m}$

Answer: C

Solution:

Equating magnetic force to centripetal force,

$\frac{mv^2}{r} = qvB\sin 90°$

$\Rightarrow \frac{mv}{r} = Bq \Rightarrow v = \frac{Bqr}{m}$

Time to complete one revolution,

$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$

Question234

A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected along the direction of the fields with a certain velocity then [2005]

Options:

A. its velocity will increase

B. Its velocity will decrease

C. it will turn towards left of direction of motion

D. it will turn towards right of direction of motion

Answer: B

Solution:

Due to electric field, it experiences force and accelerates i.e. its velocity decreases.

Question235

Two concentric coils each of radius equal to 2πcm are placed at right angles to each other. 3 ampere and 4 ampere are the currents flowing in each coil respectively. The magnetic induction in Weber /m^2 at the centre of the coils will be ($\mu_0 = 4\pi \times 10^{-7} W\, b/A\, .\, m$)

[2005]

Options:

A. 10^{-5}

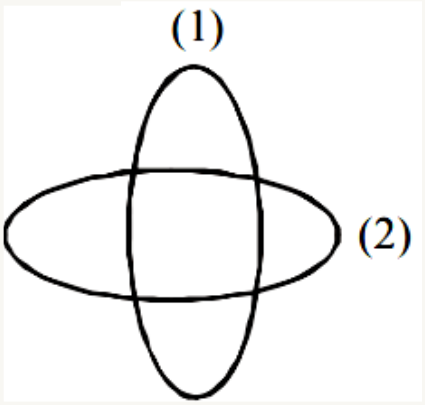
B. 12×10^{-5}

C. 7×10^{-5}

D. 5×10^{-5}

Answer: D

Solution:



The magnetic field due to circular coil (1) is

$$B_1 = \frac{\mu_0 i_1}{2r} = \frac{\mu_0 i_1}{2(2\pi \times 10^{-2})} = \frac{\mu_0 \times 3 \times 10^2}{4\pi}$$

Magnetic field due to coil (2)

Total magnetic field

$$B_2 = \frac{\mu_0 i_2}{2(2\pi \times 10^{-2})} = \frac{\mu_0 \times 4 \times 10^2}{4\pi}$$

Total magnetic field, $B = \sqrt{B_1^2 + B_2^2}$

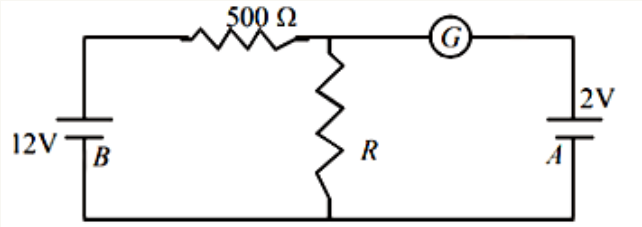
$$= \frac{\mu_0}{4\pi} . 5 \times 10^2$$

$$\Rightarrow B = 10^{-7} \times 5 \times 10^2$$

$$\Rightarrow B = 5 \times 10^{-5} W\, b/m^2$$

Question236

In the circuit, the galvanometer G shows zero deflection. If the batteries A and B have negligible internal resistance, the value of the resistor R will be -



[2005]

Options:

A. 100Ω

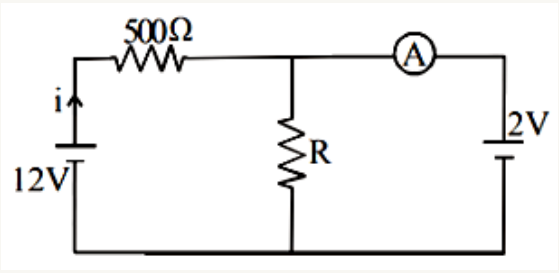
B. 200Ω

C. 1000Ω

D. 500Ω

Answer: A

Solution:



$12 - 2 = (500\Omega)i \Rightarrow i = \frac{10}{500} = \frac{1}{50}$

Again, $i = \frac{12}{500 + R} = \frac{1}{50}$

$\Rightarrow 500 + R = 600$

$\Rightarrow R = 100\Omega$

Question237

A moving coil galvanometer has 150 equal divisions. Its current sensitivity is 10 -divisions per milliampere and voltage sensitivity is 2 divisions per millivolt. In order that each division reads 1 volt, the resistance in ohms needed to be connected in series with the coil will be -
[2005]

Options:

A. 10^5

B. 10^3

C. 9995

D. 99995

Answer: D

Solution:

Resistance of Galvanometer,

$G = \frac{\text{Current sensitivity}}{\text{Voltage sensitivity}} \Rightarrow G = \frac{10}{2} = 5\Omega$

Here $i_g =$ Full scale deflection current $= \frac{150}{10} = 15\text{mA}$

$V =$ voltage to be measured $= 150$ volts
(such that each division reads 1 volt)

$\Rightarrow R = \frac{150}{15 \times 10^{-3}} - 5 = 9995\Omega$

Question238

A current i ampere flows along an infinitely long straight thin walled tube, then the magnetic induction at any point inside the tube is
[2004]

Options:

A. $\frac{\mu_0}{4\pi} \cdot \frac{2i}{r}$ tesla

B. zero

C. infinite

D. $\frac{2i}{r}$ tesla

Answer: B

Solution:

From Ampere's circuital law
$\int \vec{B} \cdot \vec{dl} = \mu_0$
$\Rightarrow B \times 2\pi r = \mu_0 i$
Here i is zero, for $r < R$, whereas R is the radius
$\therefore B = 0$

Question239

A long wire carries a steady current. It is bent into a circle of one turn and the magnetic field at the centre of the coil is B. It is then bent into a circular loop of n turns. The magnetic field at the centre of the coil will be
[2004]

Options:

A. 2nB

B. n^2B

C. nB

D. $2n^2B$

Answer: B

Solution:

Magentic field at the centre of a circular coil of radius$R$ carrying current i is $B = \frac{\mu_0 i}{2R}$
The circumference of the first loop $= 2\pi R$. If it is bent into n circular coil of radius r'.
$n \times (2\pi r') = 2\pi R$
$\Rightarrow nr' = R$ .....(1)

New magnetic field, $B' = \frac{n \cdot \mu_0 i}{2r'}$ .....(2)

From (1) and (2)

$B' = \frac{n\mu_0 i \cdot n}{2\pi R} = n^2 B$

Question240

The magnetic field due to a current carrying circular loop of radius 3 cm at a point on the axis at a distance of 4 cm from the centre is 54 μT. What will be its value at the centre of loop? [2004]

Options:

A. 125 μT

B. 150 μT

C. 250 μT

D. 75 μT

Answer: C

Solution:

The magnetic field at a point on the axis of a circular loop at a distance x from centre is,

$B = \frac{\mu_0 i a^2}{2(x^2 + a^2)^{3/2}}$

Magnetic field at the centre of loop is

$B' = \frac{\mu_0 i}{2a}$

$\therefore B' = \frac{B \cdot (x^2 + a^2)^{3/2}}{a^3}$

Put $x = 4$ & $a = 3$

$\Rightarrow B' = \frac{54(5^3)}{3 \times 3 \times 3} = 250\mu T$

Question241

Two long conductors, separated by a distance d carry current I_1 and I_2 in the same direction. They exert a force F on each other. Now the current in one of them is increased to two times and its direction is reversed. The distance is also increased to 3d . The new value of the force between them is
[2004]

Options:

A. $-\frac{2F}{3}$

B. $\frac{F}{3}$

C. $-2F$

D. $-\frac{F}{3}$

Answer: A

Solution:

Force acting between two long conductor carrying current,
$F = \frac{\mu_0}{4\pi}\frac{2I_1I_2}{d} \times l$ .....(i)
Where d = distance between the conductors
l = length of conductor
In second case, $F' = -\frac{\mu_0}{4\pi}\frac{2(2I_1)I_2}{3d}l$ .....(ii)
From equation (i) and (ii), we have
$\therefore \frac{F'}{F} = \frac{-2}{3}$

Question242

A particle of mass M and charge Q moving with velocity $\vec{v}$ describe a circular path of radius R when subjected to a uniform transverse magnetic field of induction B. The work done by the field when the particle completes one full circle is
[2003]

Options:

A. $\left(\frac{M v^2}{R}\right) 2\pi R$

B. zero

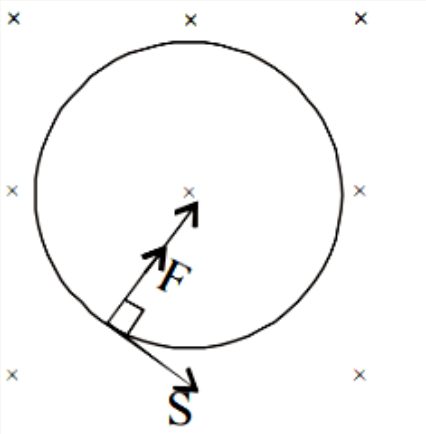
C. BQ2πR

D. BQv2πR

Answer: B

Solution:

The work done, $d W = F d s \cos\theta$
The angle between force and displacement is $90°$ Therefore work done is zero.



Question243

If an electron and a proton having same momenta enter perpendicular to a magnetic field, then [2002]

Options:

A. curved path of electron and proton will be same (ignoring the sense of revolution)

B. they will move undeflected

C. curved path of electron is more curved than that of the proton

D. path of proton is more curved.

Answer: A

Solution:

When a moving charged particle is subjected to a perpendicular magnetic field, then it describes a circular path of radius.

$r = \frac{p}{qB}$

where q = Charge of the particle

p = Momentum of the particle

B = Magnetic field

Here p, q and B are constant for electron and proton, therefore the radius will be same

---

Question244

The time period of a charged particle undergoing a circular motion in a uniform magnetic field is independent of its [2002]

Options:

A. speed

B. mass

C. charge

D. magnetic induction

Answer: A

Solution:

The time period of a charged particle of charge q and mass m moving in a magnetic field (B) is

$T = \frac{2\pi m}{qB}$

Clearly time period is independent of speed of the particle.

Question245

If in a circular coil A of radius R, current I is flowing and in another coil B of radius 2R a current 2I is flowing, then the ratio of the magnetic fields B_A and B_B, produced by them will be
[2002]

Options:

A. 1

B. 2

C. $\frac{1}{2}$

D. 4

Answer: A

Solution:

Magnetic field induction at the centre of current carrying circular coil of radius r is

$$B = \frac{\mu_0 I}{4\pi R} \times 2\pi$$

Here $B_A = \frac{\mu_0 I}{4\pi R} \times 2\pi$

and $B_B = \frac{\mu_0 2I}{4\pi 2R} \times 2\pi$

$$\Rightarrow \frac{B_A}{B_B} = \frac{I/R}{2I/2R} = 1$$

Question246

If a current is passed through a spring then the spring will
[2002]

Options:

A. expand

B. compress

C. remains same

D. none of these

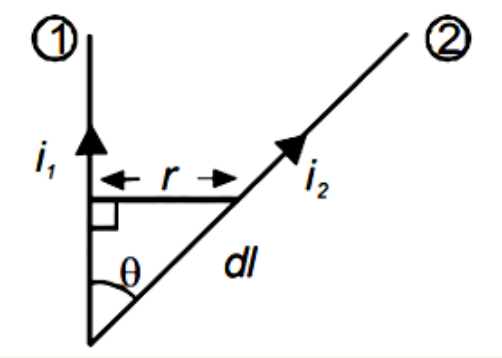
Answer: B

Solution:

When current is passed through a spring then current flows parallel in the adjacent turns in the same direction. As a result the various turn attract each other and spring get compress.

Question247

Wires 1 and 2 carrying currents i_1 and i_2 respectively are inclined at an angle theta to each other. What is the force on a small element d l of wire 2 at a distance of r from wire 1 (as shown in figure) due to the magnetic field of wire 1?



[2002]

Options:

A. $\frac{\mu_0}{2\pi r} i_1 i_2 d l \tan\theta$

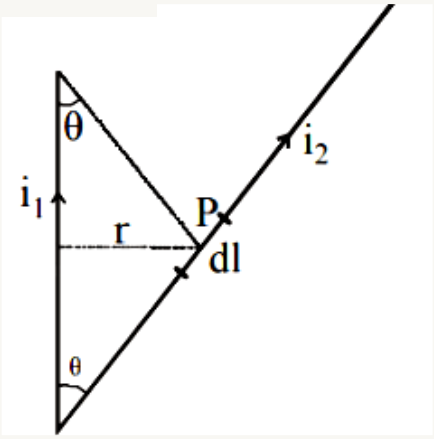
B. $\frac{\mu_0}{2\pi r} i_1 i_2 d l \sin\theta$

C. $\frac{\mu_0}{2\pi r} i_1 i_2 d l \cos\theta$

D. $\frac{\mu_0}{4\pi r} i_1 i_2 d l \sin\theta$

Answer: C

Solution:



Magnetic field due to current in wire 1 at point P distant r from the wire is $B = \frac{\mu_0}{4\pi}\frac{i_1}{r}[\cos\theta + \cos\theta]$

$B = \frac{\mu_0}{2\pi}\frac{i_1 \cos\theta}{r}$

This magnetic field is directed perpendicular to the plane of paper, inwards.
The force exerted due to this magnetic field on current element $i_2 d l$ is

$dF = i_2 d l^{1} B \sin 90°$

$\therefore dF = i_2 d l B$

$\Rightarrow dF = i_2 d l \left(\frac{\mu_0}{4\pi}\frac{i_1 \cos\theta}{r}\right)$

$= \frac{\mu_0}{2\pi r} i_1 i_2 d l \cos\theta$
